PHASE-TTC
From Motion to Time

Causal Event–RGB Fusion for Signed Time-to-Contact Estimation

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Abstract. Estimating time-to-contact from vision requires measuring how an object's apparent scale changes. Event cameras record rapid brightness changes, while RGB frames provide scene appearance; combining them requires reconciling different temporal and visual representations. We introduce PHASE-TTC, which estimates motion independently in each modality and combines those estimates in a shared phase coordinate. A compact residual head corrects the RGB estimate using phase, timing, and observation-support features from causal history. On the archived GarlTTC Test leaderboard of 28 October 2026, the submitted system ranks first among three displayed entries, with Overall Motion-in-Depth error 35.80, a 20.49% reduction relative to Garl-TTC and zero scorer-defined failures. Matched readout experiments hold the final upstream models and temporal histories fixed. Learned corrections improve on simple mixtures, but three-seed controls show no consistent transfer advantage from event access or from phase rather than inverse-TTC parameterization. A geometric extension improves local road accuracy beyond a fixed mixture, while its pooled advantage remains uncertain. These secondary evaluations reuse development-inspected recordings. A paced replay delivers all 250 predictions with numerical parity, but its 29.14 ms p95 response exceeds the 20 ms target. The evidence supports the complete system's benchmark performance and bounds the component and deployment claims.

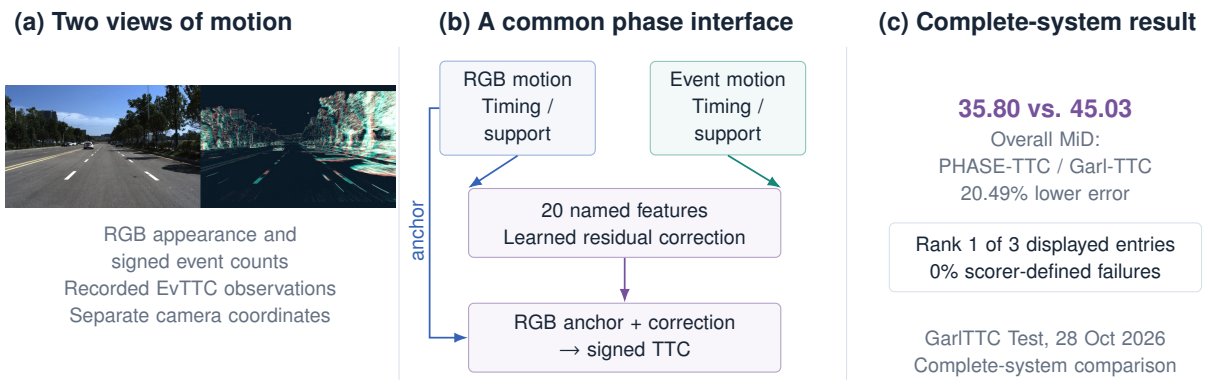


Figure 1. Estimate motion separately; combine it through phase. (a) Recorded EvTTC RGB and event observations [12]. (b) Each modality supplies phase, timing, and support features; a learned correction adjusts the RGB anchor. (c) The complete submitted system improves the archived GarlTTC Overall MiD score. The benchmark comparison does not isolate the contribution of any one component.

1. Introduction

As an object approaches a camera, its apparent size grows. For rigid frontal motion at approximately constant relative speed, the ratio of apparent size s to expansion rate $\dot{s}$ gives time-to-contact (TTC), $T \simeq s/\dot{s}$. A recognizer can identify the object while leaving this rate undetermined; estimating when contact will occur requires recovering the change in scale. Rotation, occlusion, and camera motion complicate that measurement because they can alter the image independently of closing distance.

RGB frames provide texture and spatial context, while event cameras report asynchronous brightness changes along moving contours between frames [3]. To exploit both, a fusion method must account for the timing and support of each observation as well as the different representations learned from the two streams. We address this problem by estimating motion within each modality and combining quantities with shared meanings.

In PHASE-TTC, each branch supplies phase estimates together with timing and support information to a small residual network that adjusts an RGB anchor. The signed phase coordinate gives the branches a common description of approach and recession over a reference interval, allowing each to keep its own hidden representation. We preserve apparent expansion through common object coordinates and use temporal refiners to incorporate observations available before each query.

We evaluate PHASE-TTC as a complete TTC system, then examine its readout with matched controls. The private benchmark establishes system performance. To investigate the contribution of fusion, we hold the final upstream estimators and temporal histories fixed, compare simple combinations of their outputs, and train equally sized correction heads with or without event information. A separate local experiment asks whether learned routing adds value once apparent box expansion is available.

1.1 Contributions

We present a causal event–RGB estimator with three experimentally testable contributions:

1. **A specified system and training procedure.** Nine fitted components estimate motion, refine temporal context, and combine twenty named physical and timing fields through an RGB-anchored residual head. Table 1 describes the complete computation and its supervision.
2. **A benchmark result with matched readout controls.** The submitted system achieves 35.80 Overall MiD on the archived GarlTTC Test leaderboard. We complement this complete-system result with equal-capacity, equal-history head comparisons and simple fusion baselines on fixed final-model features (Sections 4–6).
3. **An analysis of adaptation and execution limits.** Common-support geometry comparisons identify where a learned router helps beyond a fixed mixture. Timed replay measures delivery and deadline compliance separately (Sections 7–8).

The supplement retains the earlier predictive-learning, temporal, and physical-model investigations. They explain the design's development without serving as substitutes for the final-system comparisons.

1.2 Relation to prior work

Event-based TTC methods recover motion through geometric fitting and spatio-temporal registration [6]. EvTTC provides synchronized observations and a dedicated TTC benchmark [12]; DSEC illustrates event–frame acquisition in demanding driving conditions [5]. The eAP study introduces geometry-aware representations, the logarithmic motion-in-depth metric, and early and late event–RGB fusion in the Garl models [7]. Our contribution is the particular estimator interface: named motion estimates, timing, and support drive a residual correction around RGB. The phase coordinate itself comes from the benchmark geometry. Relative to feature-level late fusion, the interface exposes the meanings of its inputs and allows separate producer fitting. The matched tests below measure the utility of this readout under fixed

producers; a matched hidden-feature fusion comparison is still needed to establish superiority over that alternative.

I-JEPA predicts target representations from context [1], while VICReg regularizes representation variance and covariance [2]. These methods motivate our predictive-learning controls. Work on slow features shows that predictable distractors can dominate a learned state [11]; our experiments test how nuisance dynamics affect the information available for TTC. They support conclusions about the tested objectives and settings rather than a general impossibility of predictive learning.

2. PHASE-TTC: independent estimates, shared phase

2.1 Deployed model composition

PHASE-TTC deploys a causal chain of nine fitted components comprising four modality-specific producers, two pairwise temporal heads, two context refiners, and one fusion head. RGB and events each use a transport-trained baseline producer (A5) and a coarse-to-fine producer (C2F). The pairwise readouts extract phase features from their A5 representations, after which the context refiners incorporate temporal history. The fusion head anchors its correction on the refined RGB estimate. Figure 2 traces this path; the five auxiliary rotations serve training only.

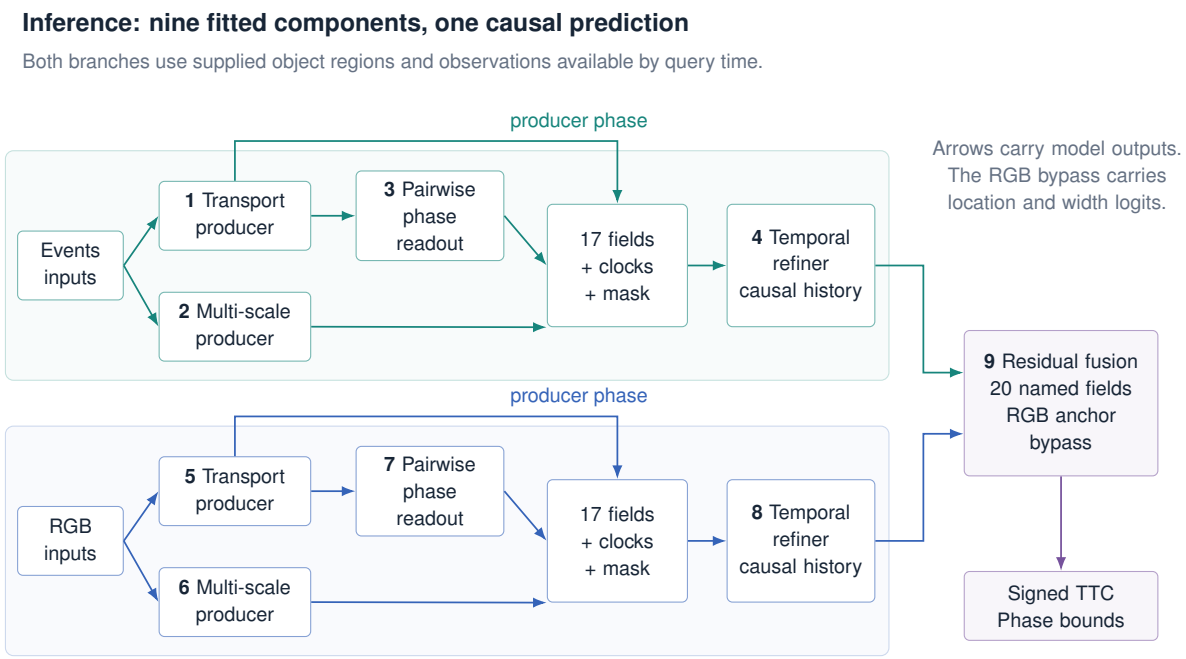


Figure 2. Causal nine-component inference graph. Supplied object regions feed parallel RGB and event producers, pairwise temporal heads, and temporal refiners. Twenty named fields enter the final RGB-anchored fusion head.

2.2 Object coordinates and causal history

Each query includes supplied object coordinates. A common spatial frame covers both endpoint regions and preserves apparent scale change, which independent tight crops can erase. The evaluated system therefore assumes external localization.

Both branches use observations available by query time t, with up to 650 ms of causal history. Runtime begins at its declared measurement boundary and ends when the host receives the complete TTC output. Figure 3 separates the sensing interval from response time; its observation marks are schematic.

Table 1. The nine fitted components in the submitted inference chain. RGB and event branches share the listed functional roles but have separate fitted weights. All targets and privileged supervision belong to fitting; inference uses sensor observations and supplied object regions.

Components	Inputs	Architecture and outputs	Fitting signal
Two transport producers	Shared-region temporal crops; observed endpoint intervals	Convolutional endpoint encoder, local feature correlation and scale residual; dense features, geometry and TTC	TTC, foreground/box geometry and RGB-derived relational distillation
Two coarse-to-fine producers	Same modality-specific crop contract	Two-scale local transport with a learned gate; alternate motion estimate	Producer objective with geometric and relational supervision
Two pairwise readouts	133 fields derived from each transport producer's endpoints	LayerNorm, width-128 GELU MLP and scalar phase output	Smooth-L1 phase regression
Two context refiners	17 fields per observation, four timing fields and validity mask	Two-layer width-160 GRU; phase correction around current expert median and phase widths	Group-weighted phase error, quantile losses and expert-relative costs
One fusion head	Ten named fields per modality; RGB anchor and width logits	Two width-32 layers with LayerNorm/SiLU; location and width corrections	Group-weighted phase error and quantile losses on out-of-fold features

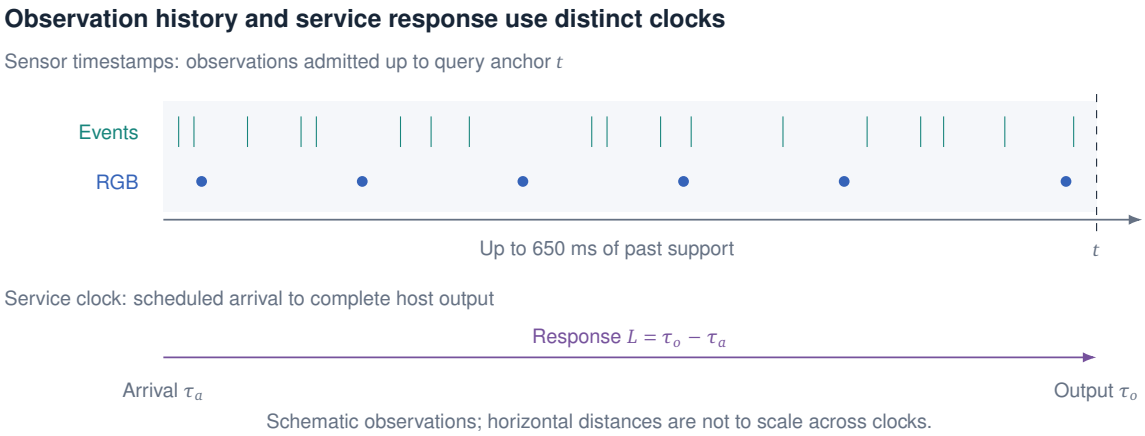


Figure 3. Causal support and response-time boundaries. RGB and event observations precede the sensor-time query anchor t. A separate service clock measures from scheduled arrival τ_a to complete host output τ_o; the two axes are schematic.

2.3 Phase as a shared motion coordinate

Let T denote signed TTC, positive for approach and negative for recession. With reference interval $d_0 = 0.1$ s, we use

$$\phi(T) = -\log(1 - d_0/T), \qquad T < 0 \text{ or } T > d_0. \tag{1}$$

For large $|T|$, $\phi(T) \simeq d_0/T$: slow approach and recession lie near zero. Sensitivity increases toward the positive domain boundary $T = d_0$. This is a shared signed coordinate, not a removal of the near-contact singularity. The benchmark evaluates error in this coordinate; Section 3 defines its weighting.

2.4 A residual correction around RGB

Each modality $m \in \{R, E\}$ exposes a vector $\mathbf{x}_m \in \mathbb{R}^{10}$: a raw phase location, the emitted point phase, three expert phases, four timing fields, and a valid-history count normalized by eight. The hidden state stays within its own branch. Let $D = \mathrm{diag}(\alpha^{-1} I_5, I_5)$, with $\alpha = 0.03$. The fusion input is

$$\mathbf{z} = [D\mathbf{x}_R; D\mathbf{x}_E] \in \mathbb{R}^{20}. \tag{2}$$

Thus corresponding coordinates have the same meaning across modalities, while timing and support keep their native scaling. Appendix E.7 defines the timing fields precisely.

The residual network $g_\theta : \mathbb{R}^{20} \to \mathbb{R}^3$ uses two width-32 layers with LayerNorm and SiLU activations:

$$\mathbf{c} = g_\theta(\mathbf{z}). \tag{3}$$

Its 1,955 parameters include the affine maps and both LayerNorms. Given the RGB raw-location anchor ℓ_R and width logits $\mathbf{w}_R$, the corrected location and widths are

$$\ell_F = \ell_R + \alpha c_0, \tag{4}$$

$$[\sigma_l, \sigma_u]^\top = \alpha\,\mathrm{softplus}(\mathbf{w}_R + [c_1, c_2]^\top). \tag{5}$$

Zero initialization of the final affine layer's weights and bias starts optimization at the RGB anchor. The factor α scales the residual; it imposes no bound on c_0. The widths parameterize phase-interval endpoints and require empirical calibration before a probabilistic interpretation.

For a finite emitted prediction, we first clamp ℓ_F to $[\phi_{\min}, \phi_{\max}]$, where $\phi_{\min} = -\log 2 + 10^{-6}$ and $\phi_{\max} = 8$, then compute

$$q_F = \frac{1 - \exp(-\mathrm{clip}(\ell_F, \phi_{\min}, \phi_{\max}))}{d_0}, \qquad \widehat{T} = \frac{\mathrm{sgn}_+(q_F)}{\max(|q_F|, 1/60\,\mathrm{s}^{-1})}. \tag{6}$$

Here $\mathrm{sgn}_+(0) = 1$, so zero phase emits +60 s. The rate floor limits the magnitude of emitted TTC near zero expansion. Figure 4 shows the anchor and correction paths; Appendix A gives the exact array layout, stable numerical inverse, and interval bounds.

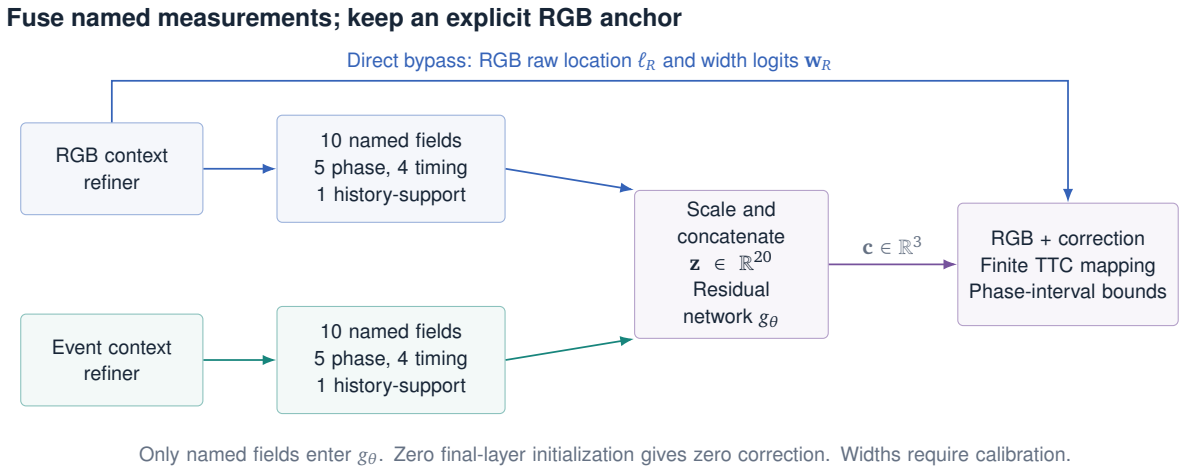


Figure 4. Shared measurements and an RGB anchor. The two ten-field descriptors feed the residual network. RGB location and width logits also reach the output directly; only the learned correction passes through the network. Hidden features do not enter the fusion input.

2.5 Submission coverage and benchmark status

The official submission contains finite predictions without fallback for 6,762 query tokens across twelve sequences: 5,805 physical queries and 957 aliases. Aliases identify repeated physical queries and do not represent additional inference work.

Section 4 reports the test result for this graph. Section 6 holds the final upstream features fixed and refits matched correction heads to test their use of event information.

3. Training and evaluation

3.1 Grouped out-of-fold training and final refitting

Five rotations of eight-sequence blocks generate leakage-controlled inputs for the upper stages. In each rotation, twenty-four groups fit the four producers and two pairwise temporal heads, eight different groups fit the context refiners and normalizers, and the remaining eight groups generate out-of-fold

features. A query group never contributes to supervised fitting of the models that generate its upper-stage features.

Within each auxiliary rotation, the context refiners fitted on its eight context groups generate twenty-field fusion features for its eight held-out groups. These fold-local feature sets are pooled to train the final fusion head. Separately, pooled seventeen-field RGB and event context inputs train the context refiners and normalizers used in the deployed forty-sequence chain; these final context models do not generate the fusion head's training inputs. Final pairwise temporal heads use representations from their corresponding forty-sequence transport-trained producers, so the out-of-fold claim applies to the specified context and fusion interfaces rather than every PAIR fit.

We pool only named quantities whose meanings agree across rotations. Targets are joined after held-out feature generation. The two sets of pooled features fit different stages: context inputs fit the deployment refiners, while fold-local refined outputs fit fusion. Figure 5 separates these paths. Appendix A specifies their exact storage layout.

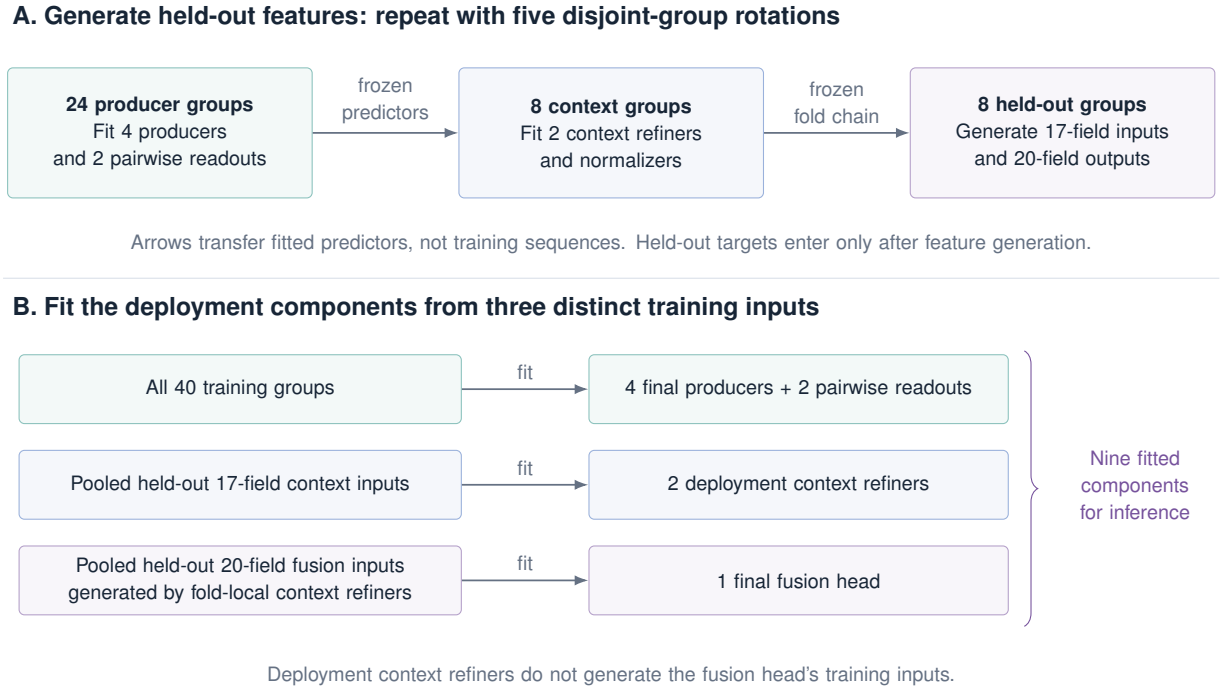


Figure 5. Five-rotation fitting scheme. Each rotation separates producer fitting, context fitting, and held-out feature generation. Final refitting produces one forty-sequence inference chain.

This grouped stacking procedure separates feature generation from supervised fitting at each stage. Deployment uses one chain comprising four producers, two pairwise temporal heads, two context refiners, and the fusion head, with producers refitted on all forty groups. Auxiliary producer fits use twenty-four groups per rotation. The resulting change in producer-training population and prior architecture-selection exposure are considered when interpreting development results.

Producer training combines relational DINO teacher signals, foreground geometry supervision, and task-specific adaptation. The deployed chain no longer evaluates the generic teacher, but its supervision remains part of the training recipe.

3.2 Metrics and target coordinates

TTC error in seconds, inverse-TTC error, benchmark phase error, and detection of urgent risk measure different aspects of behaviour. We evaluate phase using Equation 1. Negative TTC denotes recession.

The interval $[0, 0.1]$ s is outside this transform's domain. The official benchmark scorer defines motion-in-depth (MiD) error as

$$\mathrm{MiD}(\widehat{T}, T) = 10^4 \left| \log\left(1 - \frac{0.1}{T}\right) - \log\left(1 - \frac{0.1}{\widehat{T}}\right) \right| = 10^4 |\phi(\widehat{T}) - \phi(T)|. \tag{7}$$

MiD measures phase displacement. At large positive T, the first-order approximation $\phi(T) \simeq 0.1/T$ compresses differences in seconds, while sensitivity rises near the positive domain boundary. MAE in seconds complements MiD by retaining the physical time scale.

The scorer groups target TTC into four right-closed intervals: crucial $c : (0, 3]$ s, small $s : (3, 6]$ s, large $l : (6, 10]$ s, and negative $n : (-10, 0]$ s. The grouping includes zero in the negative band, although the phase transform is undefined at zero; metric validity is checked separately. Targets outside these bands do not contribute to the band-weighted score.

The scorer computes the band-weighted score

$$\text{overall_MiD} = 0.5\,\mathrm{MiD}_c + 0.3\,\mathrm{MiD}_s + 0.1\,\mathrm{MiD}_l + 0.1\,\mathrm{MiD}_n, \tag{8}$$

and a sample-weighted mean MiD. Missing bands remain missing, without weight redistribution. The complete band-weighted score requires targets in all four bands. The source archive fixes the scorer version and includes checks of exact, imperfect, and missing-band cases.

Metric reductions retain failures explicitly. The wrapper records invalid predictions by band and leaves its strict mean unavailable when any row is invalid. The official failure rate differs from the false-negative rate for urgent collision warnings. Prediction coverage, invalid-MiD count, and risk recall accompany the corresponding error metrics. A finite saturated TTC may still carry an unsupported-measurement flag.

3.3 Official benchmark and exploratory evaluations

Table 2. Evaluation populations and the quantities each supports. Historical46 and Extended-new share three sequence groups but no query identities. The 434-query inference snapshot and corrected local comparison share identities but answer different questions.

Population	Size	Interpretation
Official GarlTTC Test	6,762 scored tokens	Private targets; band-weighted MiD and scorer-defined failures.
Published Garl EvTTC	3 sequences	Published RTE; target/query correspondence to this run unverified.
EvTTC competition snapshot	434 queries; 10 sequences	Original 28 October predictions, coverage and disagreement; targets not verified at that snapshot.
Corrected local EvTTC	Same 434 queries	Original versus adapted PHASE-TTC error; historically exposed, local scoring, Slider-750 official equivalence unresolved.
Historical EvTTC transfer	468 targets; 32 sequences	Original target assignment, superseded for Extended-new. Historical46 is unaffected by the correction.
EvTTC Historical46	46 targets; 3 sequences	Original target assignment, superseded for Extended-new. Historical46 is unaffected by the correction.
Corrected EvTTC transfer	460 targets; 32 sequences	All 508 fixed candidates reassociated; development-exposed, supplied ROI.
Cadence development	2,608 queries; 25 recordings	Recording-macro paired comparison; 24 road recordings and Slider-500; historically exposed.
Native eAP replay	5,805 physical queries; 6,762 template tokens	Local output sensitivity and coverage; private Test12 targets not scored for the candidate.

The official GarlTTC Test submission covers twelve sequences and uses forty-sequence event–RGB inputs with up to 650 ms of causal history. Section 4 reports the public ranking as of 28 October 2026 over

6,762 scored tokens, including aliases of repeated physical queries [4]. The evaluator scores submitted TTC values without standardizing temporal support, localization, supervision, or runtime across systems. Separate development experiments probe mechanisms within their tested configurations.

3.4 Local targets, scoring, and development reuse

For local adaptation on EvTTC, we evaluate 434 corrected query correspondences across ten sequences using query-weighted MAE and mean relative error. We score the signed predictions and, separately, apply the organizer-described offline interpolation of negative predictions between valid outputs to both models. Their archived outputs contain two and one negative values, respectively. Interpolation may use later valid predictions and belongs only to offline scoring. Section 7 reports the paired results and the sequence-specific scoring discrepancy.

The final quality-router fit excludes the ten evaluation recordings from its optimizer updates, but those recordings informed earlier design choices. The 460-query head comparisons likewise reuse inspected EvTTC recordings, while fitting only on source-domain eAP features. We interpret both as development evidence. Appendix D retains the separate cadence and downstream selection experiments.

3.5 Independence and paired comparisons

Splits and resampling units are specified separately for each experiment. Historical V5–V7 intervals resample sequence–track clusters; later studies use complete sequence groups, a sequence-to-track hierarchy, or scenario families. A sample-weighted sequence bootstrap carries all rows from each sampled group into the replicate and recomputes the ratio of summed losses to summed counts; sequence-macro summaries weight groups equally. A sequence-macro point estimate does not imply that its archived interval used sequence-level resampling. All reductions inherit any development exposure. Appendix F identifies the study-specific units and gives the lineage, causality, and latency checks.

Three head seeds quantify optimization variability conditional on their parents; they do not independently replicate the upstream representation. The additional transfer bootstrap is post hoc: it resamples saved paired predictions without changing a model, split, or historical interval (Appendix G).

For the new matched readout and geometry studies, 10,000 complete-recording resamples use seed 20261111. We specify arms, budgets, fixed mixture weights, and reductions before executing these analyses. This freezes the retrospective comparison; it does not remove prior exposure to its evaluation recordings.

4. Official GarlTTC benchmark result

4.1 Public ranking and bandwise accuracy

As of 28 October 2026, PHASE-TTC ranks first among the three displayed submissions on the official GarlTTC Test leaderboard. Submission 1016512, listed as *PHASE-TTC (TRAIN40, 650 ms causal history)*, records Overall MiD 35.80 over 6,762 scored tokens with a 0% scorer-defined failure rate. The next displayed score is 45.03 for the Garl-TTC reference. [4]

The scorer returns Overall MiD 35.8015 at full precision, consistent with the rounded public entry. Relative to Garl-TTC's published 45.03, PHASE-TTC reduces the ranking metric by 20.49%. All four TTC bands improve (Table 3); the large positive-TTC band has the largest relative reduction and the negative band the smallest. Percentage reductions combine the unrounded PHASE-TTC values with Garl-TTC's published two-decimal scores.

Table 3. Official GarlTTC Test scores on 28 October 2026 (lower is better). PHASE-TTC scores are shown to four decimals and Garl-TTC at published precision. Relative reductions use unrounded PHASE-TTC scores. Both entries have 6,762 scored tokens and 0% scorer-defined failures.

Metric	PHASE-TTC	Garl-TTC	Decrease
Overall	35.8015	45.03	20.49%
Crucial	43.5190	53.13	18.09%
Small	28.2568	37.58	24.81%
Large	25.7179	40.61	36.67%
Negative	29.9314	31.29	4.34%

4.2 Where the weighted difference comes from

For band b, the additive contribution is $w_b(\mathrm{MiD}_b^{\mathrm{Garl}} - \mathrm{MiD}_b^{\mathrm{PHASE}})$. The crucial and small bands contribute 82.4% of the Overall difference under weights 0.5 and 0.3. The large band's 0.1 weight limits its contribution despite its larger relative reduction; the negative band contributes least.

4.3 Metric and evaluation population

The four TTC bands cover $(0, 3]$, $(3, 6]$, $(6, 10]$, and $(-10, 0]$ seconds. MiD measures error in the motion-in-depth coordinate defined in Section 3. The ranking uses the weighted Overall MiD; the scorer also reports `mean_MiD`, an unweighted average over finite per-query errors. [9]

The 6,762 scored tokens represent 5,805 distinct physical queries and 957 aliases in the submitted package. Aliases reuse predictions at the same sensor query. The official failure criterion marks nonfinite TTC predictions or predictions with $|\widehat{T}| < 0.1\,\mathrm{s}$. Band scores average finite MiD values. The available aggregate outputs support bandwise analysis; studying individual errors requires per-query scores.

4.4 Scope of the comparison

The official result establishes the accuracy of the complete forty-sequence event–RGB system with up to 650 ms of causal history. Its gains span all four TTC bands; the high weights on imminent and short-horizon contact make these bands responsible for 82.4% of the weighted improvement. Systems on the leaderboard retain their own localization, supervision, and runtime choices. Section 6 examines modality and temporal dependencies through controlled development comparisons.

The official result measures the submitted system on private test targets. The exploratory EvTTC transfer compares frozen routes on development-exposed, positive-TTC sequences with annotated object regions. Section 8 evaluates execution time separately, including deadline misses.

5. When does fusion improve on RGB?

The frozen-modality comparison tests whether the shared phase interface improves errors in seconds beyond the benchmark setting. Fusion has lower point MAE and p95 than RGB alone on the revised cohort, whereas RGB retains lower median and relative error. Paired sequence-bootstrap intervals for mean, median, and p95 differences include zero.

5.1 Evaluation population

We evaluate frozen predictions on 460 eligible queries from 32 EvTTC sequences. The registered sampling contains 508 candidate queries; eligibility requires a public TTC label within 50 ms of the query, with ties assigned to the earlier timestamp. All eligible targets are positive. The primary annotated object supplies the query region. These sequences had been used during development, as specified in Section 3.

A source audit identified a reversed low/medium target assignment for two recordings in our earlier analysis. Matching the public download identities and source hashes corrected this association without changing sensor queries or predictions. We report only the corrected cohort here; Appendix A.3 records all candidate counts, exclusions, and the superseded assignment.

5.2 Corrected error distribution

Relative time error remains $100|\widehat{T} - T|/|T|$. Pooled summaries weight queries equally; sequence-macro summaries weight each sequence group equally. Temporal support differs across complete systems: PHASE-TTC uses up to 650 ms of history, while Garl uses two observations separated by 0.1 s.

Cohort / model	n	MAE	Median	p95	RTE (%)	Macro MAE	Macro RTE (%)
Corrected extended: PHASE-TTC	460	0.813	0.446	2.742	29.112	0.827	30.016
Corrected extended: RGB context	460	0.851	0.390	3.038	29.017	0.859	29.368
Corrected extended: Event context	460	1.376	0.844	3.707	58.154	1.377	57.764
Corrected extended: Garl full	460	3.166	0.635	4.781	131.268	3.041	126.366
Historical46: PHASE-TTC	46	0.141	0.136	0.250	7.748	0.159	8.419
Historical46: RGB context	46	0.113	0.099	0.264	6.387	0.114	6.246
Historical46: Event context	46	0.286	0.241	0.692	15.822	0.289	15.566
Historical46: Garl full	46	0.250	0.186	0.561	13.327	0.282	14.649

Table 4. Corrected-label transfer: 460 eligible of 508 extended candidates across 32 sequences. Historical46 remains separate (46 queries, three sequences). Absolute errors are seconds; macro gives equal weight to sequences.

Fusion MAE is 0.813 s against RGB's 0.851 s; its empirical p95 is 2.742 s against 3.038 s. RGB retains lower median error, 0.390 versus 0.446 s, and slightly lower RTE, 29.02% versus 29.11%.

5.3 Exceedance and paired uncertainty

The full corrected distribution retains signed predictions, including large errors. Appendix K shows the fraction at or above each absolute-error threshold. Maximum absolute error is 11.40 s for fusion and 15.90 s for RGB. These are query-level errors under the corrected labels, not scorer-defined invalid predictions.

The PHASE-minus-RGB mean difference is -0.038 s with interval $[-0.147, +0.061]$ s; the p95 difference is -0.296 s with interval $[-0.657, +0.193]$ s. All three paired intervals include zero. The procedure uses 10,000 complete-sequence resamples with linear quantiles and the same frozen models. Appendix G gives complete sequence results and distinguishes this corrected analysis from the historical one.

Cohort	Paired contrast	Estimand	Difference	95% interval
Extended460	PHASE minus RGB	mean	-0.038	[-0.147, +0.061]
Extended460	PHASE minus RGB	median	+0.055	[-0.042, +0.136]
Extended460	PHASE minus RGB	p95	-0.296	[-0.657, +0.193]
Extended460	PHASE minus Garl	Macro MAE_s	-2.214	[-4.951, -0.447]
Extended460	PHASE minus Garl	Macro RTE_pct	-96.350	[-211.946, -16.629]
Historical46	PHASE minus RGB	mean	+0.028	[+0.005, +0.103]
Historical46	PHASE minus RGB	median	+0.036	[+0.026, +0.068]
Historical46	PHASE minus RGB	p95	-0.014	[-0.098, +0.109]
Historical46	PHASE minus Garl	Macro MAE_s	-0.123	[-0.178, -0.084]
Historical46	PHASE minus Garl	Macro RTE_pct	-6.230	[-8.682, -4.767]

Table 5. Whole-sequence paired percentile intervals. PHASE–RGB uses 10,000 draws, seed 20261028, for pooled query errors; PHASE–Garl uses 2,000 draws, seed 7, for equal-sequence means. Linear quantiles; negative favors PHASE. MAE, median and p95 use seconds; RTE uses percentage points. Exposed development cohorts.

6. Matched tests of the final fusion readout

The official result compares complete systems. Here we isolate readout choices while holding the final RGB and event producers, temporal refiners, supplied regions, and causal histories fixed. We use the same 460 corrected transfer queries throughout. These recordings informed earlier development; the new comparisons quantify conditional effects on that cohort.

6.1 Simple combinations of identical branch outputs

We first replace the released fusion head with equal mixtures of the two final branch predictions, combining them in phase, inverse TTC, or seconds. Each mixture receives the same branch outputs and temporal support. All emit finite TTC under their specified numerical policies. Table 6 shows that learned fusion improves mean error over these fixed combinations. Its comparison with the strong RGB branch remains uncertain (Section 5).

Frozen readout	MAE (s)	Median (s)	p95 (s)
Released fusion	0.813	0.446	2.742
RGB context	0.851	0.390	3.038
Event context	1.376	0.844	3.707
Equal phase mixture	0.997	0.519	2.759
Equal inverse-TTC mixture	1.007	0.518	2.762
Equal TTC mixture	1.050	0.614	3.192

Table 6. Frozen final-branch readouts on the identical 460 queries. Equal mixtures have no fitted parameters; the released head has 1,955. This comparison holds upstream information fixed but does not match readout capacity.

6.2 Equal-capacity heads with matched fitting

To distinguish event access from added correction capacity, we fit three heads: the original phase readout; an identical head whose ten event fields are zero during fitting and evaluation; and a residual readout expressed in inverse TTC. All contain 1,955 parameters, start with a zero final affine map, and retain the RGB anchor. The zero-event head still receives RGB timing and support. Upstream weights are frozen throughout.

Each arm uses the same 88,744 source-domain out-of-fold feature rows from forty eAP recordings, the same seeded minibatch sequence, and 12,044 AdamW updates at batch size 128. We retain the original warmup/cosine schedule, weight decay, clipping, and phase-error plus quantile objective. Seeds 7, 13, and 21 vary only the correction-head fit. EvTTC queries and targets are excluded from fitting and checkpoint selection; every arm uses its final update. The inverse-TTC arm replaces phase inputs with normalized rates and predicts an RGB rate correction before conversion to the common phase loss. Appendix W specifies the transforms and archived source bindings.

We also fit a single scalar phase-mixture weight on the source training rows alone. Searching the specified 101-point grid gives event weight 0.31. This is a train-calibrated simple baseline, evaluated once on EvTTC.

Readout	MAE (s)	Relative error (%)
Phase correction, real events	0.847 ± 0.038	30.92 ± 2.05
Phase correction, zero events	0.838 ± 0.021	29.12 ± 1.16
Inverse-TTC correction, real events	0.847 ± 0.041	30.80 ± 2.14
Source-calibrated scalar phase mixture	0.996	36.23

Table 7. Matched final-head fitting. Values are the mean and sample standard deviation of the three seed-level scores; predictions are not ensembled. The scalar mixture has one fitted coefficient and no seed-dependent optimization.

6.3 What the controlled comparison establishes

Real events improve MAE over the zero-event control for one seed and worsen it for two. The mean zero-minus-real difference is −0.0094 s, with a descriptive recording-bootstrap interval spanning zero. Inverse-TTC and phase corrections are also indistinguishable at this resolution. We average paired losses over the three fixed seeds before resampling complete recordings, preserving query correspondence; seeds do not multiply the number of independent recordings.

Paired contrast	ΔMAE (s)	95% interval (s)
Zero events minus real events	-0.0094	[−0.0780, +0.0630]
Inverse TTC minus phase	-0.0003	[−0.0061, +0.0058]
Calibrated mixture minus phase	+0.1492	[+0.0064, +0.3762]

Table 8. Paired differences after averaging query losses over fixed head seeds; 10,000 complete-recording resamples across 32 recordings. These intervals describe the inspected cohort and condition on the shared upstream models.

The learned readouts improve on the calibrated scalar mixture, but the tests give no consistent incremental benefit from event access and no observed advantage of phase over the tested inverse-TTC parameterization. A useful RGB correction, rather than event complementarity alone, can therefore account for the readout's gain on this cohort. The prior positive event control used a different adapted model; Appendix I preserves that result without transferring its attribution to the submitted system. A matched hidden-feature fusion model and genuinely unused recordings remain necessary to compare broader fusion designs.

7. Does learned routing add value beyond box geometry?

The geometric extension combines the frozen fused, RGB, and event estimates with apparent expansion from supplied box history. We fit inverse TTC from width, height, and square root of area over the preceding 650 ms, using median pair slopes and a query-time intercept. A 688-parameter gate uses fit quality and expert agreement to weight the four inverse-TTC estimates; the delivered adapter averages

three independently seeded gate outputs in seconds. Its fit uses 2,608 queries from 24 road recordings and Slider-500. Appendix J gives the geometry, 33 input fields, training recipe, and fallback rules.

7.1 Matched experts, timing, and coverage

We compare the original model, geometry alone, geometry with original-model fallback, a fixed half-and-half inverse-TTC mixture, and the released router. All use the same saved causal inputs and 434 corrected target correspondences. The fixed weight is specified before calculating these comparisons. Geometry is available on 424 queries, including 357 of 365 road queries. Table 9 reports both available coverage and identical geometry-valid support; an abstention never silently changes a comparator's denominator. We retain signed outputs and the shared finite emission policy. Organizer-style interpolation appears only in the separate supplementary analysis.

Estimator	Coverage	Available	Common	Road common
Original fusion	434/434	0.824	0.793	0.282
Geometry only	424/434	1.158	1.158	1.343
Geometry with original fallback	434/434	1.181	1.158	1.343
Fixed inverse-TTC mixture	434/434	0.333	0.289	0.221
Learned router	434/434	0.347	0.304	0.159

Table 9. Matched adaptation errors in seconds. Available-support MAE uses the displayed coverage; the common column uses the same 424 queries for every arm, and road common uses the same 357. Predictions retain their signs.

On common support, geometry alone has lower median error but large outliers, yielding mean error 1.158 s. A fixed mixture reduces MAE to 0.289 s, compared with the router's 0.304 s; the fixed-minus-router paired MAE difference is −0.0146 s, with a 95% recording-bootstrap interval of $[-0.1704, 0.0943]$ s. The road-only comparison changes that conclusion: the router reduces MAE from the fixed mixture's 0.221 to 0.159 s, with a fixed-minus-router difference of 0.062 s and a recording-bootstrap interval of $[0.014, 0.120]$ s. Excluding Slider-750 also favors the router, by 0.057 s with interval $[0.008, 0.111]$ s across nine recordings. These are descriptive comparisons on development-used data.

Thus adding geometry explains much of the pooled improvement, while learned routing adds value on the road subset by controlling unreliable geometric estimates. The evidence supports this conditional benefit; the pooled result alone would conceal it. The original model's locally reconstructed organizer-style score still disagrees for Slider-750. Appendix J.4 gives the exact discrepancy, without attributing it to an unverified physical cause. Independent recordings with detector-produced boxes are needed to test whether the routing benefit persists under new geometry errors.

8. Runtime and causal service

A 50 Hz response budget allows 20 ms for the complete prediction path, including preparation, scheduling, neural computation, and delivery to the host. We evaluate that path with a five-second paced replay: all 250 updates complete with numerical parity, and scheduled-arrival response reaches 29.14 ms at p95. Even when timing begins after RAW acquisition, copy, and payload sealing, p95 is 25.25 ms. Meeting the deadline therefore requires optimizing the remaining service path as well as input preparation. The measurements below separate this persistent replay from the sensor-preparation path used by the local adapter.

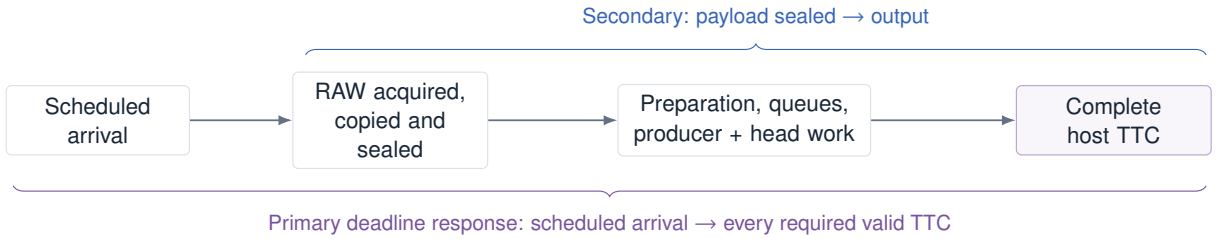


Figure 6. Response clocks for one query. Deadline response starts at scheduled arrival; the secondary clock starts after RAW acquisition, copy, and payload sealing. Both stop when all required TTC outputs reach the host.

8.1 Latency measurement boundaries

Earlier timing experiments measured different parts of the computation. V18 used warm raw-sensor queries with its original producer batching. The final test12 diagnostics timed a prepared eager path, and the EvTTC transfer measured adapter wall time, including preparation and cache behavior. Hardware and inputs also varied. The measurements retain their original boundaries; the deadline test below uses the scheduled-arrival clock. Per-arrival timestamps record scheduled arrival, observation availability, execution start, and completion, allowing the response distribution and deadline counts to be reconstructed.

Table 10. Separate runtime contracts. Both runs use an RTX 5070 Ti Laptop GPU, Python 3.11.15, PyTorch 2.11.0+cu128/CUDA 12.8, FP32, TF32 disabled, and two PyTorch CPU threads. Adapter host CPU: Ryzen 9 9955HX (16 cores/32 logical processors). The clocks and workloads differ.

Run	Measured boundary and population	Result and exclusions
Submitted adapter	434 EvTTC queries; sensor/cache preparation plus synchronized native graph and geometry/router	Mean 1.740290 s/query; excludes one-time loading, hashing/indexing, and ledger writes
Paced base system	250 single-object updates over five seconds; scheduled arrival through all required host TTC outputs	p95 29.143155 ms; RAW preloaded, disk I/O excluded; initialization precedes arrivals

For the submitted adapter, the mean decomposes into 1.158 s preparation, 0.5801 s native neural graph, and 0.001853 s geometry/router. The bounded lossless input cache holds at most 64 MiB of events and sixteen RGB frames. The original submission's observed mean of 3.268 s/query comes from another run, so the difference is descriptive rather than a controlled cache or router speedup. These component means belong to the adapter run. The paced-service measurements below use a separate persistent path.

The service reduced repeated work by preparing RGB within the object region, extending event prefixes incrementally, and reusing reverse-transport summaries exactly. It also combined top-k normalization with native query graphs and parallel branches, overlapping event execution with RGB preparation. RGB features were reused; event encodings were recomputed. A separate diagnostic verified all 462 recorded calls to the prepared transport component.

Numerical checks rejected several faster alternatives. Compiled transport arithmetic changed the final TTC on sensitive examples despite agreeing on most others. Processing smaller event packets preserved the tested state buffers, representations, and final predictions, but that experiment did not time the computation or run a 200 Hz source through the paced service. It established a viable numerical path for a later timing experiment.

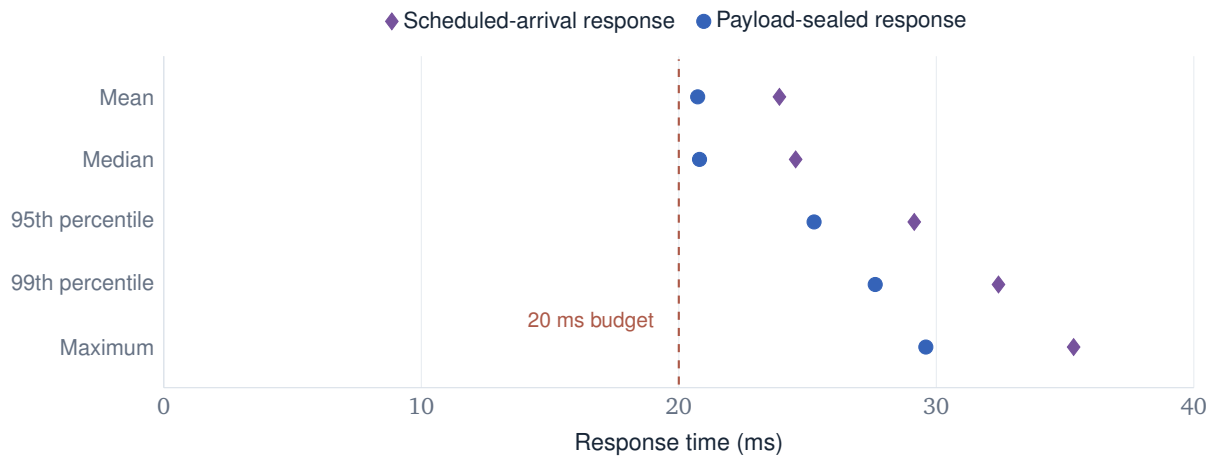


Figure 7. Response-time summaries for 250 completed updates. Scheduled-arrival latency has a p95 of 29.14 ms. Of the 250 responses, 201 (80.4%) meet or exceed 20 ms. The second clock begins after RAW acquisition, copy, and payload sealing.

8.2 Paced-service latency at 50 Hz

The replay ran on an NVIDIA GeForce RTX 5070 Ti Laptop GPU with Python 3.11.15 and PyTorch 2.11.0+cu128, built with CUDA 12.8. Computation used FP32 with TF32 disabled, two PyTorch CPU threads, eight RGB preparation workers, and a queue capacity of two. Model loading, synthetic shape priming, and initialization from preceding history finished before measured arrivals. Table 39 gives these setup intervals and the parity tolerance.

The five-second replay processed 250 single-object updates from preloaded RAW data accessible only to the CPU feeder. All updates completed with zero drops and zero source errors, and independent post-run checks confirmed TTC parity. At 200 scheduled arrivals, earlier work was still unfinished. Of the 250 responses, 201 met or exceeded 20 ms. Scheduled-arrival latency reached 29.14 ms at p95, 32.41 ms at p99, and 35.33 ms at maximum. Starting the clock after RAW acquisition, copy, and payload sealing gave a lower p95 of 25.25 ms.

The measurement excludes disk I/O and covers one object over five seconds. It therefore leaves physical-camera input, sixty-second operation, and larger object counts for further testing. Failed replays remain in separate reports, including dropped updates and rapid error acknowledgements. Latency statistics include completed TTC outputs; failed-request acknowledgements are recorded separately.

9. Discussion and limitations

9.1 System performance and component attribution

The private benchmark supports PHASE-TTC as a complete causal TTC system: its archived score improves all four TTC bands relative to the displayed Garl-TTC reference. Our matched transfer tests narrow the explanation. Learned residual readouts outperform equal mixtures and the source-calibrated scalar mixture, yet the 95% confidence interval for the paired MAE difference between real-event and zero-event heads includes zero, and phase and inverse-TTC corrections give nearly identical mean errors. We therefore treat named phase fields as an implementable design choice whose universal advantage has not been established.

The interface still offers a concrete engineering property: each coordinate has a defined meaning across separately fitted producers and out-of-fold rotations. This permits grouped fitting without requiring hidden coordinates to align. Testing whether that property improves accuracy over a matched feature-level fusion model requires an additional comparison. The current benchmark difference also includes changes in upstream estimators, supervision, and temporal history.

9.2 Adaptation depends on the error regime

The geometry experiment separates two effects. Adding apparent expansion allows a fixed mixture to recover much of the pooled gain. Learned routing further improves the road subset, where geometric outliers make an unconditional geometric estimate unreliable. Neither aggregate should stand in for the other. The supplied boxes and simplified intersensor projection remain part of this evaluation contract; detector jitter, occlusion, and calibration errors may change the result on new acquisitions.

9.3 Generalization, reproducibility, and execution

The private benchmark supplies the independent test result. The secondary experiments reuse development-inspected recordings, including the new matched controls. Their three head seeds share the source features and upstream weights. We have no confirmed unused synchronized event–RGB recordings with compatible object-level TTC labels for a new independent test. Additional fits or resampling cannot replace those acquisitions.

For the new 460-query comparison, replaying the released head on recovered features agrees with its saved predictions within 2.87×10^{-6} s. This local check has a narrower scope than historical full-system reproduction: 504 of 5,805 native-replay queries exceed the original tolerance, with a maximum difference of 0.4968 s. The original submission and its score remain separately archived. The replay discrepancy still requires investigation of the original execution environment and per-query intermediate states; repeatability on the current platform alone cannot resolve it.

The paced five-second replay delivers all 250 predictions, but 201 responses meet or exceed 20 ms and p95 is 29.14 ms. This falls short of the declared deadline. Sustained physical-camera operation, multiple objects, upstream detection, and calibrated warning probabilities remain outside the demonstrated result.

Earlier predictive-learning studies are informative negative experiments, rather than a mechanism explaining the benchmark gain. The recovered scratch, random-frozen, JEPA, and shuffled-future controls appear together in Appendix L. They support the reported failure to demonstrate a JEPA benefit in those settings; they do not identify background appearance as its cause on real data.

10. Conclusion

PHASE-TTC combines modality-specific motion estimates and temporal context in a compact RGB-anchored readout. Its complete submitted system improves the archived GarlTTC ranking metric by 20.49%. Matched final-head experiments show that a learned correction helps relative to simple mixtures, while event access and phase parameterization offer no resolved advantage over their matched controls on the inspected transfer cohort. Geometry-based adaptation adds a separate conditional benefit: learned routing improves road performance beyond a fixed mixture, although the pooled comparison remains uncertain. These results define an evaluated system and identify the measurements needed to establish broader fusion and deployment claims.

References

[1] Mahmoud Assran, Quentin Duval, Ishan Misra, Piotr Bojanowski, Pascal Vincent, Michael Rabbat, Yann LeCun, and Nicolas Ballas. Self-supervised learning from images with a joint-embedding predictive architecture. *arXiv:2301.08243*, 2023. URL https://arxiv.org/abs/2301.08243.

[2] Adrien Bardes, Jean Ponce, and Yann LeCun. Vicreg: Variance-invariance-covariance regularization for self-supervised learning. In *ICLR*, 2022. URL https://arxiv.org/abs/2105.04906.

[3] Guillermo Gallego, Tobi Delbruck, Garrick Orchard, et al. Event-based vision: A survey. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2020. URL https://arxiv.org/abs/1904.08405.

[4] GarlTTC Benchmark. GarlTTC Test public leaderboard. https://www.codabench.org/competitions/17289/, 2026. Accessed 28 October 2026. PHASE-TTC submission 1016512.

[5] Mathias Gehrig, Willem Aarents, Daniel Gehrig, and Davide Scaramuzza. Dsec: A stereo event camera dataset for driving scenarios. *IEEE Robotics and Automation Letters*, 2021. URL https://arxiv.org/abs/2103.06011.

[6] Jinghang Li, Bangyan Liao, Xiuyuan Lu, Peidong Liu, Shaojie Shen, and Yi Zhou. Event-aided time-to-collision estimation for autonomous driving. In *ECCV*, 2024. URL https://www.ecva.net/papers/eccv_2024/papers_ECCV/papers/07043.pdf.

[7] Jinghang Li, Shichao Li, Qing Lian, Peiliang Li, Xiaozhi Chen, and Yi Zhou. Toward deep representation learning for event-enhanced visual autonomous perception: the eap dataset. *arXiv:2603.16303*, 2026. URL https://arxiv.org/abs/2603.16303.

[8] Lorenzo Mur-Labadia, Matthew Muckley, Amir Bar, Mido Assran, Koustuv Sinha, Mike Rabbat, Yann LeCun, Nicolas Ballas, and Adrien Bardes. V-jepa 2.1: Unlocking dense features in video self-supervised learning. *arXiv:2603.14482*, 2026. URL https://arxiv.org/abs/2603.14482.

[9] NAIL-HNU. Garl-TTC official benchmark evaluator. https://github.com/NAIL-HNU/Garl-TTC/blob/main/garl_ttc_benchmark/codabench/scoring_program/score.py, 2026. Accessed 28 October 2026.

[10] Oriane Siméoni, Huy V. Vo, Maximilian Seitzer, Federico Baldassarre, Maxime Oquab, et al. Dinov3. *arXiv:2508.10104*, 2025. URL https://arxiv.org/abs/2508.10104.

[11] Vlad Sobal et al. Joint embedding predictive architectures focus on slow features. *arXiv:2211.10831*, 2022. URL https://arxiv.org/abs/2211.10831.

[12] Kaizhen Sun, Jinghang Li, Kuan Dai, Bangyan Liao, Wei Xiong, and Yi Zhou. Evttc: An event camera dataset for time-to-collision estimation. *IEEE Robotics and Automation Letters*, 2025. URL https://arxiv.org/abs/2412.05053.

Methods, controls, and complete evidence

Detailed methods, controlled interventions, sequence-level comparisons, and source mappings accompany the main results. Separate appendices document historical experiments, official evaluation, transfer analysis, and frozen competition inference, followed by the corrected local evaluation and subsequent controls.

Original evidence snapshot: 28 October 2026. Research coverage through 11 November 2026, including the matched readout and geometry controls.

The build uses saved evidence; separate later model fits retain their own protocols and exposure history.

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A. Implementation and source correspondence

The main text uses descriptive component names. The source archive uses A5 for the transport-trained producer, C2F for the coarse-to-fine producer, PAIR for the pairwise temporal readout, and TRAIN40 for the final forty-sequence fit. F0 identifies the frozen base fusion model; F1 identifies a separately fine-tuned fusion head. PHASE17 denotes the context input, whereas PHASE20 denotes the twenty-coordinate fusion interface. These identifiers name different interfaces and are not interchangeable.

A.1 Final training population and optimization

Final fitting uses 88,744 TRAIN40 queries from forty sequences. The former P, H, and V groups all contribute to this refit, so their earlier scores characterize development rather than independent validation of the deployed chain. Each producer receives 49,932 updates, equivalent to eighteen epochs at effective batch size 32. The event producers carry forward the earlier TRAIN40 weights; RGB training resumes to the same endpoint. [L43]

Training comprised 690,934 auxiliary endpoint updates and 58,660 final-head updates across forty-five fits. Final PAIR endpoints used 11,264 updates; each context model and the final head used 12,044. Schedulers and warmup scaled with the fitting population. Update counts describe optimization exposure rather than wall-clock GPU effort.

A.2 Exact fusion layout and emission policy

Each context refiner is represented for final fusion by a constructed 170-coordinate block. Coordinates 0–159 are its learned hidden state. Coordinates 160–169 are an explicit interface containing, in order, one raw location, one emitted phase, three expert phases, four current timing values, and one valid-history count divided by eight. Concatenating the RGB block followed by the event block gives the 340-coordinate container

$$\mathbf{s} = \left[\mathbf{s}_R^{170}; \mathbf{s}_E^{170}\right] \in \mathbb{R}^{340}. \tag{9}$$

Fusion reads $\mathbf{s}_{160:170}$ and $\mathbf{s}_{330:340}$, yielding twenty named phase, timing, and support inputs while the 160-dimensional hidden states remain within their modality branches. Within each ten-coordinate block, the first five phase coordinates are divided by 0.03; timing and support keep their native scaling. The shared interface aligns quantities with consistent meanings across the separately fitted refiners. [M22]

The selected vector passes through

$$20 \to 32\ [\text{LayerNorm}, \text{SiLU}] \to 32\ [\text{LayerNorm}, \text{SiLU}] \to 3. \tag{10}$$

Zero initialization of the last affine layer makes the initial fusion output equal to the RGB anchor. If c_0, c_1, c_2 are the three head outputs, ℓ_R is the RGB raw-location anchor, and $\mathbf{w}_R$ is the separately supplied pair of RGB width logits, the deployed computation is

$$\ell_F = \ell_R + 0.03 c_0, \tag{11}$$

$$\boldsymbol{\sigma}_F = 0.03\ \text{softplus}(\mathbf{w}_R + [c_1, c_2])\,, \tag{12}$$

$$\widehat{\text{TTC}} = \text{phase_to_ttc}(\ell_F), \tag{13}$$

$$q_{10}^{\phi} = \text{clip}(\ell_F - \sigma_{F,0},\ \phi_{\min}, \phi_{\max}), \tag{14}$$

$$q_{90}^{\phi} = \text{clip}(\ell_F + \sigma_{F,1},\ \phi_{\min}, \phi_{\max}). \tag{15}$$

The native bounds are $\phi_{\min} = -\log 2 + 10^{-6}$ and $\phi_{\max} = 8$. For the point prediction, the inverse map clamps ℓ_F to these bounds, computes $q = -\,\text{expm1}(-\phi)/0.1$, and emits $\text{sgn}_+(q)/\max(|q|, 1/60)$ seconds, where $\text{sgn}_+(0) = 1$. Thus the emitted magnitude is at most 60 s and zero phase maps to +60 s. The reported point phase is recomputed from this finite TTC; the learned-width phase-interval endpoints are clamped separately. The rate floor keeps inversion finite near zero expansion, imposing a nominal signed horizon of 60 s. [M22, R02]

The compact fusion head contains 1,955 trainable parameters: 672 + 64 + 1,056 + 64 + 99 across three affine layers and two LayerNorms. Figure 4 summarizes the named-field and anchor paths. [M22]

A.3 Transfer target association and eligibility

The historical transfer registered sixteen uniformly spaced queries in each of thirty-two EvTTC sequences, together with forty-six earlier queries. Four duplicate identities were removed, leaving 554 saved candidates: 46 Historical46 and 508 Extended-new. The associated object is the primary annotated object matched to the scenario's scalar TTC series; the public projection does not establish arbitrary per-object TTC correspondence.

The original assignment interchanged the low-100 and medium-100 TTC sources for the two CCRs-1 recordings. The later public-source audit fixes their identities by download mapping and SHA-256. An independent RGB-identity record links the competition extracts to the same native recordings, matching all 50 low and 43 medium frames with constant competition-to-native offsets. Transfer queries already use native microseconds, so no additional offset is fitted or applied. [GT01]

All 508 Extended-new candidates are reassociated, including the forty previously excluded queries. The matching rule remains nearest public TTC within 50 ms, with the earlier timestamp breaking ties. The low and medium groups each contributed fifteen previously scored queries; they now contribute ten and twelve. Historical46 does not contain these groups and retains its 46 eligible queries.

Table 14 lists the candidate and eligibility counts for every sequence group.

The revised Extended-new denominator is therefore 460, with 48 exclusions across the original 508 candidates. All retained targets are positive. Every query anchor and every model prediction remains unchanged. The historical 468-query ledger is preserved in `data/transfer_queries.csv`; current results use `audit_20261104/corrected_transfer_candidates.csv`. The associated receipt records the original and corrected source hashes and the neighboring label timestamps for each reassociation. [GT01]

B. Competition inputs: historical coverage and disagreement

28 OCTOBER INFERENCE SNAPSHOT / 434 query identities / 10 sequences / coverage and prediction disagreement

Three released Garl-TTC variants were evaluated on the same ten-sequence EvTTC queries as the frozen PHASE-TTC reference. The 434 queries span eight road and two Slider sequences, with sequence, row, frame index, and timestamp matched across models. The comparison contains 1,302 Garl outputs and 434 PHASE-TTC outputs. [T01]

At the time of this frozen inference comparison, no verified numeric TTC targets mapped to the 434 query identities. The analysis therefore measured coverage and paired prediction disagreement:

$$d_i^{(m)} = \widehat{T}_{i,m}^{\mathrm{Garl}} - \widehat{T}_i^{\mathrm{PHASE}}, \qquad a_i^{(m)} = |d_i^{(m)}|. \tag{16}$$

The differences in Equation 16 compare model outputs rather than prediction error. This population differs from both the labeled transfer in Section 5 and Garl's published three-sequence RTE evaluation. Appendix H gives the adapter, timing measurements, and ten sequence traces.

The corrected target audit reported on 1 November supersedes the earlier absence of local labels: it recovers all 434 targets after resolving the public source assignment for two road sequences. Section 7 uses those targets to compare original and adapted PHASE-TTC under one documented local evaluator. The results below preserve the original Garl-versus-PHASE outputs and their coverage rather than retroactively presenting them as a reproduced error ranking. The published three-sequence result remains a separate protocol, and the Slider-750 discrepancy prevents claiming official equivalence for the new local scores. [T01, T02, N01]

Table 11. Predictions on 434 EvTTC competition queries. The table reports finite-output coverage, signed TTC range, and negative counts. Median and P95 summarize absolute differences from PHASE-TTC where both models produce finite outputs. [T01]

Model	Finite	Range (s)	< 0	Median (s)	P95 (s)
PHASE-TTC	434/434	[-42.225, 51.869]	2	–	–
Garl full	424/434	[0.907, 52.705]	0	0.261	1.098
Garl event-only	434/434	[-42.211, 406.095]	10	0.681	5.623
Garl visual-only	426/434	[-23.832, 38.767]	5	0.236	1.524

B.1 Frozen checkpoints and native input support

The released full, event-only, and visual-only state dictionaries load strictly and run in float32 with frozen weights. RGB-dependent routes use two real causal frames, while event-only uses two observed causal intervals. Startup queries without the required observations remain unavailable. PHASE-TTC uses up to 650 ms of context, so the models share query identities under different temporal support. [T01]

Supplied boxes and calibration condition each route. Rotation-only projection omits translation because object depth is unavailable. For Slider, the adapter derives rotation from the supplied camera poses because the data provide no named event-to-RGB extrinsic. These input adaptations preserve the released checkpoint weights.

B.2 Coverage, sign, and disagreement tails

PHASE-TTC and event-only Garl each emit 434 finite values; full Garl emits 424 and visual-only 426. The first query in each road sequence lacks two causal RGB frames, creating eight gaps in both RGB-dependent variants. Full Garl has two further Slider-750 gaps because the earlier event endpoint precedes recorded coverage. All invoked models return finite values; nulls correspond to unavailable inputs.

	PHASE-TTC	Garl full	Garl event-only	Garl visual-only
CCRs1-low	50/50	49/50	50/50	49/50
CCRs1-medium	43/43	42/43	43/43	42/43
CCRs1-high	30/30	29/30	30/30	29/30
CCRs2-low	46/46	45/46	46/46	45/46
CCRs2-medium	55/55	54/55	55/55	54/55
CCRs2-high	59/59	58/59	59/59	58/59
CCRm-low	42/42	41/42	42/42	41/42
CCRm-medium	40/40	39/40	40/40	39/40
Slider-750	51/51	49/51	51/51	51/51
Slider-1000	18/18	18/18	18/18	18/18

Counts show finite predictions per sequence; tinted cells mark queries with insufficient causal input.

Figure 8. Finite prediction coverage by model and sequence. Tinted cells mark the eight missing road RGB pairs and two additional early Slider-750 event gaps in full Garl. [T01]

On shared finite rows, median absolute disagreement from PHASE-TTC is 0.26 s for full Garl, 0.68 s for event-only, and 0.24 s for visual-only. Their p95 disagreements are 1.10, 5.62, and 1.52 s. Full Garl emits no negative TTC, compared with two negative PHASE-TTC outputs, ten event-only outputs, and five visual-only outputs.

Slider-750 contains the largest excursions: event-only reaches 406.10 s, with p95 absolute disagreement of 241.68 s. The native height-ratio conversion becomes ill-conditioned as the two predicted endpoint

heights approach equality. Figures 16 and 17 preserve the sign and magnitude of these outputs across all ten sequences.

Timing remains descriptive because the variants ran once in fixed order with shared cache and warm-state effects, and PHASE-TTC used a different adapter and longer history. [T01]

C. Adaptation lineage and calibration controls

C.1 Adaptation lineage and rejected candidates

The name "adapted PHASE" refers to several checkpoints with different fitting populations. F0 denotes the original nine-component TRAIN40 graph. The first F1 adaptation updates its original 1,955-parameter fusion head while freezing the eight preceding components and their normalizers. Its all-32-recording refit uses 88,744 TRAIN40 OOF states and 3,571 admitted EvTTC states, for 92,315 training rows; 257 native queries lack compatible targets. Each of seeds 7, 13, and 21 completes two epochs and 1,444 updates, with seed 7 designated for delivery before scoring. These are training-fit results. [A01]

The preserved all-32 F1 cache confirms that its supervision consumed the old CCRs1-low/medium source assignment. Its preparation script reads each query's `ttc_csv_path`, hashes the file, and joins its TTC curve before writing supervised shards. The preserved `data_v3/TARGET_LEDGER.json` records the old low-file hash beginning `780ee30f` and medium-file hash beginning `1c5adba8`; under the corrected identity map those files belong to medium and low, respectively. It admits 98 low and 141 medium queries, with two and eight additional unsupported targets. Its completion receipt binds both that ledger and the merged training array, and the three seeds' start receipts and delivered seed-7 bundle bind the same completion receipt. Thus 239 admitted supervised rows used the old map, and the ten exclusions also require a new eligibility check. Rescoring retained outputs cannot repair those F1 weights: a corrected all-32 F1 requires rebuilding the supervised cache and refitting seeds 7/13/21. No corrected all-32 F1 training is reported here. The grouped F1 campaign below excludes the eight overlapping competition recordings, including these two recordings, and restarts from F0 in each fold. This particular source swap does not invalidate that grouped comparison or establish a dependency in the final 688-parameter router, whose optimizer population excludes those recordings. [A01,A02]

That grouped campaign comprises 21 head fits: seven scenario families times three seeds, with 2,531 queries from twenty-four road recordings. It retains TRAIN40 during each fit. F1 reduces recording-macro RAE from F0's 30.600% to 25.468/26.188/25.938%, but increases query MAE from 0.8494 s to 0.8585/0.8925/0.8653 s for seeds 7/13/21. All three family-bootstrap intervals for the RAE difference include zero. For targets above 4 s, seed 7 increases MAE from 1.2542 to 1.5496 s. On Slider-500, outside this optimizer fit but historically exposed, F1 increases RAE from 83.887% to 87.578/91.022/87.330%; each variant returns 76 of 77 outputs. Lower road relative error therefore did not establish uniform improvement. [A02]

The next quality-fusion campaign freezes all nine F0 components and fits three 401-parameter gates. Its named `evaluated` checkpoint has separate fitting, calibration, and confirmation roles; `all_data_refit` instead consumes all 2,677 eligible queries from twenty-four road and three Slider recordings. The latter's reported errors are resubstitution. On 1,747 road OOF queries, the quality gate gives RAE 12.834% and MAE 0.4224 s, versus 22.081% and 0.7544 s for an equal-parameter control without quality descriptors. Both receive geometry. This supports quality information for that gate and population. [A03]

The preserved bundles close the upstream-weight distinction: the 401-parameter gate, startup router, and submitted router all bind the original F0 head. Hash checks of their thirteen upstream components, including normalizers and seal/recipe records, agree with their manifests. None substitutes the affected all-32 F1 heads; complete checkpoint identities and cache bindings are retained in the additional source index. The gate preparation explicitly selects F0, RGB context, and event context as native experts; startup reuses those experts, and submission refits the router while copying the verified original upstream

bundle. This establishes a separate weight lineage, while retaining the historical data exposure of all three gate campaigns. [A01,A03,A04,A07]

The startup campaign then executes 117 fits, including Q/QA/NAcap controls, seconds- and inverse-coordinate trials, and all-data refits. It retains the 688-parameter inverse-coordinate router as a candidate after rejecting the seconds-coordinate trial. Road OOF MAE changes from the previous gate's 0.4224 to 0.4049 s, while CCRm confirmation MAE worsens from 1.020 to 1.036 s. Its descriptive road RAE-difference interval, resampling five families, is $[-1.5176, 0.8001]$ percentage points. Development now includes Slider-750/1000, so this comparison changes both architecture and exposure. The final submitted recipe below returns to a 2,608-query optimizer population; earlier use of those Slider recordings still counts as design exposure. [A04]

A further 312 completed small-head fits test sign/log/geometry variants (96), supervised cost (48), correlated expert perturbations (24), native-context resets (48), support versus capacity (48), and support with resets (48). Eight family-held-out folds, seeds 7/13/21, and 300 updates per fit use the 2,608-query population. Promotion requires full coverage and no worsening of both recording-macro RAE and MAE in either road or Slider relative to IQ. None passes all four error criteria. The six preserved fits from an earlier failed launch do not enter the 312 total. The 447 context resets cover 149 prepared queries; they do not represent 447 cold sensor starts. These fits are separate from the 21 F1 fits, the 117 startup fits, and the later 48 canonical-router fits. [A05]

C.2 Measured interval coverage by variant

The grouped F1 experiment also measures substantial undercoverage. Its nominal 80% intervals cover 67.88/65.78/67.72% for seeds 7/13/21. Rescaling with each fold's training residuals raises coverage only to 70.17/68.63/70.01%; that procedure is not split conformal and was not bundled. The earlier 401-parameter `evaluated` gate uses separate CPNAO calibration records and attains 79.83% CCRm coverage. The startup `evaluated` candidate reports 83.10% CCRm coverage after separate CPNAO calibration. These are descriptive results on historically exposed groups. Their all-data refits consume those calibration labels and carry no validated equivalent interval; neither does the final 2,608-query submitted router. Finite-output coverage and probabilistic interval coverage describe different properties. [A02,A03,A04,A07]

C.3 Geometry controls and support differences

F0 versus the final router changes both the available experts and the learned combination. The older 401-parameter capacity control establishes a benefit of quality descriptors only within its own experiment. Its Slider-750 geometry observer has MAE 0.1976 s on 49/51 outputs, versus 3.3157 s for that gate on 51/51; Slider-1000 gives 0.0374 s on 16/18 versus 0.2757 s on 18/18. Different masks prevent a direct attribution from these aggregates. The new common-support comparison in Section 7 now evaluates geometry, fallback, fixed mixture, and the final router on these saved inputs. It supplies the matched attribution that the earlier aggregate records lacked. [A03,N01]

D. Canonical cadence as a controlled intervention

FROZEN-GRAPH INTERVENTION / 149-query screen and 2,608-query development population / candidate not promoted

The metric phase uses a fixed interval, whereas the sensor supplies observations at measured intervals. These two time quantities must remain separate. The canonical-PAIR experiment asks whether normalizing the learned readout to a reference interval improves its consistency, then tests whether any local improvement survives the downstream contexts and fusion. The intervention introduces no optimizer updates into the original chain. [N02]

Table 12. Frozen F0 with canonical PAIR timing on the exposed development population. Metrics average recordings equally within each domain; availability is explicit. Road ΔMAE has a descriptive 95% recording-bootstrap interval [−0.0005047, 0.0000931] s. Slider-500 is one previously used recording.

Domain	Paired/all	MAE orig.	MAE canon.	RAE orig.	RAE canon.
Road	2531/2531	0.810620	0.810490	30.6004	30.5910
Slider-500	76/77	1.178156	1.120809	83.8859	82.4420

D.1 Recomputing the readout before rescaling

The intervention first rebuilds the 133-coordinate PAIR input with its three time-derived fields recomputed for $d_0 = 0.1$ s: duration, log-duration, and inverse duration. The frozen PAIR network processes this input and emits raw phase ϕ_c. Given the actual interval Δ, the adapter then computes

$$q_c = -\operatorname{expm1}(-\phi_c)/d_0, \tag{17}$$

$$q_\Delta = q_c d_0/\Delta, \tag{18}$$

$$\phi_\Delta = -\log(1 - d_0 q_\Delta). \tag{19}$$

Only after this transformation does it apply the original emission policy. The implementation checks finiteness and $1 - d_0 q_\Delta > 0$; unsupported transformations remain explicit failures. When Δ equals the FP32 representation of 0.1 s, a direct identity path preserves the original bytes. This is not a division of an already emitted TTC, nor a rescaling of the original phase with its time-conditioned features unchanged. [N02]

Weights, normalizers, observation cutoffs, and producer batches remain fixed. The run reconstructs the complete PHASE17 feature vector from corrected PAIR features, including disagreements, and reruns temporal contexts and the final fusion head. Predictions and missing-output status are sealed before a separate process joins targets. Each new query first reproduces a paired original execution on the same platform; an old cache is not substituted as the control.

A small time-warp diagnostic motivated this population test. It altered prepared observations and their clocks, rather than reselecting fixed-duration windows from a time-scaled raw stream. Under speed factor a, the ideal comparison is $aT_{\text{warped}}/T_{\text{original}} = 1$. Canonicalization improved all four tested F0 contrasts, yet the road event-context ratio at $a = 0.5$ moved from 0.6738 to 1.891. The downstream regression prevents an interpretation in which every intermediate correction must benefit the complete estimator. [N02]

D.2 Fixed screen and modality localization

The initial screen contains 72 road queries and 77 Slider-500 queries. A later four-arm localization control holds this population fixed and corrects events only, RGB only, both, or neither. In that paired CPU control, road macro MAE is 0.7489 s for the original and event-only arms, and 0.7487 s for RGB-only and both. The observed road change therefore comes from the RGB intervention in this control. On the 76 available Slider outputs, the corresponding values are 1.178, 1.163, 1.142, and 1.121 s. Both modalities contribute there. This localization experiment diagnoses a fixed candidate; it does not select a new recipe after inspecting the four results. [N02, N05]

D.3 Complete development population

The full experiment reuses the sealed screen and adds 2,459 road queries, yielding 2,531 road queries across twenty-four recordings and 77 Slider-500 queries. The original F0 and its canonical counterpart both return all road predictions and 76 Slider predictions. The previous Slider absence remains absent. Neither route introduces a new missing prediction or a sign change. [N02]

Road recording-macro MAE changes from 0.8106 to 0.8105 s, a reduction of 0.13 ms. A paired bootstrap over complete road recordings, using 10,000 resamples and seed 7, gives a descriptive 95% difference

interval of $[-0.50, 0.09]$ ms. The interval includes zero. Slider-500 improves from 1.1782 to 1.1208 s, a 4.87% relative reduction on one acquisition already inspected in the screen. Table values retain the original precision.

The original fused estimator, denoted F0, satisfies the fixed numerical development criteria: preserve coverage, avoid worsening either error metric in either domain, and improve at least one metric. The small road effect and reuse of development recordings motivate the downstream router and refitting comparisons below.

D.4 Propagation through the quality router

The frozen adapted router improves aggregates on the complete population but fails the earlier road screen. That failed criterion remains binding. A subsequent paired refit evaluates eight folds, three seeds, and original/canonical feature arms, totaling 48 fits. The canonical ensemble improves full-population road MAE from 0.5396 to 0.5378 s and Slider MAE from 0.3650 to 0.3565 s. On the fixed screen, however, road MAE worsens from 0.4848 to 0.5118 s and relative error from 16.72% to 17.43%. No seed passes both evaluation views. The fixed-screen degradation excludes this refit under the combined selection criteria, so the delivered PHASE-TTC-Adapt recipe is retained. The comparison concerns out-of-fold ensembles; the delivered router is fitted on the complete adaptation population. [N03]

D.5 Native replay: output sensitivity versus accuracy

The same-platform eAP comparison covers 5,805 physical queries and their 6,762 submission-template tokens, including aliases. Both arms preserve availability and sign, and the event context remains identical. For F0, median absolute output change is 1.4305×10^{-5} s, p95 is 7.6771×10^{-5} s, and the maximum is 0.006496 s. These are differences between model predictions, not errors relative to targets. The private Test12 labels were not used, so the experiment establishes no accuracy or ranking conservation. [N03]

Historical parity is a separate issue. The recaptured original is exactly repeatable locally, but every one of the 5,805 queries fails some comparison with historical intermediate states, and 504 fail the historical TTC tolerance. Local repeatability cannot remove those failures or assign their cause. The paired original/canonical design isolates the new intervention within the present platform while retaining the older reproducibility discrepancy.

Across those 5,805 original-output comparisons, the absolute historical-to-local TTC difference has median 0.0001855 s, p95 0.001094 s, p99 0.002797 s, and maximum 0.4968 s. The 504 TTC failures represent 8.68% of queries. These magnitudes describe reproducibility drift without reading private targets; they are not accuracy errors and do not show that drift caused the local canonical change. The maximum also prevents extrapolating the small same-platform canonical deltas to numerical conservation of the submitted historical run. [A08]

A separate 120-query warm-weight raw-path benchmark measures an original p95 of 963.353 ms and a canonical p95 of 1,008.124 ms. It uses supplied boxes and potentially warm operating-system caches; it excludes detection, physical capture, and sensor transport. These measurements cover a different execution path from the five-second paced service in Section 8. Their percentiles cannot be subtracted from precomputed-input percentiles to recover preprocessing cost. The result provides no sustained 50 Hz evidence and adds a runtime cost to the proposed intervention. [N07]

E. Technical foundations and historical method families

PHASE-TTC grew through four model families: predictive embedding, geometry-constrained endpoint estimation, phase-domain temporal refinement, and RGB-anchored event fusion. Each family tests a different mechanism with its own objective and inference path. [M01--M14]

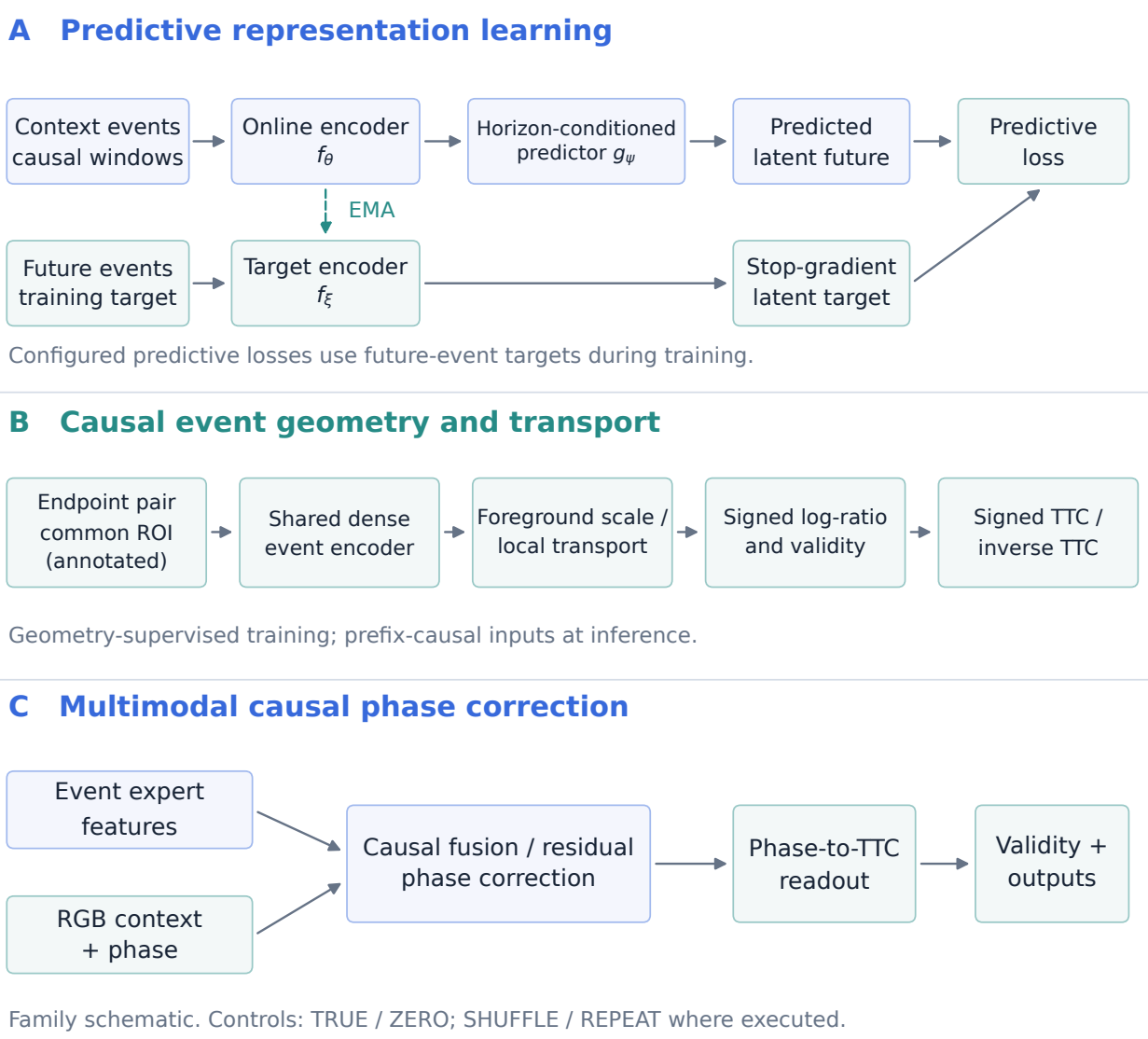


Figure 9. Selected model families. (A) Predictive latent learning uses a training-only target branch. (B) Causal scale and transport map endpoint geometry to signed motion. (C) Phase-domain fusion combines event and RGB evidence. Panel A contains the JEPA objective; Sections E.6 and 2 detail the later temporal and fusion paths.

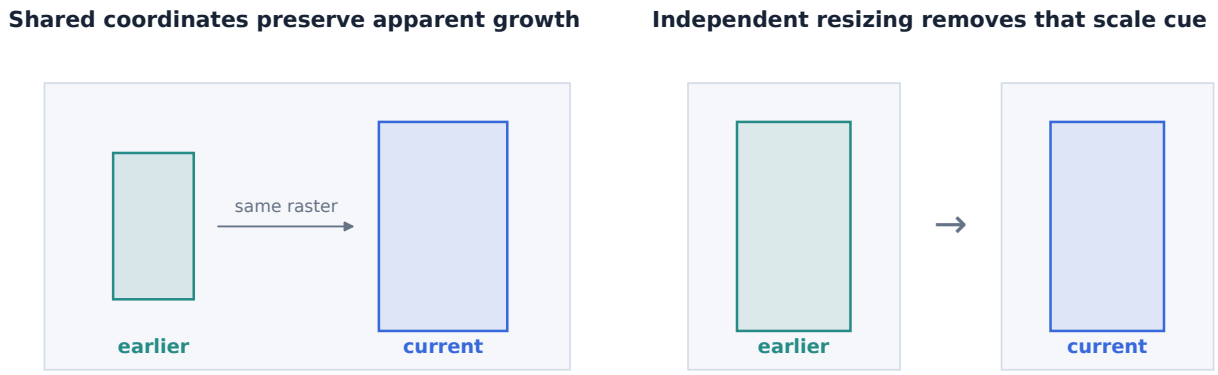
E.1 Events, temporal windows, and object coordinates

An event stream is a sequence $\mathcal{E} = \{(x_i, y_i, t_i, p_i)\}_{i=1}^{N}$, with integer microsecond timestamps and normalized polarity $p_i \in \{-1, +1\}$. `EventBatch` stores aligned NumPy arrays for these fields. Each context window is anchored at its end time. Indexing rejects nonpositive durations, mismatched reference times, and future windows that begin before context ends. Causality tests cover crops, features, checkpoints, and calibration inputs. [M01]

The dense voxel representation distributes an event between adjacent temporal bins. For K bins, duration Δt, and normalized time $u_i = (K-1)(t_i - t_0)/\Delta t$, the unnormalized representation is

$$V_{p,k}(x, y) = \sum_i \mathbf{1}[x_i = x, y_i = y, p_i = p] \max(0, 1 - |u_i - k|). \tag{20}$$

Separate polarity yields $2K$ channels. Two flattened weighted histograms accumulate the adjacent-bin contributions without constructing a buffer with twice the event count. Optional robust scaling divides occupied voxels by the 95th percentile of their absolute magnitude; empty voxels remain zero. Median centering is omitted because equal occupied counts would collapse to zero. Event count, time surface, and sparse tokens remain separate representation alternatives. [M02]



Conceptual common-ROI construction; boxes denote object regions in a shared raster.

Figure 10. Shared coordinates preserve apparent expansion. A common crop maps both endpoints at one spatial scale, whereas independent tight crops normalize away part of the size change. The diagram is schematic.



Figure 11. Event and RGB appearance for one object region. FCWD1:0007 illustrates the cross-modal alignment problem. The orange region is the original visualization overlay.

Object-centred experiments use a common square region that covers both endpoint boxes, expands by a fixed margin, and maps both event windows into one raster coordinate system. Tight independent crops can erase apparent scale change. The resulting tensor is *oracle-box-conditioned*: the event backbone assumes external localization even though its forward signature has no box argument. Causal-scale tests confirm that only the designated endpoint boxes affect the crop. [M03]

The eAP adapter reconstructs local targets from public three-dimensional tracks, whereas the benchmark uses private smoothed labels. Results therefore keep reconstructed eAP targets, EvTTC labels, and official GarlTTC labels in separate evaluation populations. [M01]

E.2 Real temporal crops and event input planes

Figure 12 follows the same EvTTC vehicle at three recorded queries. Each row shows three causal endpoints in a shared object-centred coordinate system for its modality. The RGB square covers the last two supplied boxes with a 25% margin; event crops use the manifest's event-camera square. Coordinates are shared across endpoints within each modality, not across the two physical cameras. The displayed triplets are input excerpts, not the complete 650 ms context history.

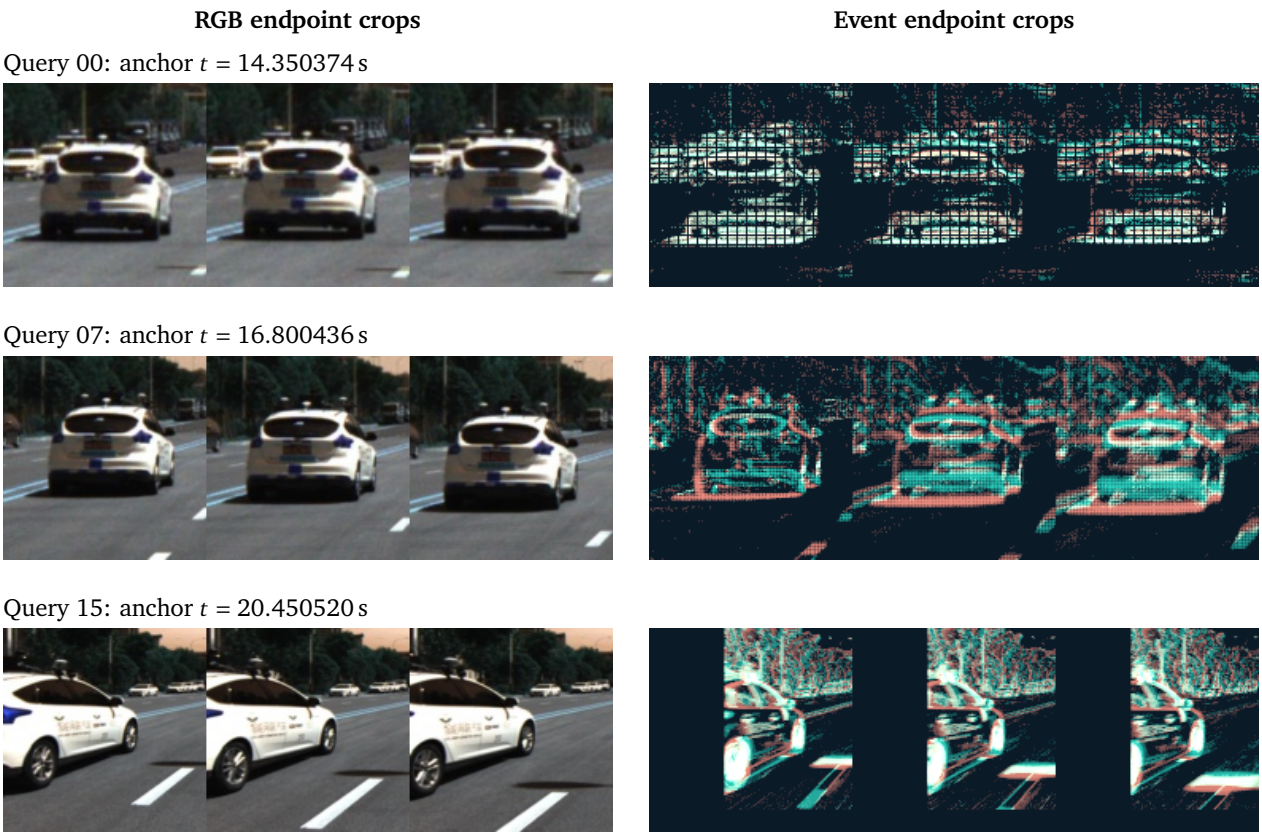


Figure 12. Actual temporal inputs from the EvTTC road excerpt. Each strip contains three 128×128 crops, ordered from oldest to newest. Event windows end at $t-200$ ms, $t-100$ ms, and t, where t is the stated anchor; each accumulates the preceding 100 ms. The RGB frame timestamps differ from the corresponding event endpoints by at most 7 μs in these examples; exact times are preserved in the source sidecar. Turquoise and coral denote positive and negative event counts with per-crop log display scaling. RGB crops preserve the frozen runtime's stored channel order; JPEG images are visual encodings, not lossless tensors. Source: EvTTC [12], `CCRm-low-0-overlap-0`; queries are selected illustrations from the previously exposed transfer cohort.

Figure 13 separates the latest event crop of query 07 into the planes produced by the runtime encoder. This distinguishes the polarity display used above from the actual twelve-channel representation. The voxel planes retain temporal structure within the accumulation window; count and rate planes provide additional support measurements. These are preprocessing outputs, not learned feature maps or neural activations.

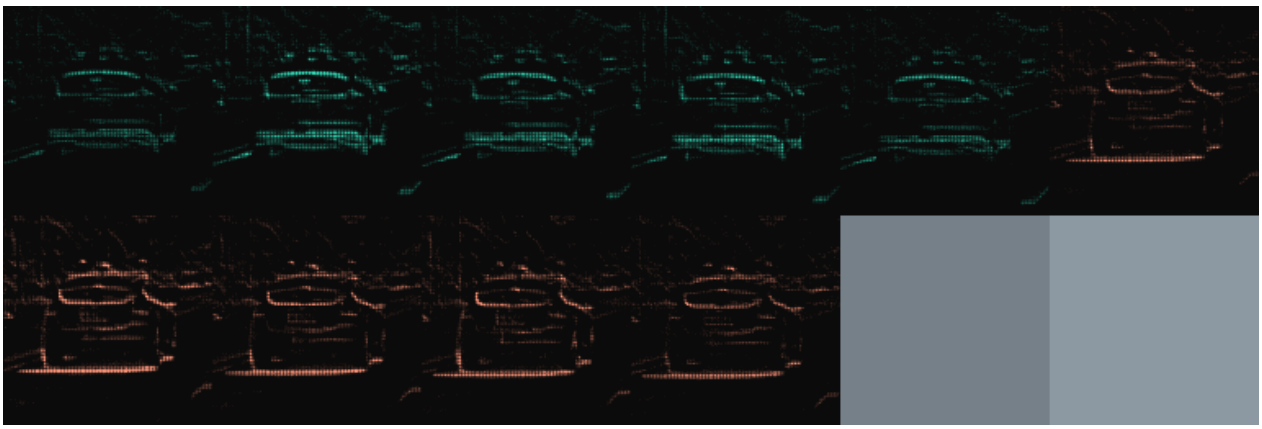


Figure 13. Twelve event input channels from a real measurement. Query 07, window $[16.700436, 16.800436)$ s. Reading left to right across the upper row gives positive voxel bins 1–5 and negative bin 1; the lower row gives negative bins 2–5, log event count, and log event rate. Each tile is 128×128 pixels. Display values are clipped to $[0, 2]$ for voxel planes and $[0, 20]$ for count/rate planes, then colorized; the saved sidecar retains the unclipped per-plane minimum, maximum, and mean. The two support planes appear nearly uniform at this display scale. Source: EvTTC [12]; extracted with the frozen runtime's `map_roi_lut` and `encode_mapped_lut`.

The source package preserves the website's extraction script, exact query and sensor timestamps, original source-byte hashes, runtime-code hashes, asset checksums, and dataset attribution in `data/real_sensor_excerpt/`. The excerpt derives from `2024-08-08-10-49-53.hdf5`; raw HDF5 and annotations are not included. It reproduces the existing website material without running inference or selecting a new model. The supplied EvTTC attribution and Clear BSD license notice accompany the images; dataset rights remain with their authors.

E.3 Predictive embedding learning

The generic JEPA implementation encodes a context representation with an online encoder f_θ and predicts a future latent conditioned on a horizon identifier:

$$z_t = f_\theta(V_t), \qquad \widehat{z}_{t+h} = g_\psi(z_t, e_h), \qquad z^\star_{t+h} = \mathrm{sg}\big(f_\xi(V_{t+h})\big). \tag{21}$$

The target starts from the online weights and is updated by an exponential moving average,

$$\xi \leftarrow m\xi + (1-m)\theta. \tag{22}$$

For multihorizon samples, a validity mask excludes unavailable future pairs from the alignment objective. The trainer supports normalized Smooth-L1 alignment, dense token prediction, spatial or tubelet masking, and variance or `visreg` regularization. Each experiment records its active loss configuration. [M04]

The V8 attribution experiment copies the `CausalScaleTTC` endpoint encoder and predicts three views of an unlabeled third endpoint from two earlier endpoints: dense features, a global geometry token, and foreground logits. Its objective is

$$\mathcal{L}_{\mathrm{V8}} = \tfrac{1}{3}\left(\mathcal{L}_{\cos}(\widehat{D}_2, D_2^\star) + \mathcal{L}_{\cos}(\widehat{g}_2, g_2^\star) + \mathcal{L}_{\mathrm{SL1}}(\widehat{F}_2, F_2^\star)\right). \tag{23}$$

The scalar foreground-logit view uses Smooth-L1 because one-dimensional cosine similarity is almost everywhere constant. A gradient-stopped, EMA-updated target encoder processes the future endpoint in evaluation mode under `no_grad`. TTC, boxes, masks, category, metadata, geometry losses, contrastive losses, VICReg, and router outputs remain outside this objective. [M05]

V8 schedules EMA momentum by cosine interpolation between configured endpoints. It measures dispersion, effective rank, covariance trace, intersample similarity, and norm for the dense, global, and foreground views. A persistent-collapse rule aborts training when these diagnostics remain below their thresholds. Downstream probes then measure TTC utility.

V8 includes strict encoder transfer, random-frozen and scratch controls, partial-finetuning allowlists, and a shuffled-future control. The shuffled control uses a deterministic cross-track derangement that preserves the future marginal while holding the batch schedule, model, optimizer, update count, masking, augmentation, and EMA schedule fixed. Track identities construct the derangement and are excluded from model inputs. The declared gates require improvement over these controls before attributing a gain to predictive temporal structure. [M05]

E.4 Spatial structure and temporal memory

The high-resolution tubelet path preserves tokens as $[B, T, P, D]$, with distinct temporal and spatial patch axes. Images are padded on the right and bottom to a ceiling-based patch grid; invalid temporal positions and spatial padding carry masks. Windowed attention mixes spatial positions within a frame before a temporal mixer receives each patch trajectory. This separation prevents flattened raster order from being interpreted as time. [M06]

A block-causal Transformer permits attention among all spatial patches at the same instant and all earlier instants, while masking later frames. If token i belongs to time t_i and token j to time t_j, its attention mask is $A_{ij} = \mathbf{1}[t_j > t_i]$. A triangular token mask would impose an arbitrary left-to-right spatial ordering within each frame, so the implementation supplies the temporal block mask explicitly.

The recurrent delta-memory alternative maintains a separate matrix state per spatial patch and attention head. Writing the channelwise retention as α_t, the implemented update is

$$\widetilde{S}_t = \operatorname{diag}(\alpha_t) S_{t-1}, \tag{24}$$

$$e_t = v_t - k_t^\top \widetilde{S}_t, \tag{25}$$

$$S_t = \widetilde{S}_t + \beta_t k_t e_t^\top, \qquad o_t = q_t^\top S_t. \tag{26}$$

Invalid patch steps retain the previous state and emit zero. State accumulation uses FP32, and the output returns to the input value dtype. Chunked execution passes explicit state between chunks and returns a detached final state, avoiding state shared between sequences. The causal recurrence uses no custom CUDA kernel. Tests establish equivalence between chunked and unchunked computation; runtime experiments determine deployment speed. [M06]

Before allocation, the implementation estimates quadratic attention-score memory, $BH(TP)^2 b$ bytes for H heads and b bytes per value, and rejects layouts above the configured safety limit. Runtime evaluation reports observed memory, preprocessing time, and throughput.

E.5 Causal scale, local transport, and TTC anchors

The geometry-constrained model learns a foreground probability field and differentiable visible extent for each endpoint. It forms geometry tokens from the encoder token, log-height, centroid, foreground fraction, and sensor support. The analytic log-scale difference is supplemented by a bounded residual,

$$r = \log h_2 - \log h_1 + \delta(a, b), \qquad \delta(a, b) = r_{\max} \tanh\left(\frac{u(a, b) - u(b, a)}{2}\right). \tag{27}$$

With deterministic evaluation and reversed transport inputs, the residual changes sign under endpoint reversal. The primary output remains geometry-bound; the auxiliary direct inverse-TTC head does not feed into it. [M07]

The timestamp anchor matters. Under constant relative speed and apparent size inversely proportional to distance, the *current-endpoint* inverse TTC is

$$q_2 = \frac{e^r - 1}{\Delta t}, \qquad T_2 = \frac{\Delta t}{e^r - 1}. \tag{28}$$

The learned-height-ratio reference implements

$$T_1 = \frac{\Delta t}{1 - h_1/h_2} = \frac{\Delta t}{1 - e^{-r}} = T_2 + \Delta t, \tag{29}$$

which is anchored at the earlier endpoint under the same assumptions. Each route uses the convention determined by its label timestamp and evaluator. The benchmark phase defined below is a metric coordinate distinct from the current-endpoint log-scale difference. [M07, M10]

The transport extension derives event-native motion evidence from dense features D_1, D_2. It builds cosine correlations in a local neighbourhood $\mathcal{N}(p)$ and obtains expected displacement by

$$P(\delta \mid p) = \frac{\exp\big(\langle \bar{D}_1(p), \bar{D}_2(p+\delta)\rangle/\tau\big)}{\sum_{\delta' \in \mathcal{N}(p)} \exp\big(\langle \bar{D}_1(p), \bar{D}_2(p+\delta')\rangle/\tau\big)}, \quad d(p) = \sum_{\delta \in \mathcal{N}(p)} P(\delta \mid p)\delta. \tag{30}$$

Out-of-bounds candidates are masked before softmax. The resulting summaries include translation, anisotropic and isotropic divergence, flow magnitude, confidence margin, entropy, and forward–reverse cycle error, with additional foreground-weighted variants. Coarse-to-fine models mix measurements from two scales through a learned gate. Shifted views replace a $[B, C, K, H, W]$ unfolded tensor to reduce intermediate memory. [M08]

Training-time RGB distillation introduces privileged visual supervision. The A4 DINOv3 teacher [10] supplies local cosine-similarity maps at six fixed feature-grid offsets. An L1 loss transfers these spatial relationships without requiring equal channel dimensions; slicing avoids circular wraparound. The student consumes events alone at inference, making this route privileged distillation rather than label-free self-supervision. [M09]

E.6 Phase-domain routing and temporal refinement

The later benchmark coordinate is

$$\phi(T) = -\log\left(1 - \frac{d}{T}\right), \qquad d = 0.1 \text{ s}, \quad T < 0 \text{ or } T > d. \tag{31}$$

Its inverse uses $q = -\operatorname{expm1}(-\phi)/d$. The implementation clamps phase to its configured support and saturates TTC magnitude at 60 seconds; zero phase maps to +60 seconds. Saturation defines finite numerical output, while signed negative values represent recession. [M10]

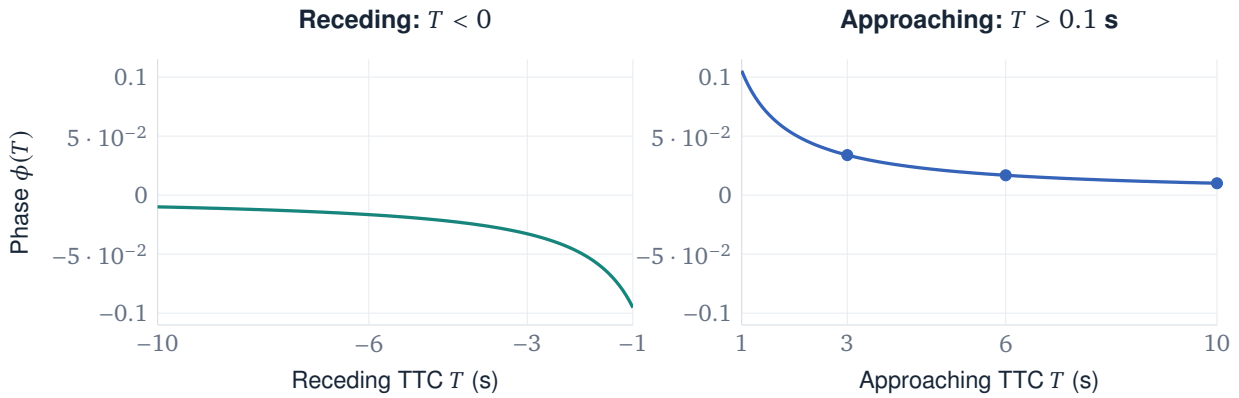


Figure 14. Signed phase geometry. Analytical curves show $|T| \in [1, 10]$ s on common vertical scales, with approaching-band markers at 3, 6, and 10 s. The zoom excludes $[0, 0.1]$ s and the adjacent singularities.

The Simplex risk router selects among frozen expert TTCs by predicting relative expected absolute-phase costs. For three sorted expert phases, convexity and the 1-Lipschitz property of $\mathbb{E}|a - Y|$ imply ordered secant slopes $-1 \le s_0 \le s_1 \le 1$. Softmax increments parameterize these constraints; integrating the slopes across expert-phase gaps yields relative costs. Hard selection returns one original expert prediction under a fixed tie order. The learned quantities are relative selection costs rather than calibrated collision probabilities. [M11]

The temporal refiner predicts a correction from a bounded feature history and four timing channels using a masked two-layer GRU; a causal Transformer provides the matched control. Histories require a

contiguous valid suffix, and missing prefixes leave recurrent state unchanged. Residual mode anchors the phase at the median of the *current* experts,

$$\widehat{\phi} = \mathrm{median}(\phi_1, \phi_2, \phi_3) + 0.03\, a_\theta(H). \tag{32}$$

Separate heads predict phase-quantile widths and relative costs for the original experts; the latter do not estimate risk for the corrected output. Training combines phase absolute error, pinball losses, and relative-cost regression. Globally defined query mass supplies the weights without minibatch renormalization. Sparse-history variants retain selected slots from a longer history and recompute timing gaps from integer anchors, preserving cold starts and the original query population. [M12]

E.7 Physical fields and temporal coordinates at the fusion interface

For each modality, the PHASE-ONLY input selects ten ordered fields from its context state: raw phase location (`raw_location`), emitted point phase (`point_phase`), the current A5, C2F, and PAIR expert phases, four current timing fields, and the number of valid history slots divided by eight. The twenty selected fields concatenate RGB first and events second. The first five fields are dimensionless signed-phase coordinates. The 160 learned hidden coordinates per modality are outside this input. Raw location precedes the native output guard, while emitted phase includes that guard. [M22, P04]

Let a_j be the integer-microsecond anchor of retained observation j, v_j its availability timestamp, and q the current query. The four history timing fields, in order, are

$$\left(\frac{a_q - a_j}{10^6}, \quad \frac{v_q - v_j}{10^6}, \quad \frac{a_j - a_{j^-}}{10^6}, \quad \frac{v_j - a_j}{10^6}\right) \quad \text{seconds.} \tag{33}$$

The third field is zero for the first retained observation; missing prefixes are zero-padded and masked. Differences are formed as integers before conversion to seconds. The fusion interface takes the fields at the current slot, so its first two timing values are zero and its third describes the gap to the previous retained anchor. Availability denotes the latest admitted sensor or annotation dependency, not measured detector runtime; the final field is availability minus the observation's label anchor, without consuming a TTC value. [P04]

Context heads receive PHASE17 fields standardized by per-coordinate means and population standard deviations fitted on unique valid observations in the allowed fitting population; a standard deviation at most 10^{-6} is replaced by one. Padding remains zero. Their four timing channels are separate, expressed in seconds. At the subsequent PHASE-ONLY input, each modality's first five phase-valued fields are divided by 0.03; the four timing fields remain in seconds and support remains valid-count/8. No fold-specific hidden-state standardization aligns latent coordinates. The RGB raw location also bypasses the correction MLP as its residual anchor, and the two native RGB width logits bypass it separately as quantile-width anchors. Zero correction therefore reproduces the RGB context prediction and widths. [M22, M23, P04]

E.8 Multimodal complement and uncertainty boundaries

The later joint-measurement complement anchors on RGB phase and learns a correction from RGB features, event features, and the RGB phase itself:

$$\widehat{\phi} = \phi_{\mathrm{RGB}} + s\, g_\theta([z_{\mathrm{RGB}}, z_{\mathrm{event}}, \phi_{\mathrm{RGB}}]). \tag{34}$$

The final layer starts at zero, so the initial prediction reproduces the RGB anchor up to native phase conversion. A matched ZERO control removes the complete event-feature block, including admitted event-clock features, while retaining network capacity. When RGB is absent, the model copies the event TTC without applying correction or averaging. The paired TRUE–ZERO comparison isolates the contribution of event information to the learned RGB correction. [M13]

Uncertainty has multiple meanings across these families. The causal-scale route predicts log-variance in log-ratio space and propagates it through the local Jacobian. For $q = \mathrm{expm1}(r)/\Delta t$,

$$\mathrm{Var}(q) \approx \frac{e^{2r}}{\Delta t^2}\,\mathrm{Var}(r), \qquad \mathrm{Var}(T) \approx \frac{\Delta t^2 e^{2r}}{(e^r - 1)^4}\,\mathrm{Var}(r). \tag{35}$$

The singularity near zero expansion requires separate finite-output and `known` indicators. The implementation marks unsupported or near-zero log-ratio predictions unknown and derives model-based threshold probabilities from an inverse-TTC Gaussian approximation. Empirical calibration requires measurements of coverage, calibration error, and corruption response. The temporal refiner's phase quantiles carry a different meaning from collision-class probabilities. [M07, M12]

E.9 Optimization and execution semantics

Component losses keep their true counts over each effective batch, and tail top-k selection uses the complete effective batch instead of separate microbatches. Activation recomputation preserves random-number-generator state, keeping gradient accumulation and recomputation equivalent to the stated objective. The joint-measurement loss includes teacher relations, unlike the label-free JEPA loss. [M14]

Deployment optimizations include bounded event history, chunked memory, persistent input state, reusable feature caches, and graph execution. Because each changes state lifetime, numerical-parity tests accompany every admitted path. Runtime measurements state the associated preprocessing and inference boundary.

F. Evaluation details and causal checks

This appendix describes data separation, causal checks, and runtime measurement in detail.

F.1 The unit of independence

Windows from one trajectory share appearance, geometry, object identity, and adjacent event history. Complete sequence groups define the grouped partitions, while the resampling unit depends on the study. Public metadata establish separation by sequence identifier but do not establish independent physical acquisition sessions for every distinct identifier. Small disjoint splits remain development evidence when the project has reused their sequences for diagnosis. This limitation does not itself demonstrate leakage. [M16, M23, P02]

The corrected grouped-recovery protocol keeps learned parents fold-local. Outer-development rows are excluded from encoder fitting, teacher-target retrieval for training, training losses, fitted feature statistics, and expert-parent fitting. A shared immutable DINO teacher cache may contain rows that a fold never retrieves. Cache storage and fold-specific consumption therefore have different boundaries. A frozen parent still carries its prior exposure, so the analysis follows sample membership through preprocessing, checkpoints, retrieved targets, normalization, and model selection. [P02]

A deterministic event representation can be reused when preprocessing and sample identity stay fixed. A checkpoint fitted on a larger population changes the exposure of every downstream result. File identity verifies which representation or model was used; training membership determines the strength of the resulting claim. [M17]

F.2 Study-specific paired inference

Historical V5 outer-development and V7 comparisons use 422 clusters, each defined by a sequence identifier and a track identifier. They use 5,000 paired bootstrap replicates over 8,192 queries from nine sequences. Their seed-7 fits do not replicate representation training across seeds. The earlier public-validation paired audit instead uses 108 sequence–track clusters over 2,048 queries. These historical intervals retain their original units even when the reported point metric averages sequences. [P02]

Table 13 distinguishes these intervals from the later sequence and hierarchical analyses. The source protocol's occasional label "acquisition group" denotes the recorded sequence grouping; independence between physical acquisition sessions remains unverified.

Table 13. Resampling units for selected paired analyses. Counts describe the recorded protocols, not independent acquisitions or independent pretraining runs. Archived numerical intervals are unchanged. [P02,L03,N02,R01,A02--A04,H_V16]

Study	Resampling unit	Replicates	Seeds and reduction
Public-validation paired audit	108 sequence–track clusters; 2,048 rows	5,000	Fixed fitted checkpoints; paired rows, sequence-macro MiD.
V5–V7 grouped development	422 sequence–track clusters; nine sequences, 8,192 rows	5,000	Seed-7 models; bootstrap seed 20260813. Seeds are not extra clusters.
V12 H16–H8 replication	Nine sequences, then tracks within sampled sequences	8,192 accepted hierarchical draws	Per-query losses averaged across fixed matched head seeds; separate exact sequence comparison.
V16 PHASE-ONLY selection	Eight V sequence groups	4,000	Bootstrap seed 20261012; group-macro bucket MiD contrasts separately for head seeds 7/13/23.
V18 TRUE/ZERO fusion	Eight V sequence groups	2,000	Bootstrap seed 12345; groupwise MiD averaged over fixed seeds 7/13/23; no TTC ensembling.
Grouped F1 adaptation	Seven scenario families; 24 recordings	2,000	Bootstrap seed 7; recording-macro RAE differences separately for head seeds 7/13/21.
401-parameter gate confirmation	Four CCRm recordings within one family	2,000	Bootstrap seed 7; query-weighted RAE difference for the fixed three-seed ensemble.
Startup router comparison	Five road scenario families	5,000	Bootstrap seed 7; query-weighted RAE differences for fixed three-seed ensembles.
Historical Extended-new snapshot	32 complete sequence groups; 468 eligible rows under the former target map	10,000	Seed 20261028; pooled-query mean, median, and p95; superseded target assignment.
Corrected Extended-new analysis	32 complete sequence groups; 460 eligible rows after target reassignment	10,000	Seed 20261028; paired saved predictions and corrected eligibility; no model refit.
Canonical cadence	24 road recordings; Slider-500 separate	10,000	Seed 7; recording-macro differences. One Slider recording has no between-recording interval.
Later JEPA pilot closure	Eight complete sequence groups	Not recorded in the distributed summary	Seeds 7/13/21 kept together; descriptive post hoc sequence intervals.

Comparisons pair the same eligible rows and sequences. For per-row metric differences d_{si} in sequence s, the sample-weighted estimate is

$$\widehat{\Delta} = \frac{\sum_s \sum_{i=1}^{n_s} d_{si}}{\sum_s n_s}. \tag{36}$$

A paired sequence bootstrap samples sequence identities with replacement and carries every row from each selected sequence into the replicate. Sample-weighted resampling recomputes the ratio of summed differences to summed counts; the sequence-macro calculation averages sequence means. The scorer-wrapper comparison uses the sample-weighted form with a fixed bootstrap seed, complete sequence clusters, and no removal of invalid MiD rows. This wrapper does not redefine the historical recovery bootstraps. [M15]

Intervals based on few sequences have limited precision and inherit development exposure and experiment multiplicity. Random seeds measure optimization variation for a fixed split, whereas out-of-fold partitions vary the held-out groups. Results report seeds, folds, and sequences separately.

F.3 Causality tests

Prefix-invariance tests determine whether later inputs alter an earlier output. The causal-scale suite checks foreground logits, geometry tokens, pair tokens, analytic and residual log-ratios, and TTC. A legacy symmetric smoother provides a positive control by violating prefix invariance; the causal-left mode must remain invariant under the same perturbation. [M18]

Network masks alone cannot establish causality. Tests also perturb crop coordinates, normalization inputs, teacher frames, and labels. Self-supervised samples must remain unchanged when labels change, and crop tests restrict geometry to the designated endpoint boxes.

Streaming tests cover state reset and update semantics. The bounded object-ROI estimator rejects timestamp rollback until reset, validates aligned packets and coordinates, and discards events outside its history. It consumes ROI-local event coordinates and supplied object observations. Paced replay then adds acquisition, queuing, temporal state, multiple objects, and host-visible output. [M03,M18,M19]

F.4 Matched controls and the scope of attribution

Architectural ablations match rows, seeds, training budgets, parent exposure, selection rules, and model capacity. A zeroed-input model removes modality information while retaining capacity. A shuffled-future control disrupts temporal correspondence while preserving the future marginal and optimization schedule. A frozen random encoder measures what the downstream head can recover without learned features.

A hard router selects among fixed expert predictions; a residual phase model can correct them. Teacher distillation measures the effect of privileged RGB supervision on an event backbone. Failed and vetoed arms remain in the analysis because they constrain mechanism-level explanations.

F.5 Latency, memory, and sustained service

A tensor-forward benchmark starts after tensor preparation. The paced runner instead includes a RAW producer, absolute arrivals, a bounded queue, and host-visible output. Completion occurs when the consumer returns every required prediction. Fixed arrivals expose backlog, while the results record drops, acquisition errors, incomplete histories, and invalid outputs. [M20]

The primary deadline-to-output response is

$$L_i = t_i^{\text{host output}} - t_i^{\text{scheduled arrival}}. \tag{37}$$

A secondary clock starts after RAW acquisition. Deadline evaluation reports tail quantiles, maximum response, coverage, backlog, and drops because mean response can hide missed arrivals. Warmup uses data available before the first admitted anchor. The 50-Hz milestone requires sustained cadence, numerical parity, and complete finite outputs. [M20]

The available RAW sequences are shorter than the sustained-service milestone. The replay therefore characterizes short-run latency for its hardware, population, numerical tolerance, and duration.

Implementation checks and their scope are reported in Appendix S. The official leaderboard score evaluates prediction accuracy; it provides no measurement of sensor-to-output latency or sustained service. [M21]

G. Corrected transfer uncertainty and historical record

The scientific revision retains the original prediction ledger and its historical numerical derivations while introducing an additive corrected target projection. The correction changes both target values and eligibility; current statistics use all 460 eligible Extended-new queries derived from the 508 registered candidates. Historical46 is unaffected by the low/medium correction. [GT01]

Table 14. All 508 extended candidates retained. Nearest TTC within 50 ms inclusive, earlier ties; corrected labels yield 460 eligible queries. Historical46 retains all 46 and is scored separately.

Sequence	Candidates	Historical eligible	Corrected eligible
CCRm-low-0-overlap-0	16	16	16
CCRm-low-100-overlap-100	16	13	13
CCRm-low-50-overlap-50	16	15	15
CCRm-medium-0-overlap-0	16	13	13
CCRm-medium-100-overlap-100	16	14	14
CCRm-medium-50-overlap-50	16	15	15
CCRs-1-high-0-overlap-0	16	15	15
CCRs-1-high-100-overlap-100	16	15	15
CCRs-1-high-50-overlap-50	16	15	15
CCRs-1-low-0-overlap-0	16	13	13
CCRs-1-low-100-overlap-100	16	15	10
CCRs-1-low-50-overlap-50	16	16	16
CCRs-1-medium-0-overlap-0	16	14	14
CCRs-1-medium-100-overlap-100	16	15	12
CCRs-1-medium-50-overlap-50	16	15	15
CCRs-2-high-100-overlap-100	12	11	11
CCRs-2-low-100-overlap-100	16	16	16
CCRs-2-medium-100-overlap-100	16	15	15
CCRs-3-low-100-overlap-100	16	13	13
CCRs-3-medium-100-overlap-100	16	16	16
CCRs-side-high	16	16	16
CCRs-side-low	16	16	16
CCRs-side-medium	16	16	16
CPLA-high	16	15	15
CPLA-low	16	15	15
CPLA-medium	16	15	15
CPNA-high	16	15	15
CPNA-low	16	12	12
CPNA-medium	16	13	13
CPNAO-high	16	14	14
CPNAO-low	16	16	16
CPNAO-medium	16	15	15

G.1 Target reassociation versus metric recomputation

The source audit first reproduces every historical target or exclusion for all 554 saved candidates under the old source assignment. It then replaces only the two verified low/medium curves and applies the unchanged nearest-label rule at the unchanged native timestamps. The projection records the corrected source SHA-256, target time, time distance, and before/after label neighbors. The separate metric builder verifies the projection hash, all prediction fields against the historical CSV, and the nearest-neighbor decisions before recomputing statistics.

This provides distinct checks of recording/source identity and arithmetic. The original raw sensor streams are not needed for saved-prediction scoring. Repeating the full association audit additionally requires the source curves and original query manifest pinned in `audit_20261104/transfer_reassociation.json`; no target-driven clock fitting is involved.

G.2 Paired complete-sequence bootstrap

Each of 10,000 resamples draws 32 sequence-group identities with replacement and includes all eligible queries from each selected group, repeating queries when their group is drawn more than once. PHASE-TTC and RGB share the draw. Pooled mean, median, and p95 differences use NumPy's linear quantiles with seed 20261028. The three intervals contain zero (Table 5). They condition on the saved models and historically exposed recordings.

The mean difference is -0.038 s, median difference +0.055 s, and p95 difference -0.296 s. Complete-precision estimates and intervals reside in `audit_20261104/corrected_transfer_statistics.json`. Sequence-macro MAE/RTE comparisons against Garl use their original 2,000 group resamples and seed 7; they are separate estimands from the pooled PHASE–RGB comparison.

G.3 Sequence-level comparisons

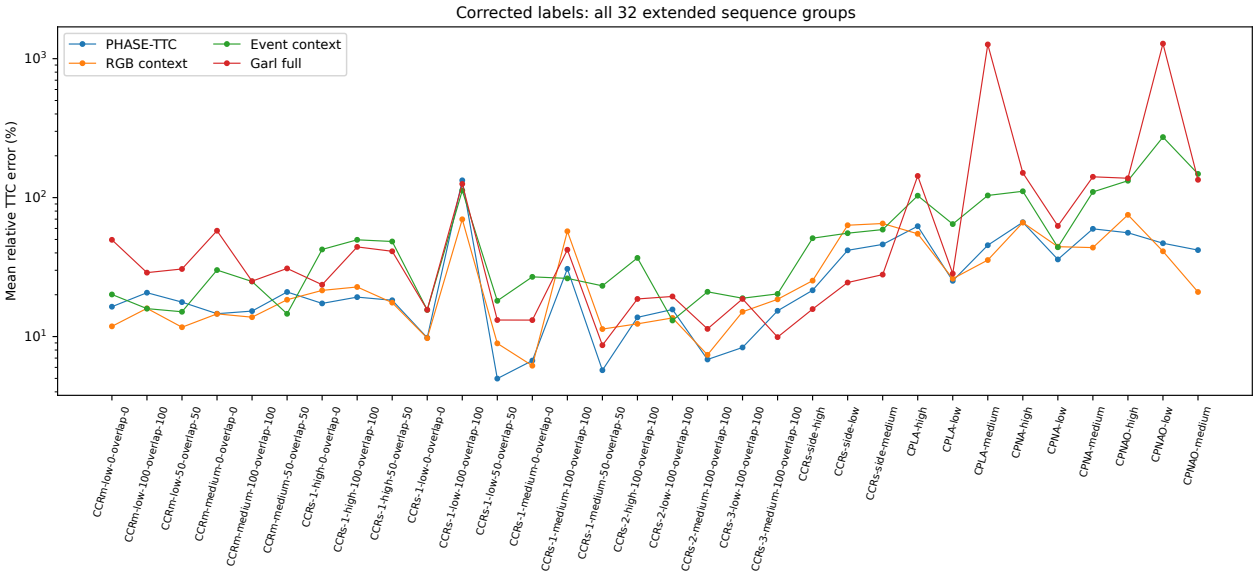


Figure 15. Corrected sequence-level relative errors. Curves show each model's mean relative TTC error on a logarithmic scale; lower is better. The thirty-two sequence groups retain their native identities. Table 15 gives the corrected eligible query counts and errors.

Table 15. Corrected extended-cohort errors by all 32 sequence groups. MAE in seconds; RTE in percent.

Sequence	n	PHASE MAE	RTE	RGB MAE	RTE	Event MAE	RTE	Garl MAE	RTE
CCRm-low-0-overlap-0	16	0.535	16.401	0.350	11.845	0.548	20.089	1.143	49.694
CCRm-low-100-overlap-100	13	1.241	20.687	0.976	15.960	1.056	15.888	1.607	28.848
CCRm-low-50-overlap-50	15	0.873	17.699	0.563	11.675	0.846	15.084	1.483	30.650
CCRm-medium-0-overlap-0	13	1.183	14.607	1.097	14.565	1.707	30.105	4.746	57.778
CCRm-medium-100-overlap-100	14	1.308	15.253	1.225	13.787	1.800	24.898	2.020	25.036
CCRm-medium-50-overlap-50	15	1.632	20.935	1.495	18.408	1.457	14.583	2.412	30.928
CCRs-1-high-0-overlap-0	15	0.581	17.324	0.691	21.492	1.246	42.358	0.715	23.613
CCRs-1-high-100-overlap-100	15	0.436	19.227	0.473	22.748	1.033	49.698	1.005	44.193
CCRs-1-high-50-overlap-50	15	0.495	18.253	0.441	17.546	1.337	48.337	0.992	41.092
CCRs-1-low-0-overlap-0	13	0.457	9.808	0.364	9.735	0.539	15.563	0.622	15.576
CCRs-1-low-100-overlap-100	10	1.882	133.204	1.144	69.806	2.124	112.589	2.062	125.459
CCRs-1-low-50-overlap-50	16	0.107	4.983	0.217	8.934	0.348	18.090	0.379	13.154
CCRs-1-medium-0-overlap-0	14	0.177	6.715	0.157	6.170	0.638	26.896	0.353	13.138
CCRs-1-medium-100-overlap-100	12	1.165	30.732	1.915	57.257	1.106	26.229	1.256	42.152
CCRs-1-medium-50-overlap-50	15	0.102	5.726	0.202	11.323	0.447	23.179	0.206	8.664
CCRs-2-high-100-overlap-100	11	0.443	13.747	0.400	12.346	1.185	36.812	0.588	18.652
CCRs-2-low-100-overlap-100	16	0.979	15.666	0.802	13.616	0.869	13.056	1.091	19.437
CCRs-2-medium-100-overlap-100	15	0.164	6.839	0.177	7.389	0.501	20.995	0.290	11.356
CCRs-3-low-100-overlap-100	13	0.359	8.348	0.595	15.077	0.779	18.878	0.876	18.667
CCRs-3-medium-100-overlap-100	16	0.531	15.324	0.629	18.539	0.598	20.285	0.294	9.908
CCRs-side-high	16	0.466	21.539	0.570	25.262	1.010	51.047	0.334	15.771
CCRs-side-low	16	1.207	41.760	2.146	63.283	1.460	55.465	1.104	24.485
CCRs-side-medium	16	1.507	46.059	1.889	65.107	1.893	58.878	1.611	27.916
CPLA-high	15	1.051	62.285	0.908	54.915	1.723	103.255	2.274	143.221
CPLA-low	15	0.775	25.190	0.949	26.090	1.829	64.513	0.704	28.375
CPLA-medium	15	0.731	45.430	0.622	35.531	1.796	103.663	15.406	1265.982
CPNA-high	15	1.038	66.503	1.072	66.142	1.543	111.085	2.122	150.708
CPNA-low	12	1.133	35.904	1.467	44.345	1.275	43.942	2.334	62.536
CPNA-medium	13	0.924	59.614	0.712	43.684	1.669	109.875	2.212	141.205
CPNAO-high	14	1.432	55.864	2.038	75.177	3.032	132.298	3.517	137.892
CPNAO-low	16	1.035	46.955	0.956	41.095	4.935	272.685	39.943	1283.264
CPNAO-medium	15	0.511	41.916	0.248	20.936	1.732	148.122	1.615	134.360

G.4 Preserved historical analysis

The files `revision/derived_transfer_analysis.json`, `tables/revision_bootstrap.tex`, the earlier numerical-case table, and the original transfer figures reconstruct the 468-query analysis under the prior assignment. Their matching arithmetic does not establish correct sensor–target correspondence. In particular, the former p95 pair 3.80/3.24 s, the former ordering of mean errors, and the extreme numerical cases from that population are superseded by the corrected analysis above. These files remain available as historical evidence and are not mixed into the revised tables or figures. [R01, L46, GT01]

H. Competition adaptation and complete prediction traces

This appendix details input construction and timing for the original 434-query competition inference snapshot of 28 October. Linear and symmetric-log views show the full signed prediction traces. The corrected target audit and quality-adapted PHASE-TTC comparison added on 1 November appear in Section 7; they do not alter the frozen predictions shown here. [T01, N01]

H.1 Frozen models and causal input construction

The full, event-only, and visual-only routes use the public checkpoints with their released configurations and source revision. Strict state-dictionary loading and float32 inference keep each model fixed. The release identifies the checkpoint bytes but does not document complete training membership, so the evaluation represents frozen-checkpoint adaptation rather than a claim of clean zero-shot transfer.

Garl-TTC predicts the ratio of object heights at two endpoints and converts it to TTC as

$$\widehat{T} = \frac{\Delta t_{\text{measured}}}{1 - \widehat{h}_0/\widehat{h}_1}. \tag{38}$$

The released configuration sets the native model field dT = 0.1 s. For Equation 38, the adapter substitutes the measured separation of each endpoint pair while leaving the released representation unchanged. The checkpoint metadata do not establish the cadence used during training.

The full and visual routes select the latest real RGB frame at or before a query, then the preceding frame closest to 100 ms earlier. Full Garl builds one causal event window per RGB endpoint with support capped at $\min(\Delta t_{\text{measured}}, 100\,\text{ms})$. Event-only Garl uses two consecutive 100 ms windows after startup and splits the observed causal prefix into two intervals for earlier queries. Each endpoint uses the released 20-plane, polarity-agnostic time volume. Queries without sufficient history remain gaps.

PHASE-TTC uses its declared 650 ms causal history. The shared query population therefore compares the frozen systems under their respective temporal support.

Supplied boxes use zero-based, half-open event-plane coordinates. The adapter selects the latest box available at each endpoint; if the endpoint precedes all query-causal boxes, it uses the earliest such box and records availability at query time. Road sequences pair right Blackfly RGB with right Prophesee events. Slider uses its supplied streams and a rotation derived from camera poses because its calibration lacks a named event-to-RGB extrinsic. Calibrated intrinsics, distortion, and rotation-only ray projection map the event box into RGB; unavailable object depth precludes translational correction. A common square edge spans the endpoint boxes, with endpoint-specific event centers and the current center applied to both RGB frames.

H.2 Published three-sequence comparison

Table VI of the eAP paper reports Garl-TTC on three EvTTC sequences, with an unweighted mean RTE of 10.60% [7]. That study uses a separate labeled population; shared scenario names do not establish shared recordings or query identities. [T02]

Table 16. Garl-TTC's published zero-shot relative TTC error on three EvTTC sequences, as reported in the eAP paper, Table VI. [T02]

Published sequence	RTE (%)
CCRs2-medium	8.31
CCRs2-high	10.56
CCRm-medium	12.93
Unweighted average	**10.60**

Table 17. Garl–PHASE prediction comparisons for all ten sequences. Each entry gives shared-finite coverage and median absolute prediction difference in seconds. [T01]

Sequence	Full n/N; med.	Event n/N; med.	Visual n/N; med.
CCRs1-low	49/50; 0.465	50/50; 0.776	49/50; 0.213
CCRs1-medium	42/43; 0.308	43/43; 0.863	42/43; 0.200
CCRs1-high	29/30; 0.207	30/30; 0.519	29/30; 0.111
CCRs2-low	45/46; 0.221	46/46; 0.339	45/46; 0.243
CCRs2-medium	54/55; 0.178	55/55; 1.045	54/55; 0.254
CCRs2-high	58/59; 0.188	59/59; 0.682	58/59; 0.223
CCRm-low	41/42; 0.270	42/42; 0.703	41/42; 0.283
CCRm-medium	39/40; 0.151	40/40; 0.839	39/40; 0.350
Slider-750	49/51; 0.408	51/51; 0.470	51/51; 0.258
Slider-1000	18/18; 0.288	18/18; 0.119	18/18; 0.067

H.3 Runtime boundary and interpretation

Table 18. Offline processing time in seconds per query. Model time covers the synchronized forward pass and host output; full time includes tensor preparation. The Garl variants ran in full/event/visual order; PHASE-TTC used its own adapter and history length. [T01]

Model	Model med.	Model P95	Full med.	Full P95
PHASE-TTC	0.753	7.820	2.687	9.073
Garl full	0.039	0.045	0.542	1.009
Garl event-only	0.007	0.011	0.350	0.806
Garl visual-only	0.007	0.011	0.158	0.221

Timing separates tensor preparation from the synchronized model-to-host call and excludes loading, indexing, calibration, source hashing, and result serialization. The variants run once in full, event-only, visual-only order, so operating-system and HDF5 caches and GPU warm state can carry between runs. Table 18 reports these observations alongside PHASE-TTC, whose adapter and temporal support differ.

Independent verification accounts for each planned record and replays the float32 height-to-TTC conversion. Full and visual Garl remain partial because of input gaps; event-only contains all 434 rows. Across the ten sequences, the four routes exhibit distinct signed behavior and disagreement tails. [T01]

H.4 Target audit and evaluation boundary

The later audit recovers all 434 local TTC targets after correcting the public source mapping for CCRs1-low and CCRs1-medium. It preserves the recorded clock offsets rather than fitting timing shifts to match errors. Earlier 423-target summaries are superseded. The corrected original-versus-adapted comparison applies the same documented handling of negative predictions to both columns and reports complete population counts. The historical signed traces below retain their original values and gaps. [N01]

For Slider-750, the corrected local evaluator gives 283.9% relative error for the original PHASE-TTC predictions, compared with 132% in the organizer-provided result. The other nine rows agree after rounding. The adapted local value of 76.071442% therefore cannot be compared directly against the

organizer's 132% as if both used an equivalent scoring contract. The unresolved target/evaluator correspondence and the cohort's historical development exposure remain explicit limitations. This appendix supplies no new error ranking for the three Garl variants. [N01]

H.5 All ten sequences, with missing inputs preserved

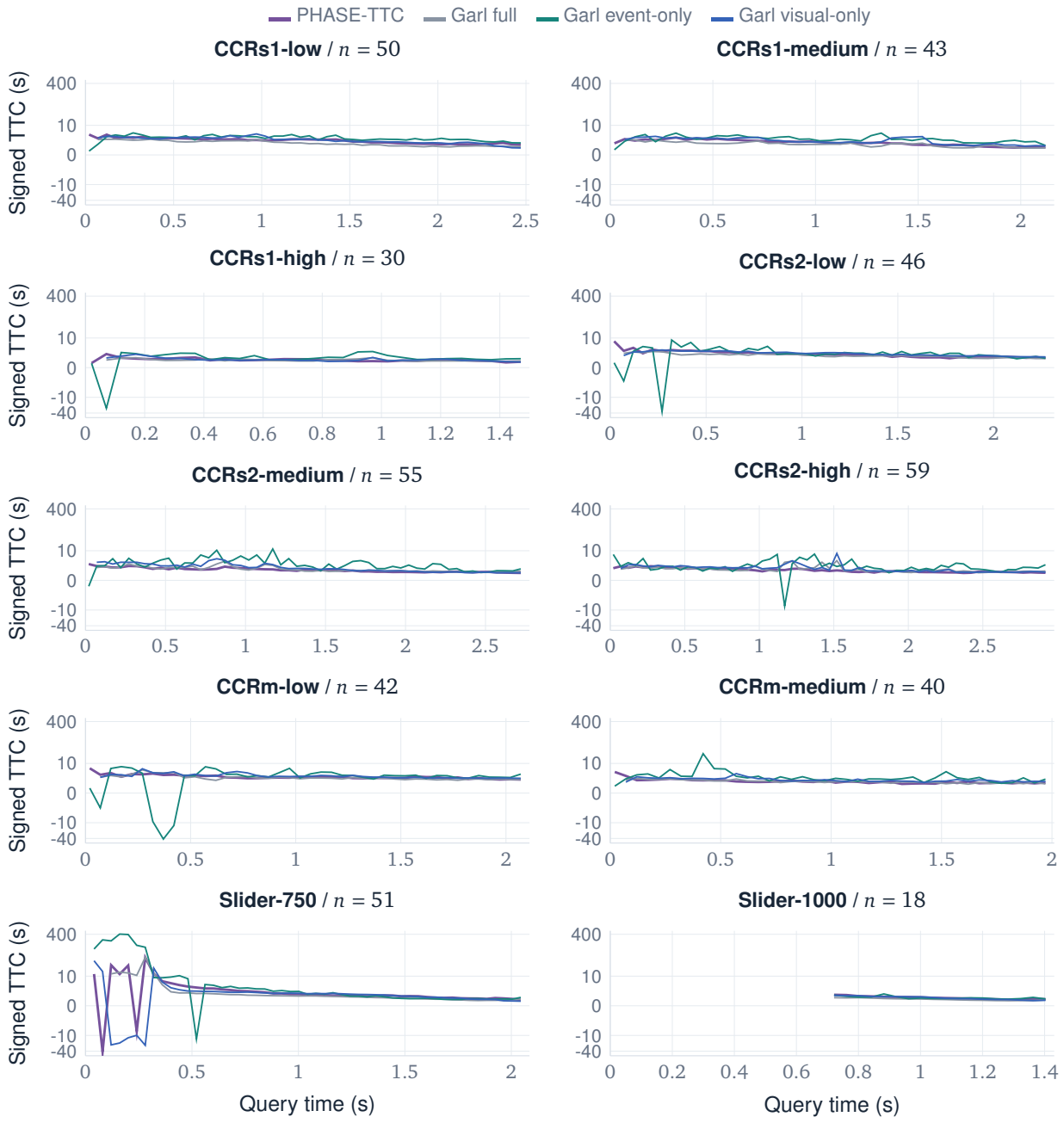


Figure 16. Signed predictions on a symmetric-log scale. PHASE-TTC is purple, Garl full gray, event-only teal, and visual-only blue. All panels are linear inside $[-2, 2]$ s and logarithmic outside; gaps mark unavailable inputs. [T01]

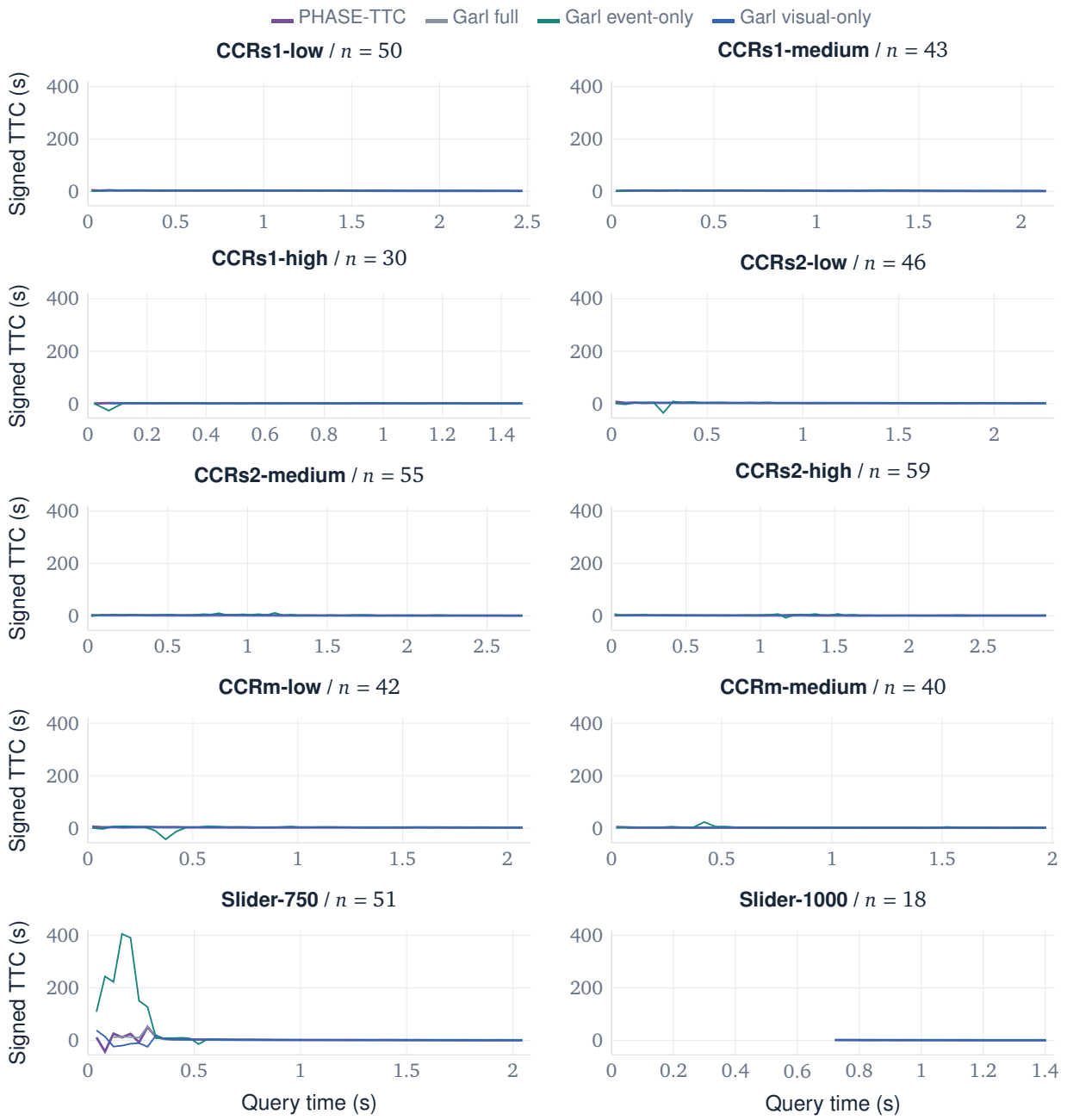


Figure 17. Signed predictions on a common linear scale. The same outputs as Figure 16 show the full excursion range, which compresses the central values. [T01]

I. Historical design analysis and controlled ablations

Three findings shaped PHASE-TTC: preserving scale change mattered more than representation stability alone; correcting an expert's phase could improve on selecting an expert; and changing the loss often traded one TTC range against another. The studies below used different data splits and pretrained models. Appendices L, M, and O give the full configurations and unsuccessful variants.

I.1 Preserve the motion variable before predicting it

Using a common object region preserves apparent expansion across observations. Independently resizing each endpoint can remove that signal before the network sees it (Figure 10). An explicit object query also distinguishes objects that share one full-frame input but have different TTC targets. The estimator consequently requires externally supplied localization. [M03]

The geometry experiments exposed a related difficulty: stable foreground maps and accurate absolute heights did not ensure accurate changes in scale. Some models also received RGB-derived SAM masks or DINOv3 relations during training, even when their inference inputs contained only events.

Removing paths from a trained V8 model identified which inputs its prediction used. Analytic scale alone scored 373.18 MiD, and adding the residual without transport gave 373.60. Restoring transport reduced the score to 163.23; restoring history reduced it further to 155.82. The weights remained fixed throughout these tests. [E02,E12,E25,E26,E21,E23,E24]

I.2 Predictive embeddings and dynamic scale

Predicting a future embedding and estimating expansion are different objectives. A representation can make the future easy to predict by retaining persistent appearance, yet discard the small changes needed to recover $\dot{s}/s$. Conversely, preventing collapse by maintaining variance says little about which physical variable supplies that variance. This distinction motivates evaluating TTC and geometric increments alongside prediction loss.

In the synthetic controls, a temporal-residual objective recovered more dynamics than variance regularization when nuisance features changed slowly. Making those nuisances vary independently across frames reversed the ordering. By varying nuisance dynamics independently of the target motion, this controlled construction shows how the choice of predictable signal changes the ordering of the objectives, consistent with the slow-feature concern raised by Sobal et al. [11]. Testing that mechanism on real driving data requires interventions on the nuisance signal. [E02,E12]

On real data, the V8 frozen and partially fine-tuned JEPA encoders failed the benefit criterion against scratch, random-frozen, and shuffled-future controls. A separate fixed-teacher pilot reduced future-state loss while worsening teacher-increment loss and failed to outperform both static and shuffled-future controls on annotated log-height change. Across these tests, predictive pretraining fell short of the prescribed benefit criteria. The role of background appearance in the real-data result is an open attribution question. Appendices L, E, and P give the objective definitions and controls. [E23,E24,N04]

These findings motivate measuring expansion directly at the fusion interface and checking it at the TTC output. They leave open whether other predictive targets, sampling schemes, or training budgets can learn a better motion representation.

I.3 Correct phase rather than only selecting an expert

A hard router is limited to the predictions its experts already make. SIMPLEX-T can move beyond them by learning a correction around their median phase from masked temporal context. The auxiliary cost head describes the original experts; evaluating the corrected prediction's risk requires a corresponding risk target. Phase quantiles also need empirical calibration before use as collision probabilities. [M_TEMPORAL_MODEL,M_PHASE_CONTRACT]

Extending the context from eight history slots (H8) to sixteen (H16) lowered MiD for each of three head seeds on the initial exploration cohort. The paired interval across all seeds excludes zero. However, one seed had already informed exploration; the two new seeds alone give a difference of −2.42 MiD with interval $[-5.24, 0.04]$. All heads shared the same upstream producers. Figure 25 shows both summaries; H8 remained the selected candidate. [M_SIMPLEX]

Keeping all experts produced the lowest score in the removal study: FULL-C0 reached 121.40 MiD, compared with 127.79 for A5-only, 126.70 for C2F-only, and 124.67 for A5 plus PAIR.

In the temporal-aggregation comparison, set aggregation with age gave 122.44 MiD; removing explicit time gave 123.69. Because the frozen experts can already encode temporal information, the remaining differences cannot be assigned to recurrence alone.

Transport probes improved on destructive controls, but stronger calibrated or pooled alternatives narrowed that advantage. [M_SIMPLEX,M_STAGE61,M_STAGE62]

I.4 Adaptation trades error between TTC ranges

Changes to the readout often improved imminent-contact estimates at the expense of receding motion. The band-reweighted phase-loss arm (V14 M1–L0) reduced crucial MiD from 156.83 to 134.47 but increased negative MiD from 61.38 to 84.21. V17's geometry bypass lowered the aggregate score on the development-used V split from 106.545 to 102.762 while worsening MAE; its head seeds shared encoders. Across V14–V17, loss reweighting, residual range, sampling, numerical precision, clipping, geometry bypasses, and height readouts failed to meet all acceptance criteria. Invalid height-derived predictions were retained in scoring. The next experiments therefore allowed the measurement model itself to adapt. [L02,L50--L52]

V18 full adaptation reduced aggregate MiD on H from 112.29 to 78.92 and crucial MiD from 157.27 to 95.85, while negative MiD rose from 65.52 to 87.09. The model passed the overall sign criterion but remained inaccurate in the magnitude of receding TTC. V19 increased the training weight on negative targets and recovered some negative-range accuracy and MAE, at the cost of aggregate and crucial-range error. [L10,L24,L25,L30]

I.5 Events help in the matched V18 comparison

Supplying real events improved V18 C1 relative to its equal-capacity ZERO control. TRUE-event fusion achieved aggregate MiD 87.29 ± 0.87, compared with 92.47 ± 0.16; dispersion is the sample standard deviation of the scalar scores across seeds. The paired comparison over complete groups also favored TRUE. Each seed repeated C1 adaptation and head fitting from the same A5 initialization and RGB context model. This experiment uses V18 C1, whereas the final forty-sequence system derives from V16. [L03,L31--L33]

Section N.5 separates the historical V16 selection and its four-fold context cross-fit from the later five-rotation TRAIN40 refit. The V18 crucial-band gain remained below its registered threshold. Its comparison with the preserved V16 model gave an interval of approximately $[-3.69, 0.14]$, and no V18 candidate met all acceptance criteria. Section 6 reports new equal-capacity final-head controls on fixed source-domain out-of-fold features. Those controls keep the temporal support of the final refiners; they do not compare against retrained Garl producers.

J. Geometric adapter implementation and earlier evaluation

LOCAL ADAPTATION / 2,608 fitting queries / 434 corrected evaluation queries / historical exposure retained

The base fusion model supplies several estimates of the same signed quantity: its final fused TTC, the RGB context estimate, and the event context estimate. Their errors need not agree, and their reliability changes with temporal support and apparent motion. PHASE-TTC-Adapt leaves all nine upstream

components frozen and learns a small router over those three outputs and a causal box-expansion estimate. The experiment adapts the fusion rule over fixed representations; Section D evaluates temporal normalization as a separate intervention. [N01]

The adapter uses the frozen base model, denoted F0 in the source archive. Earlier fine-tuned heads and quality-gate candidates are separate models; Appendix C documents their populations, unsuccessful controls, and interval coverage. In particular, the delivered router does not use the fine-tuned weights affected by the earlier target-source assignment.

J.1 Causal box geometry and its quality descriptors

For each query, the geometric branch admits supplied boxes with timestamps in the preceding 650 ms, rejects nonfinite or nonpositive extents, and requires strictly increasing observation times. It fits three apparent sizes: width, height, and square root of area. Let s_i denote one such size, s_n its latest value, and u_i time relative to the query in seconds. The branch constructs $\rho_i = s_n/s_i$ and estimates a line using median pair slopes,

$$b = \operatorname{median}_{i<j} \frac{\rho_j - \rho_i}{u_j - u_i}, \qquad a = \operatorname{median}_i(\rho_i - b u_i), \tag{39}$$

$$q_s = -b/a \quad (a > 0). \tag{40}$$

The median of the three q_s values supplies the geometric inverse TTC. This is an apparent-scale model with a query-time intercept; it does not recover metric depth or compensate for arbitrary rotation, occlusion, or annotation jitter. At least two boxes spanning 10 ms are needed for a fit. Near-zero expansion, nonfinite descriptors, or an endpoint older than 150 ms makes the geometric estimate unavailable. A two-point zero residual is not interpreted as zero uncertainty. [N01]

Twenty descriptors expose the conditions of that fit: the three axis/area rates, their median and short-history counterpart, rate disagreement, residual and residual-to-motion ratio, temporal span, normalized box count, admitted-box fraction, endpoint age, quantization support, short/long disagreement, aspect ratio and its change, center motion, and slope scatter. Absolute coordinates and absolute clock time are absent. We use supplied object regions to evaluate the fusion rule under known localization. Autonomous deployment will require testing a causal detector together with the estimator, including the effects of its box errors and delay.

The intersensor projection uses calibrated rotation without depth-dependent translation, and Slider extrinsics remain provisional. The annotated-region contract and these projection limits also apply to the adaptation comparison; they should accompany any interpretation of its geometric input. [A03, A07]

J.2 Router architecture and exact output policy

Each quality router receives 33 values. Thirteen describe the three neural magnitudes, their availability, pairwise sign agreements, geometric availability, and neural/geometric sign agreement. The remaining twenty are the geometric quality descriptors, with the first five rate coordinates replaced by their magnitudes. This design lets the weights respond to agreement and support without supplying a sequence identifier or a target-derived error. After fitted normalization and componentwise clipping to $[-8, 8]$, the gate computes $g(\mathbf{x}) = W_2 \tanh(W_1\mathbf{x} + \mathbf{b}_1) + \mathbf{b}_2$, with hidden width 18 and four output logits. Its two affine maps contain 688 trainable parameters. The final affine weights start at zero with bias $(2, 0, 0, 0)$, favoring F0 when other experts are available without forcing an exact initial F0 output. [N01]

Let m_j indicate an available expert and let $g_j(\mathbf{x})$ be its gate logit. The router forms normalized masked weights

$$w_j = \frac{m_j \exp g_j(\mathbf{x})}{\sum_k m_k \exp g_k(\mathbf{x})}, \qquad \widehat{q} = \sum_{j:m_j=1} \frac{w_j}{\widetilde{T}_j}, \tag{41}$$

where $\widetilde{T}_j$ preserves the expert sign and limits its magnitude to $[0.1, 60]$ s. The implementation excludes zero-valued experts from inverse-coordinate fusion. It applies a rate-magnitude floor of $1/60\,\mathrm{s}^{-1}$ before

Table 19. Corrected local EvTTC comparison on historically exposed data. MAE is in seconds; RAE is a percentage. Both columns apply the organizer-described interpolation of negative predictions for scoring only. Values are query-weighted local scores.

Population	n	MAE orig.	MAE adapt.	RAE orig.	RAE adapt.
All queries	434	0.8250	0.3359	49.894	17.250
Road	365	0.3054	0.1852	11.917	7.469
Excluding Slider-750	383	0.3296	0.1933	18.730	9.417
Slider-750	51	4.5448	1.4071	283.931	76.071

inversion, yielding a finite signed output with magnitude at most 60 s. Exact cancellation, $\widehat{q} = 0$, returns the original fused prediction F0 rather than choosing an approach direction.

The delivered ensemble contains routers trained with three random seeds, for $3 \times 688 = 2,064$ learned router parameters. Each router performs Equation 41 and emits seconds; the ensemble then averages those three TTC predictions in seconds. It does not average their inverse rates. The ensemble returns F0 exactly when no valid box history is supplied, when endpoint age is nonfinite, or when the latest box is older than 150 ms. A valid geometric estimate is a separate requirement: a fresh but too-short box history can leave geometry unavailable while allowing the router to combine available neural estimates. These distinctions preserve the implemented fallback semantics. [N01]

J.3 Fitting and local evaluation

The router fit uses 2,608 queries: 2,531 from twenty-four road recordings and 77 from Slider-500. Each of three seeds runs 300 AdamW updates with learning rate and weight decay both 0.01. The evaluation recordings are excluded from these optimizer updates, but they informed earlier development; the comparison therefore measures local adaptation under the protocol in Section 3.

J.4 Corrected evaluation of the delivered adapter

The evaluation contains 434 corrected target correspondences. We report both signed runtime predictions and the organizer-described offline interpolation of negative outputs. The latter may use later valid predictions and is never part of the causal forward path. Appendix A.3 and the source records document target correction; the scoring policy is fixed in Section 3. [N01]

Both scoring contracts favor the adapter. With the original signed outputs retained, F0/adapted MAE is 0.8244/0.3470 s and RAE is 49.73/17.79%. After organizer-style interpolation, the corresponding pairs are 0.8250/0.3359 s and 49.89/17.25%. The MAE reductions are 57.91% and 59.28%, respectively. [N01]

Over all 434 queries, MAE decreases from 0.8250 to 0.3359 s and mean relative error from 49.89% to 17.25%. The respective relative reductions are 59.28% and 65.43%. Both errors decrease in each of the ten sequences under this evaluator. On the 365 road queries, MAE falls from 0.3054 to 0.1852 s, showing improvement across the road domain as well as the pooled population. These query-weighted reductions use a different aggregation from the recording-macro cadence summaries in Appendix D.

Comparison with organizer scores. Our reconstruction matches the organizer-provided original relative error for nine of ten sequences after rounding. For Slider-750, we obtain 283.9% rather than the organizer's 132%; the adapted model scores 76.1% under our local evaluator. The available records leave the source of this discrepancy unresolved. Reconciliation requires the organizer-aligned targets and exact evaluator version, so we report the adaptation gains under the local scoring contract. Appendix H records the target correspondence and evaluation boundary.

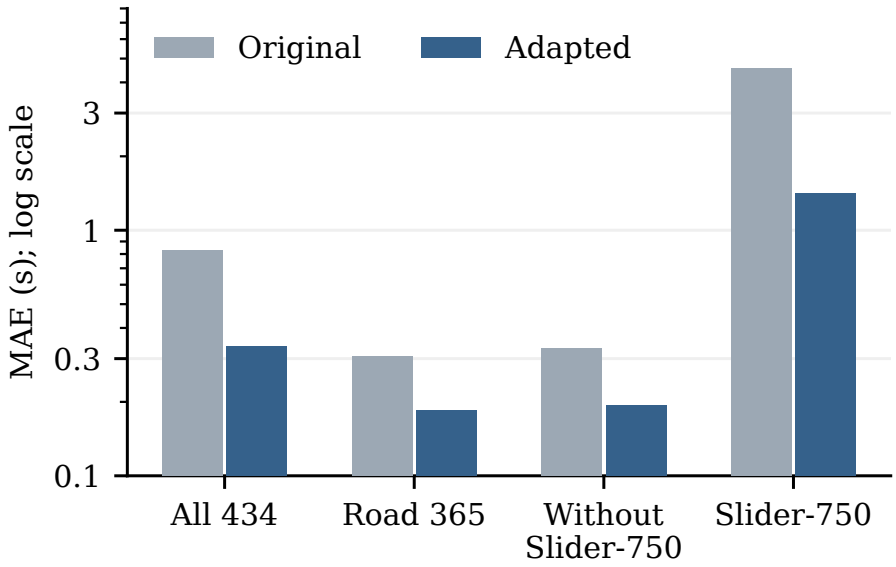


Figure 18. Local effect of quality adaptation. The corrected evaluator compares the original and adapted outputs on the same queries. A logarithmic vertical axis separates the road and Slider error scales. This figure reports exposed development evidence, with organizer-style negative-output interpolation applied to both variants. [N01]

J.5 Matched attribution follow-up

Section 7 adds geometry-only, fallback, fixed-mixture, and router comparisons on identical query support. The earlier protocol above is retained to distinguish raw signed inference from organizer-style offline interpolation.

K. Additional sensor and benchmark illustrations

The following figures supply the sensor detail and score decompositions behind the compact main-text tables.

K.1 Real sensor observations

Figure 19 shows the appearance of the two sensor streams on an actual EvTTC road recording [12]. The RGB frame records scene appearance, whereas the event view accumulates signed brightness changes during the preceding 100 ms. Vehicle outlines, road markings, and background motion therefore appear differently in the two modalities. These are recorded observations rather than schematic renderings. The side-by-side views retain the coordinates of their respective cameras.

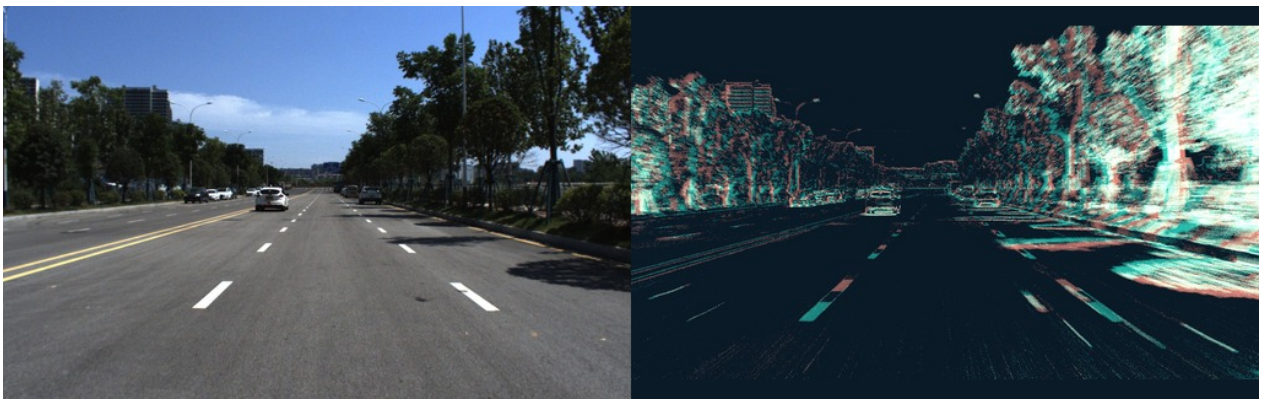


Figure 19. Real RGB and event observations from EvTTC. Left: Blackfly left-camera frame 287 at $t = 14.350374$ s in recording `CCRm-low-0-overlap-0`. Right: Prophesee left-camera events in $[t - 100\,\text{ms}, t)$, with positive counts in turquoise and negative counts in coral; overlap can appear pale. Counts use log scaling and an occupied-pixel 95th-percentile display scale. The RGB display swaps stored channels 0 and 2 to match the dataset publisher's visualization; the model-input crops preserve the runtime channel order. Images are resized for display and retain separate sensor coordinates. Source: EvTTC, Sun et al. [12]; the same excerpt is used in the project website.

The RGB view preserves scene appearance while event polarities reveal moving contours and contrast boundaries. The model receives localized object crops in a common coordinate frame across causal endpoints, preserving apparent scale change. Appendix E.2 shows the temporal crop triplets and twelve event input planes for this recording, with their timestamps and extraction provenance.

K.2 Benchmark score decompositions

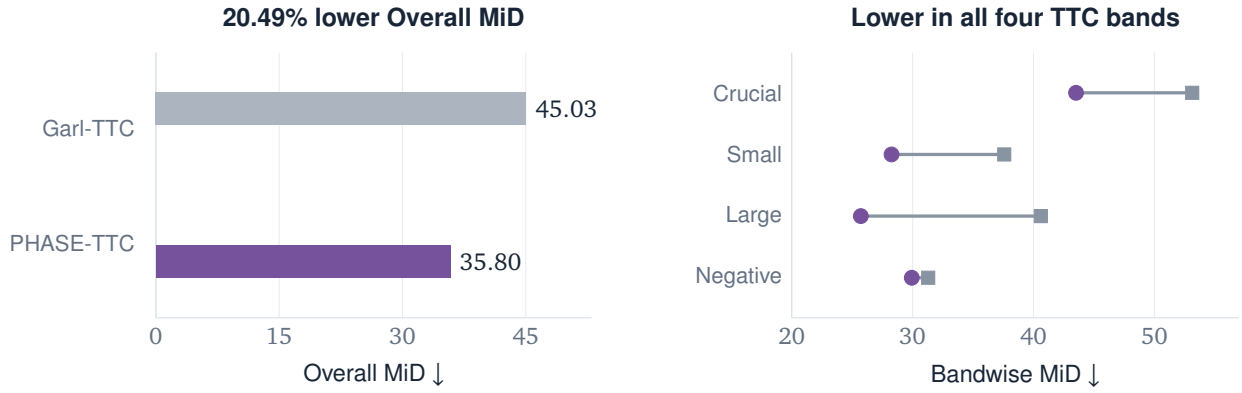


Figure 20. Archived GarlTTC Test results, 28 October 2026. Overall MiD and the four TTC bands compare the complete PHASE-TTC and Garl-TTC submissions. Both report 0% scorer-defined failures across 6,762 tokens.

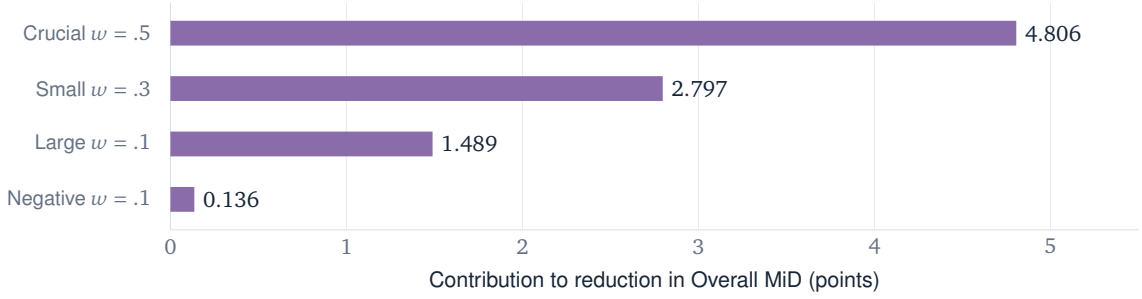


Figure 21. Bandwise contributions to the Overall MiD difference. Crucial and small TTC account for about 82% under the benchmark's 0.5/0.3/0.1/0.1 weights. Table 37 gives the rounding reconciliation. [B01, B02]

K.3 Transfer error distributions

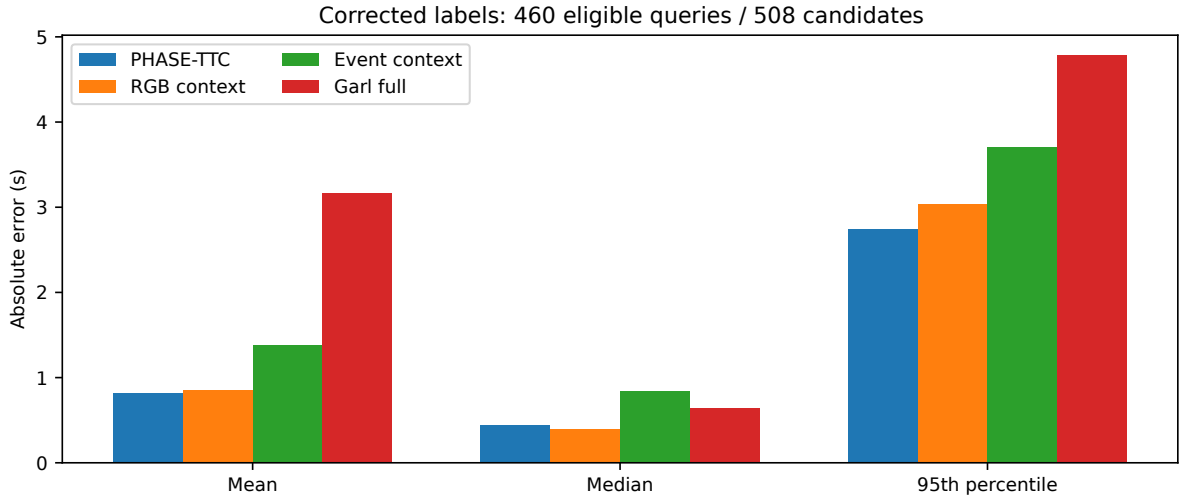


Figure 22. Corrected frozen-transfer error summaries. PHASE-TTC and RGB use the same 460 eligible queries after reassociating all 508 candidates. Lower point mean and p95 favor fusion; lower median favors RGB. Paired intervals are reported separately in Table 5.

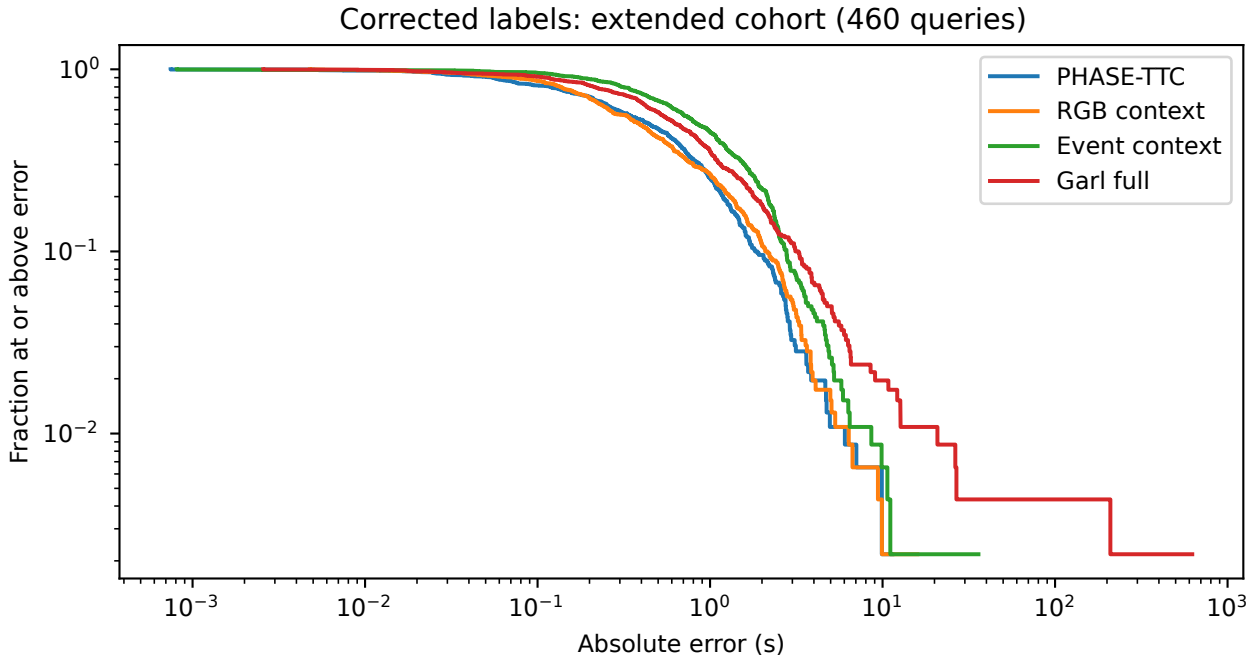


Figure 23. Corrected absolute-error exceedance. Curves include all 460 eligible transfer queries with equal query weights. No signed prediction is clipped and no extreme error is removed.

L. The initial representation-learning programme

PHASE-TTC began as E-JEPA-TTC, a study of predictive event representations for time-to-contact estimation. The programme covered pooled and spatial embeddings, apparent expansion, and transport-conditioned residual dynamics. Some models improved TTC while losing the geometric behaviour that motivated them, so the experiments assessed accuracy, geometric consistency, and representation content separately.

The historical names distinguish otherwise similar version numbers. *Causal Scale v5–v8* refers to synthetic operator tests, whereas *Scientific Recovery V5–V8* refers to later real-data studies. The EvTTC and eAP A0/A1 experiments also used different data and metrics.

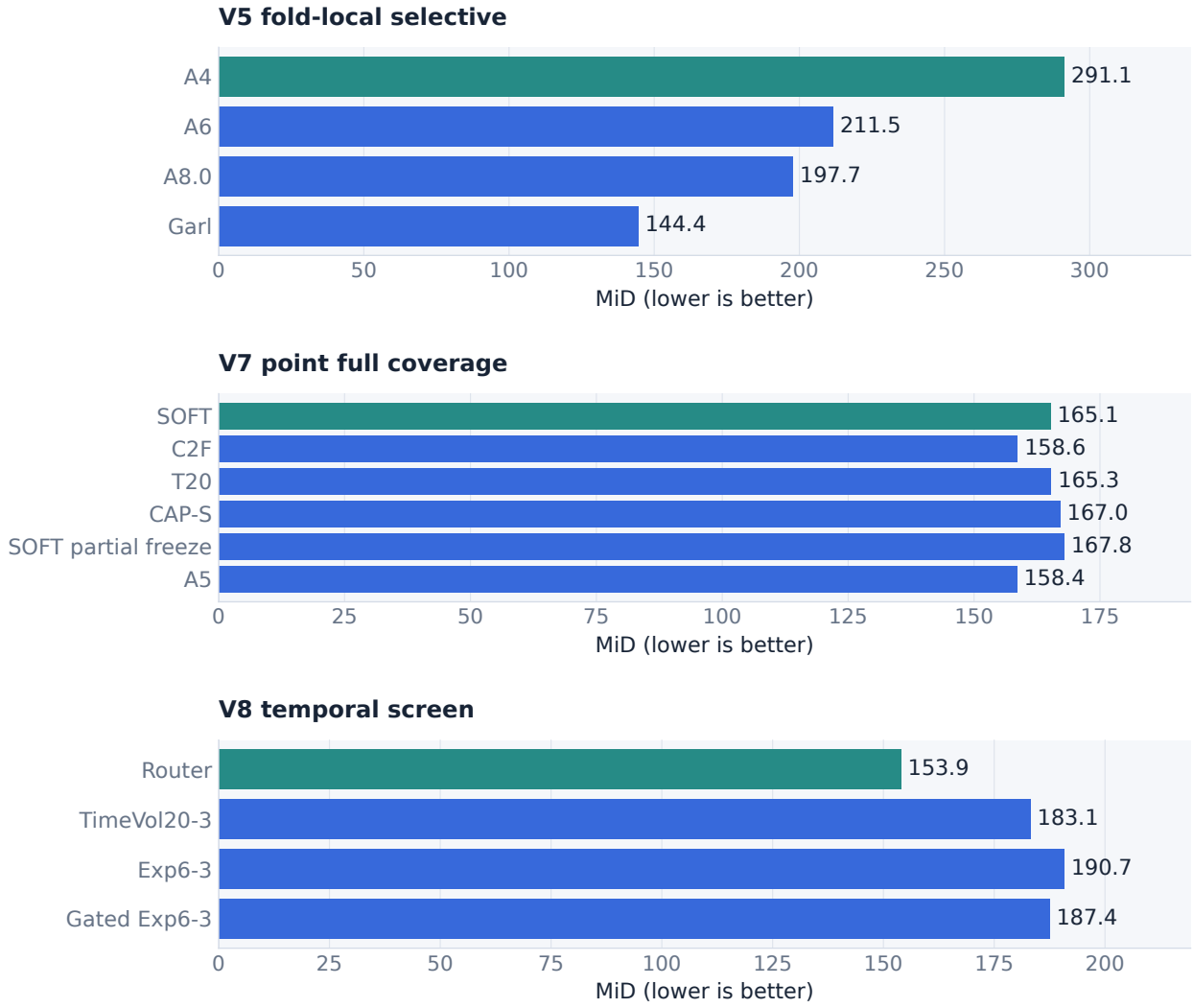


Figure 24. Early recovery results by experiment family; lower MiD is better.

L.1 Global and dense representations on EvTTC

The spatial-granularity comparison tests global pooling (A0) against a dense spatial readout (A1) under matched EvTTC conditions. Five sequence-grouped folds and seeds 7, 13, and 21 provide fifteen paired runs per model (Table 20).

Architecture	MAE (s)	Relative error (%)	Selection score
Global A0	1.0109	30.25	0.58452
Dense A1	1.0067	30.55	0.59312

Table 20. Global A0 and Dense A1 means over five EvTTC folds and three seeds.

The dense readout reduces mean absolute error while increasing relative error and the composite selection score. Its 2.16-fold latency and 1.58-fold training cost favor the compact globally pooled model under this protocol.

Attention residuals, temporal KDA/Object-KDA, object-conditioned crops, and FlowMimic-style alignment all failed their selection criteria. Because those screens used different data or criteria, they do not extend the matched A0/A1 ranking. [E01]

L.2 Representation variance and dynamic information

The temporal-residual objective captured the desired dynamics under a slow synthetic nuisance process. Its mean dynamic R^2 was about 0.72, versus 0.15 for variance regularization. The latter maintained high embedding variance while preserving nuisance information, showing that variance alone was a poor proxy for useful temporal content.

Changing the nuisance on every frame reversed the result: the residual objective's dynamic R^2 became negative, while the variance control reached about 0.74. Later probes kept both level and change features and varied the nuisance process. Neither effective rank nor per-dimension standard deviation demonstrated TTC-relevant content on its own. [E02, E12]

L.3 Object identity and crop geometry

Object-level eAP/GarlTTC requires an explicit query because several annotated objects can share one event frame while having different TTC values. Object-LHR resolved the ambiguity by cropping the selected object at both endpoints before estimating temporal scale.

Independent resizing erased much of the apparent scale change required by the geometric head. Object-Expansion v2 stabilized the output by predicting sign and log inverse-TTC magnitude; Object-Signed-Expansion v3 combined them in one expansion coordinate and added ordered temporal features plus box motion. Perturbation tests showed strong dependence on box motion and little response to events, so this accuracy could not be credited to event representation learning.

Object Event v4 placed three causal endpoints in one common crop, preserving apparent growth and separating the neural event path from the motion ablation. The object query and its coordinate system became part of the model specification. Earlier variants used different conditions and remain separate experiments.

L.4 Physical tests of the scale operator

The frozen v4.31 matcher passed representation-stability checks but failed the zoom, reversal, and translation requirements. Incomplete source provenance and missing validation records also prevented its selection. Its foreground, height, pseudoflow, and correspondence diagnostics motivated a new scale estimator.

For two apparent heights h_{t-1} and h_t separated by Δt, the Causal Scale operator defines

$$r_t = \log h_t - \log h_{t-1}, \qquad \widehat{\tau_t^{-1}} = \frac{\exp(r_t) - 1}{\Delta t}. \tag{42}$$

Under idealised constant-speed approach, this gives inverse TTC at the newer endpoint. Changes in shape, viewpoint, occlusion, or relative motion can violate the geometric assumptions. For $\hat{\tau}_t = \Delta t/(\exp(r_t) - 1)$, the sensitivity is

$$\frac{\partial \hat{\tau}_t}{\partial r_t} = -\frac{\Delta t \exp(r_t)}{(\exp(r_t) - 1)^2}. \tag{43}$$

The derivative grows without bound as r_t approaches zero. Output clipping can contain the resulting TTC value, but it cannot improve the scale measurement.

The learned foreground failed to reproduce the ideal operator on held-out synthetic data. Causal Scale v5 reached correlation 0.92 with translation leakage 0.027, missing both requirements. V7 reduced leakage to 0.0034 and symmetric TTC error to 0.25, but correlation stayed at 0.92 and the validation result did not transfer to the new synthetic group.

Causal Scale v8 selected on both mean and worst-group performance across independent synthetic groups. Fixed symmetric smoothing supported sparse endpoints, and a final CVaR loss emphasized large ratio errors. Correlation remained below the required threshold in every group. A subsequent real-data screen evaluated the operator under exploratory conditions. [E13, E14, E15]

L.5 Foreground supervision in the first eAP screens

The first eAP screens used a common object crop, three event endpoints, and oracle boxes; they did not evaluate detection. A0 learned a weak rectangular foreground target. A1 replaced it with height, width, and centre supervision. A1-FR added a full-resolution decoder, A1-DF used deeper features, and A1-DF-R supervised the pair ratio directly. A3 kept the A1 architecture and trained with RGB-derived SAM masks. All screens reused public validation data.

Arm	MiD ↓	Failure (%)
A1 geometry	346.829	9.961
A3 SAM distilled	353.635	10.889
Garl trained matched	203.634	0.000

Table 21. Selected early eAP public-validation results for A1, A3, and matched Garl. A0 and the A1 architecture variants described in the text are not tabulated here.

A3 did not improve on A1, and both scored worse in aggregate MiD than matched Garl. The diagnostics showed that estimating absolute height did not guarantee recovery of its temporal derivative. Garl achieved lower aggregate error but predicted positive TTC for receding cases, leaving signed motion unresolved.

A4 distilled local spatial relations from a frozen DINOv3 RGB teacher into an event student; A4D also learned changes in those relations across endpoints. Both used RGB during training and events at inference. [E25, E26]

The A4 scaling follow-up (A4-S1) retained the 355,118-parameter student and the same 2,048 public-validation samples. Training increased from 2,048 to 8,192 samples, while the DINO distillation weight increased from $\lambda = 4$ to 8, selected by three-fold sequence cross-validation within training. With seed 7, eighteen training epochs, and checkpoint selection at epoch 15, sequence-macro MiD fell from 322.7 to 262.8, failures from 11.08% to 8.59%, and log-scale-ratio correlation rose from 0.260 to 0.448. Matched Garl remained lower at 203.6 MiD with no failures. The 18.5% MiD reduction combines more training data with a changed loss weight; the preregistered 8,192-sample $\lambda = 4$ control remained unexecuted at that closure. Attribution to either change alone and replication across seeds therefore remained unresolved. A5 and A6 subsequently varied TTC specialization and transport while monitoring the inherited geometry. [H_A4_S1]

L.6 Fold-specific producer training

The first grouped-development run inherited an A4 parent trained on all nine development sequences, including each held-out outer group. That comparison was excluded from selection, and the relationally distilled producer (A4) was retrained inside every outer fold. The corrected study initialized A6 and A8.0 from the matching fold-local parent and trained Garl from scratch.

The comparison matched samples, targets, training budget, metric, and oracle-ROI access. The E-JEPA-family branch used three endpoints and twelve channels; Garl used two endpoints and forty channels. Checkpoint selection still used the outer-development split, and TTC buckets shaped the initial 8,192-token sample. Thus, the correction removed parent exposure without turning the study into a blind test or a target-free sample. [E03]

Model	Parameters	MiD ↓	Failure (%)
A4 parent	355,118	291.088	9.448
A6 transport	498,130	211.509	7.849
A8.0 dual transport	627,827	197.691	7.019
V6.1 radius 2	627,827	194.122	6.689
A5 causal	424,274	155.472	4.761
Garl local matched	24,674,178	144.353	0.000

Table 22. Scientific Recovery V5–V6 results on 8,192 tokens from nine sequences, three folds, and seed 7.

A8.0, which freed transport while retaining the frozen parent geometry, improved MiD over A6 by 13.8 points; the paired 95% interval was $[-17.7, -10.1]$. It still exceeded the preregistered MiD threshold of 175 and remained behind Garl, so V5 produced no candidate.

Sequence-scoped track identifiers and preserved token/target IDs made the fold-local correction auditable through the full training lineage. [E03, E05]

L.7 Transport capacity and geometric retention

Increasing the transport radius from one to two did not yield a clear V6 gain: V6.1 reduced mean MiD relative to A8.0, but the paired interval $[-8.2, 1.0]$ crossed zero. A5 achieved lower TTC error than the geometry-frozen variants while weakening their geometric dynamics, which kept accuracy and mechanism retention as separate criteria.

V7 tested soft geometric distillation, coarse-to-fine transport, extra temporal bins, greater capacity, and partial freezing. Under full-coverage point evaluation, A5 scored 158.4 MiD and served as the reference. Promotion required both lower TTC error and retained geometric behaviour. [E04, E06, E07, E08, E09, E10, E11]

Arm	Point MiD	Delta vs A5	Paired 95% interval
A5 reference	158.449	–	–
SOFT	165.116	+6.668	[+3.084, +10.547]
C2F	158.573	+0.125	[-3.025, +3.432]
T20	165.260	+6.812	[+2.290, +11.514]
CAP-S	167.025	+8.576	[+3.735, +13.630]
SOFT partial freeze	167.826	+9.378	[+5.359, +13.272]

Table 23. V7 full-coverage MiD relative to A5, using seed 7 and three outer folds.

Every V7 arm failed at least one mechanism, geometry, or confirmation requirement, and none advanced to confirmation seeds. More temporal bins, capacity, and partial freezing did not close the comparator gap. The JEPA objective was inactive in these runs, so their results provide no evidence about JEPA pretraining.

A post-hoc A5/C2F analysis found complementary errors. Separating a deployable feature-based router from a target-aware oracle motivated the subsequent prospective nested-routing comparison. [E07, E08, E09, E10, E11]

L.8 Nested routing and predictive-learning attribution

Only the nested A5/C2F router passed V8's candidate screen; all three temporal-representation interventions failed it. [E16, E17, E18, E19]

The seed-7 router improved on A5 by 4.6 MiD points. Its scores were 153.9, 153.3, and 156.0 for seeds 7, 13, and 23. All three runs reused the same development sequences and estimate optimization stability, not independent external replication. Matched Garl seed replications were unavailable.

Arm	MiD	Finite (%)	Stored gate
router	153.877	100.00	True
timevol20_3	183.092	95.25	False
exp6_3	190.711	94.10	False
gated_exp6_3	187.400	94.80	False

Table 24. V8 seed-7 MiD and finite coverage for the router and temporal-representation arms.

The stored V8 summaries support point estimates, finite coverage, and selection outcomes. They do not specify enough aggregation detail to reconstruct valid uncertainty intervals, so none are reported.

A post-training path replay located the trained model's dependence. The analytic height path had high error, and adding a residual without transport changed little. Transport plus the residual produced the main improvement; history added a smaller gain. Because the replay disabled paths without retraining, it measures use inside this trained graph rather than the effect of retraining each ablation. [E16, E20, E21]

Replay intervention	MiD ↓
analytic only	373.176
analytic residual	373.596
analytic residual transport	163.231
analytic residual transport history	155.824
analytic transport	373.176

Table 25. V8 post-training path replay showing MiD for the analytic, residual, transport, and history paths.

The H2 diagnosis was shortcut/residual dominance: the analytic height path did not explain the trained model's advantage. This motivated a measurement interface with less dependence on explicit height estimation. The diagnosis did not rule out useful geometry or transport features.

V8 tested JEPA attribution through D0–D4. The meta-router had no single transferable encoder, so A5 supplied the downstream reference. D2 froze the JEPA encoder, D3 fine-tuned part of it, and scratch, random-frozen, and shuffled-future arms controlled for downstream capacity and temporal correspondence. [E22, E24]

Arm	1%	5%	10%	25%	100%
Scratch (D0)	265.20	286.35	277.40	270.97	276.35
Random frozen (D1)	379.82	369.40	361.68	370.87	365.44
JEPA frozen (D2)	380.09	380.01	380.04	379.97	380.12
JEPA partial fine-tuning (D3)	379.51	379.40	379.41	379.26	379.62
Shuffled future (D4)	379.02	379.01	378.89	378.89	379.12

Table 26. V8 seed-7 controls on the same three-fold OOF queries, recomputed from the recovered signed prediction artifacts with one sequence-macro evaluator. All five arms and nominal label fractions appear together. Historical aggregate values for D2/D3 are retained in the source data; their small discrepancy from this recomputation is documented below.

D2 and D3 produced finite predictions with no recorded failures, but neither passed the causal-benefit criterion against scratch, random-frozen, and shuffled-future controls. V8 provided no positive JEPA attribution, and the next experiments focused on the measurement interface and feature usefulness. [E23]

The recovered aggregate matches the archived E23 file hash, but rescoring its associated predictions differs by up to 0.177 MiD for D2/D3. Table 26 therefore uses the recomputed predictions for all arms rather than mixing cached aggregates with newly calculated controls. Scratch remains substantially better than either JEPA arm at every fraction. The origin of the small cached-aggregate discrepancy is unresolved; it does not alter that ordering. Only seed 7 completed this attribution study.

M. Temporal information and phase refinement

The temporal studies (V9–V12 in the historical records) ask whether causal motion information improves phase estimates beyond the existing expert outputs. The controls removed transport fields, disrupted correspondence or order, reduced the expert set, and replaced recurrence with unordered aggregation. Prespecified comparisons used sequence-level uncertainty and kept separately trained producer families distinct.

M.1 E-Clock: incremental information in transport features

E-Clock represented signed TTC in benchmark phase. With $\Delta = 0.1$ seconds,

$$\phi(T) = -\log(1 - \Delta/T), \qquad d(\hat{T}, T) = 10^4 |\phi(\hat{T}) - \phi(T)|. \tag{44}$$

The phase exists only on the admissible signed-TTC domain, and d measures phase error rather than seconds. Historical A5 replay, raw inverse-TTC evaluation, and bounded deployment output follow different numerical conventions and are reported under their original definitions.

Transport helped X0's direct-phase model, but did not close the gap to A5. With architecture, capacity, matcher, and input structure fixed, DYN-U scored 225.0 MiD versus 270.6 for the zeroed BASE-U control. Canonical A5 and PAIR-U scored 158.4 and 158.1. X0.5 next conditioned the transport fields on A5.

X0.5 found no confirmed incremental value from nine frozen transport fields. DYN9 scored 155.2 MiD, against 158.7 for calibrated A5 and 175.0 for a target-blind within-sequence shuffle. The difference from calibration was −3.5 with hierarchical 95% interval $[-11.4, 3.5]$; the interval against raw A5 also crossed zero. Leave-one-sequence-out tuning kept normalization and ridge fitting outside the outer meta-test. DYN9 did not meet the incremental-utility criteria, so the conditional X1 adapter was not trained. [M_X0, M_X05, M_X05_GATE]

M.2 Pair routing and local motion fields

R2 gave Stage61's lowest point estimate but did not meet the incremental-utility criterion. Across 8,192 queries from nine sequences, MiD was 150.9 for R2, 152.9 for R1, 153.9 for RouterR, and 185.4 for shuffled R2. The paired hierarchical intervals comparing R2 with R1 and RouterR crossed zero.

Stage62 could not isolate a local-field benefit. LOCALFIELD scored 155.5 MiD, close to GLOBALPOOL at 155.4, SHUFFLEFIELD at 155.8, and TIMESWAP at 155.6. Although all improved on this producer family's 162.2 A5 replay, LOCALFIELD did not meet the mechanism criteria against its controls.

Table 27. Pair routing (Stage 61) and local-field experiments (Stage 62).

Arm	MiD	Queries	Finite
Stage 61			
R1	152.855	8,192	100%
R2	150.876	8,192	100%
R2_SHUFFLE	185.429	8,192	100%
RouterR	153.877	8,192	100%
Stage 62			
RouterR	153.877	8,192	100%
X2_A5_REPLAY	162.205	8,192	100%
X2_GLOBALPOOL	155.351	8,192	100%
X2_LOCALFIELD	155.463	8,192	100%
X2_SHUFFLEFIELD	155.827	8,192	100%
X2_TIMESWAP	155.585	8,192	100%

Sequence-macro MiD, evaluated on nine sequences and three folds per stage. Lower is better. [M_STAGE61, M_STAGE62]

The canonical V7 A5 and the A5 retrained inside the nested router use different checkpoints and training exclusions. Their scores remain separate despite overlap in architecture and queries. [M_STAGE61,M_STAGE62]

M.3 Temporal features, regret estimation, and geometry support

Stage63 repaired the experimental base by standardizing temporal bins and restoring one FP32 producer state. Equivalent fixed-grid pooling also made CUDA backward deterministic on the prescribed shapes.

The raw-time study compares a temporal event-rate residual (RAW-L1) with total-count redistribution across bins (COUNT-L1), a bin-permutation control, and a state-only residual with event-rate inputs zeroed (STATE-L1). Their MiD scores are 162.2, 157.2, 164.3, and 150.7, respectively. The better count and state controls leave the proposed temporal-rate benefit unsupported. A later regret-based expert router (RISK17) reaches 146.0 MiD, but omitted validation controls and their execution order prevent acceptance of that result.

Convex expert routing (SIMPLEX17) improves the point estimate by 2.0 MiD over the frozen regret router, reaching 144.0. Hierarchical and sequence-only 95% difference intervals, $[-5.5, 1.2]$ and $[-4.7, 0.3]$, both include zero. The reduced-gating, unconstrained, and PAIR-permuted alternatives score 143.9, 144.5, and 149.2. Across these four arms, twelve fits and 18,000 CPU float64 updates over three folds preserve the registered comparison; the small routing gain remains statistically unresolved.

Stage68 stopped before fitting because the normal-flow estimator supplied too little valid geometry. Only 368 of 8,192 tokens had any valid window; 583 of 16,384 windows supported the full-affine fit, and none supported the reduced-isotropic fit. Fold support rates of 4.9%, 5.6%, and 3.0% all missed the fixed 20% threshold. Insufficient support excluded the subsequent ridge and neural fits. The support limitation concerns this normal-flow estimator under the tested conditions. [M_STAGE63_REPAIR,M_STAGE64,M_STAGE65,M_STAGE69_DECISION,M_STAGE67_DIAGNOSTICS,M_STAGE67_DECISION,M_GEOMETRY]

M.4 SIMPLEX-T: residual phase refinement

Expert selection constrains the output to the available expert phases. SIMPLEX-T instead anchors a residual predictor at their label-free median,

$$\hat{\phi}_t = \Pi\big(\mathrm{median}_k\, \phi_{t,k} + 0.03\, g(h_t)\big)\,, \tag{45}$$

where h_t summarizes masked history and Π enforces the numerical output bounds. Unlike expert selection, this residual can leave the experts' convex hull. A separate cost head estimates relative losses of the original experts; it does not estimate the corrected prediction's risk. Both the recurrent and bounded-window Transformer versions mask missing history and start with zero-initialized output heads.

The primary recurrent model uses two 160-unit GRU cells. It maps expert and timing features to phase, auxiliary phase quantiles, and relative expert costs. The quantiles are not calibrated collision probabilities. History is retrospective within the current query ROI; the experiments did not verify persistent object tracking.

The original residual-context campaign (T6) completed 72 fits and 180,000 saved optimizer updates on OLD_DEV: 8,192 queries from nine sequences in three folds. H1 uses the current observation; H8 adds up to seven past slots. Each head shares the frozen expert producers. Across head seeds 7, 13, and 23, H8 reached mean MiD 121.8 versus 133.1 for H1. The mean paired-loss difference was −11.3, with hierarchical 95% interval $[-16.2, -6.6]$ conditioned on these heads and reused development sequences. This estimates a context benefit within that lineage, rather than independent external replication. [H_T6]

The trained controls refine that interpretation. Repeating the current observation through eight slots failed to match past context, while expert selection remained worse than residual correction. Training with reversed past order matched chronological H8 within the recorded uncertainty; this model could learn the reversed convention, so the result does not establish dependence on chronological order in a fixed trained model. Adding hidden producer coordinates degraded both H1 and H8, whereas removing those coordinates recovered the scalar-feature result. This negative finding concerns the tested

Original T6 head, seed 7	MiD ↓	Training condition
Current observation (H1)	132.832	One observation, 160-unit recurrent cells
Past context (H8)	121.646	Eight chronological slots, same cell width
Reversed past	121.225	Trained with reversed past order
Repeated current	131.211	Current observation repeated across eight slots
Expert selector	143.504	Selection in place of residual correction
Latent, current only	171.057	Added hidden producer coordinates, H1
Latent, past context	184.072	Added hidden producer coordinates, H8
Zeroed latent, H8	120.910	Same latent-input architecture with coordinates zeroed

Table 28. Trained residual-context controls from the original 72-fit T6 campaign on OLD_DEV (8,192 queries, nine sequences, three folds). These rows report separately trained controls, not post-training interventions on one fixed model. Historical identifiers are H1/H8-C160, PAST_REVERSED, REPEAT_CURRENT, SELECTOR, LATENT, and LATENT_ZERO. [H_T6]

Table 29. H16 and H8 temporal heads on OLD_DEV.

Head seed(s)	H8 MiD	H16 MiD	Δ MiD	Hierarchical 95% CI
7 (observed)	121.646	118.866	-2.781	[-5.879, +0.301]
13	121.841	119.697	-2.144	[-5.317, +0.712]
23	121.865	119.162	-2.703	[-5.514, -0.071]
New seeds 13/23	121.853	119.429	-2.424	[-5.244, +0.041]
All three seeds	121.784	119.241	-2.543	[-5.213, -0.111]

Nine sequences, 8,192 queries, and three folds. Delta is H16 minus H8; multi-seed rows average paired losses. Seed 7 had been evaluated previously. All heads share fixed expert producers. [M_SIMPLEX]

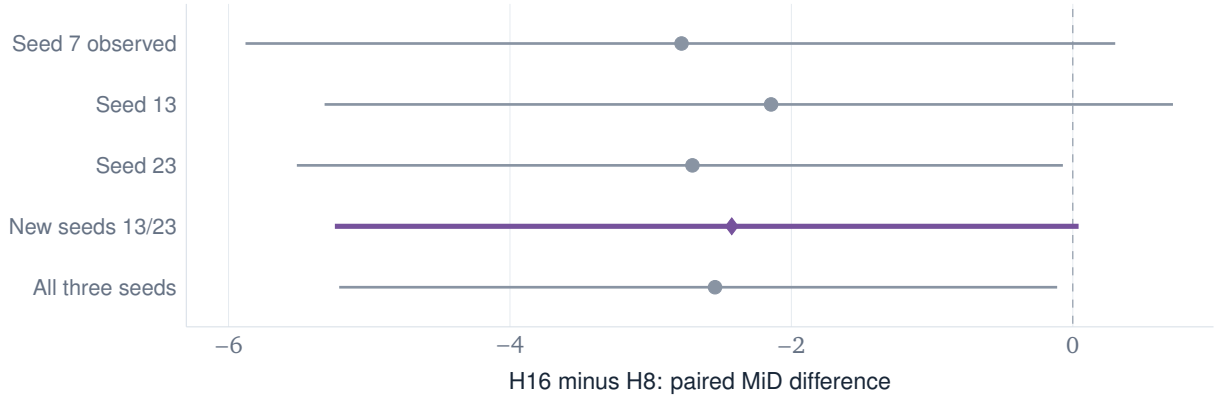


Figure 25. Paired H16-minus-H8 MiD differences for three head seeds on the historical development set (OLD_DEV).

latent interface and does not identify misalignment as its sole cause or exclude other useful latent representations. [H_T6]

A subsequent extension completed 24 fits and 60,000 saved updates. It tested H16, reduced expert sets, and set aggregation while reusing the original H8 endpoints and retaining H8 as the registered candidate. These are separate campaign totals; reused endpoints are not new fits. Seeds 7, 13, and 23 retrained only the heads over shared expert producers.

H16 reduced MiD in every head seed. Across all three, mean paired loss was 119.2 for H16 and 121.8 for H8, a difference of −2.5 with hierarchical 95% interval $[-5.2, -0.1]$. Seed 7 had already informed exploration. Seeds 13 and 23 alone gave −2.4 with interval $[-5.2, 0.04]$, which crosses zero. The analysis averages losses across shared-producer heads, rather than ensemble predictions or full-system replications. [M_TEMPORAL_MODEL, M_PHASE_CONTRACT, M_SIMPLEX]

The full expert/context model outperformed every registered reduction. FULL-C0 scored 121.4 MiD, versus 127.8 for A5-only, 126.7 for C2F-only, and 124.7 for A5 plus PAIR. Unordered aggregation with age reached 122.4; removing explicit time gave 123.7. Each reduction failed the accuracy screen or its small-excess-error bound. The experts themselves can encode temporal information, so this comparison does not isolate recurrence as the source of the gain.

The analytic EWMA/transport control lost to H8 on every sequence, scoring 155.4 versus 121.6 MiD; the hierarchical interval for the 33.8-point difference was $[17.1, 51.0]$. WIDE averaged 119.0 across three head seeds, but its intervals against both H8 and H16 crossed zero. It was not promoted. [M_SIMPLEX, M_EWMA, M_WIDE]

M.5 Capacity and teacher-feature coverage

Stage70–76 and the V11 capacity extension planned direct, pairwise-predictive, trajectory, and temporal-permutation comparisons at approximately 2M, 4M, and 8M parameters. The 4M trajectory model was primary. The proposed 30,000-query set covered 27 sequences; extra timestamps increased density, not the number of independent acquisitions.

The V11 audit found teacher features for 4,775 historical tokens and none for the other 25,225. Matching token IDs also left ROI construction, RGB preprocessing, and teacher provenance unresolved.

Stage70 ended with both completed and unfinished endpoints. The partial record supplies no capacity ranking and no evidence that unlabeled JEPA pretraining improved SIMPLEX-T. [M_CAPACITY_PROTOCOL, M_CAPACITY_AUDIT, M_STAGE70_LEDGER]

N. Transfer, output stability, and runtime

N.1 Development, training, and transfer populations

The historical development set (OLD_DEV) informed model selection, while the forty-sequence fitting set (TRAIN40) includes training examples. Transfer experiments evaluate frozen models under separate dataset-specific input construction. EvTTC and Forward Collision Warning Data (FCWD) instead evaluate frozen models with dataset-specific adapters. FCWD contains three driving sequences with paired events, frames, and LiDAR-derived TTC [6]. The expanded EvTTC set adds timestamps to the same 32 sequences rather than a new holdout.

The earlier H8/TRAIN40 route predates the final refit in Section 2. A5 and C2F each received 49,932 updates, the pair head 6,840, and each of three H8 heads 2,500. Its 88,744 training predictions diagnose fit but cannot estimate unseen-scene performance. Inference used events, while training also used an RGB teacher and geometric supervision.

The initial EvTTC evaluation had 946 valid targets among 1,024 queries from 32 sequences; the expansion had 3,640 among 3,947. FCWD contributed 597 eligible targets among 630 queries from three previously exposed sequences. Rows without ground truth remain unscored; all reported methods return finite predictions on the eligible rows. Full RGB/event Garl was omitted from FCWD because the RGB-to-event mapping had not been verified. [M_SCIENCE_AUDIT, M_V12_EVTTC_946, M_V12_EVTTC_3640, M_V12_FCWD_597]

N.2 Input and output conditions

H8 used retrospective event history and supplied oracle ROI metadata. Garl event-only used its native pair of event windows, while Garl full added RGB. The systems differed in context and input construction. EvTTC projected RGB to events with calibrated rotation but no depth-dependent parallax correction. These transfer adapters do not reproduce the published benchmark protocol.

RGB synchronization was corrected before full Garl evaluation. Selected images had to lie within 1 ms of each event endpoint and about 100 ms apart; the largest admitted offset was 453 μs. The runner strictly

Table 30. Frozen H8 and Garl models evaluated on EvTTC and FCWD.

Method	MAE (s)	Median AE (s)	RMSE (s)	RTE (%)
EvTTC: 946 eligible / 1,024 queries; 32 sequences				
H8 / head 7	1.321	0.766	3.027	62.35
H8 / head 13	1.353	0.731	3.206	62.92
H8 / head 23	1.409	0.738	3.392	68.87
Garl event	8.048	0.997	159.247	293.47
Garl RGB + event	1.868	0.692	7.875	76.77
EvTTC expanded: 3,640 / 3,947; same 32 sequences				
H8 / head 7	1.361	0.722	3.475	53.70
H8 / head 13	1.395	0.699	3.618	54.52
H8 / head 23	1.456	0.702	4.003	59.15
Garl event	4.560	0.958	83.896	161.12
Garl RGB + event	94.510	0.661	5561.622	5180.18
FCWD: 597 / 630; three sequences				
H8 / head 7	1.778	0.675	3.588	30.20
H8 / head 13	1.597	0.580	3.258	27.41
H8 / head 23	1.569	0.598	3.109	27.48
Garl event	2.146	0.602	5.072	26.89

Scores use queries with valid TTC targets; prediction coverage is 100% on these queries. H8 bounds TTC at ±60 s, while Garl retains its native signed outputs. The FCWD comparison uses event-only inputs. [M_V12_EVTTC_946,M_V12_EVTTC_3640,M_V12_FCWD_597]

loaded the released checkpoint and upstream normalization. The RGB extension followed inspection of event-only results, making this a development comparison with frozen weights.

H8 bounds TTC at ±60 seconds. Garl keeps its native scale-ratio division, which emits very large finite values when the estimated scales nearly match. Full-population metrics retain those native predictions. [M_TRANSFER_REPORT]

N.3 Transfer results and metric-dependent rankings

H8 had the lowest mean error on the first 946 EvTTC targets: its three heads reached 1.32–1.41 seconds MAE, versus 8.05 for Garl event-only and 1.87 for Garl full. Garl full had the best median, 0.69 seconds versus 0.73–0.77 for H8.

The 3,640-target expansion showed the same split. H8 achieved 1.36–1.46 seconds MAE, Garl event-only 4.56, and Garl full 94.51. Yet Garl full retained the best median at 0.66 seconds, ahead of H8's 0.70–0.72. Rare near-singular scale ratios drove Garl full's mean, so native-output and matched-clipping results remain separate.

On FCWD, H8 led in MAE at 1.57–1.78 seconds versus 2.15 for Garl event-only. Garl led in mean relative TTC error at 26.9%, compared with 27.4–30.2% for H8. These rankings weight the TTC distribution differently, and only three sequences were available for clustered uncertainty. [M_V12_EVTTC_946,M_V12_EVTTC_3640,M_V12_FCWD_597]

N.4 Head and full-route runtime

Prepared-input CPU profiling measured three rotated-order blocks of 500 batch-one calls per head. Median blockwise p95 was 9.28 ms for H1, 19.47 ms for H8, and 31.45 ms for H16. The timer covered canonical TTC emission, but not event reading, ROI construction, expert inference, detection, or tracking.

The shared-GPU route was much slower than the isolated heads. Across 1,728 supplied-ROI requests from 64 TRAIN queries, H8 p95 was 9.35 and 8.48 seconds in two warm blocks; H16 reached 13.20 and 15.30 seconds. The application reopened HDF5 inputs between cold measurements, while the operating system could retain cached pages.

Table 31. Head-only and full-pipeline runtime.

Model	MiD	Head p95 (ms)	Route 1 p95 (s)	Route 2 p95 (s)
H1_SEED7	132.832	9.28	2.944	2.902
H8_SEED7	121.646	19.47	9.349	8.479
H16_SEED7	118.866	31.45	13.201	15.304
FULL_C0	121.405	21.07	8.477	9.337
A5_ONLY_C0	127.785	21.42	7.928	7.471
C2F_ONLY_C0	126.705	19.28	8.185	8.256
A5_PAIR_C0	124.669	19.83	7.086	7.502
SET_AGE_C0	122.443	10.00	7.628	8.303
SET_NOTIME_C0	123.687	9.91	7.727	7.967

Head: median of three blockwise p95 values, CPU FP32, prepared inputs, 500 measurements per block. Route: p95 on 64 TRAIN queries with supplied ROI, shared GPU, and resident producers. Accuracy uses OLD_DEV seed 7, nine sequences and three folds. [M_HEAD_COST, M_ROUTE_COST]

ROI/voxel reading and preparation consumed 85.5% of H8's measured time and 90.4% of H16's in the second warm block. These preparation costs motivated bounded caching and transfer scheduling in the efficient-context study.

CUDA graphs reduced recorded A5 launches from 5,877 to 1,270 while preserving the tested outputs, losses, RNG state, and gradients. A later window measured 41.78 versus 32.89 updates/min, but host load and data wait also changed, so the run did not isolate graph speedup. Adaptive caching reached 26.31 updates/min against a 36.75 reference and failed its throughput test. [M_HEAD_COST, M_ROUTE_COST, M_ROUTE_REPORT, M_ENGINEERING]

N.5 Historical selection of phase-only fusion

The multimodal selection campaign (V16) completed 32 fits of 2,500 updates each, totaling 80,000 updates. Initial corrections consumed frozen RGB and event context states already supervised on the same head-training groups, denoted H. Fitting another supervised head on those states constituted in-sample stacking. Internal validation V comprised 16,432 queries in eight groups and had prior development exposure. The successive recipes were recorded before their new predictions, but the choice of subsequent experiments remained adaptive. [H_V16]

The search tested phase-unit normalization, relative-phase scaling, additional fusion states, a penalty for worsening RGB, and a fusion-preserving anchor. None passed the complete original criteria. An equal-weight phase ensemble required a relaxed 4% long-band safeguard and therefore also failed the original 2% criterion. These unsuccessful alternatives remain part of the selection history. [H_V16]

The retained phase-only head (PHASE_ONLY) removed 320 hidden-state coordinates and used twenty explicit phase, clock, and support coordinates to correct the RGB context estimate. Its three head seeds achieved internal group-and-bucket MiD 89.22, 89.20, and 89.22, with MAE 1.870, 1.859, and 1.862 seconds. Each passed the original acceptance criteria, including the 2% per-band safeguards and the paired event-minus-control requirement. Their 95% difference intervals were $[-5.74, -1.30]$, $[-5.73, -1.35]$, and $[-5.69, -1.40]$. These outcomes did not require the later 4% sensitivity threshold. Seed 7, the first registered seed, supplied the frozen candidate; all seeds shared the frozen upstream models. [H_V16]

A separate four-fold experiment on H (GROUP_CROSSFIT) fitted eight context models and two readouts, accounting for ten of the 32 fits and 25,000 updates. Its held-group features and training-group normalizers addressed the stacking limitation. Despite 90.41 MiD and a favorable event-minus-control interval, its 1.893-second MAE failed the RGB non-degradation criterion, yielding no candidate. This rejected experiment differs from the later five-rotation TRAIN40 refit in Section 2; that deployed lineage uses its own out-of-fold construction. The final protocol cannot be attributed retrospectively to historical V16 selection. [H_V16]

The controls also differ across campaigns. V16 ZERO removes five event predictive coordinates while preserving event clocks and support counts. V18 ZERO removes the entire event feature block, including its temporal coordinates, on RGB-supported rows; its event-only fallback remains active when RGB is unavailable. Neither intervention alone establishes the effect of removing events from the final deployed graph. [H_V16_ZERO, H_V18_ZERO]

O. From event representations to multimodal measurement

The later experiments assembled spatial measurement, temporal transport, signed-TTC readout, and multimodal correction into one causal estimator. PHASE-TTC combines task adaptation and teacher supervision with event/RGB producers, context models, and a compact fusion head. No matched factorial separated all of these components, so the final result does not establish a JEPA contribution.

O.1 V13: a parallel RGB route and a common temporal interface

V13 added an RGB route with the event branch's A5, C2F, PAIR, and temporal-context decomposition. The common interface made it possible to test event information after an image-based TTC estimate. Each modality still used its own checkpoints, normalization, training data, and temporal support.

Three disjoint sequence-group roles separated fitting stages. P supplied 53,890 queries from 24 sequences for producers and PAIR. H supplied 18,422 queries from eight sequences for downstream normalizers and heads, while V supplied 16,432 queries from eight sequences for internal evaluation. Earlier work had already used V, so its results remain development evidence. Distinct sequence identifiers do not establish independent physical acquisition sessions. [P02]

PHASE17 connected both modalities to their context models through named phase, timing, and support fields. Their meanings remain stable across folds, whereas independently trained encoders can rotate their latent coordinates. The explicit interface enabled the later grouped out-of-fold context and fusion training without a separate latent-alignment model.

O.2 V14: changing the emphasis of the objective

V14 could not improve short-TTC error without harming other ranges. Its six-cell factorial combined three sample-mass schemes with two losses, holding the frozen event producers, PAIR parent, and seed-7 draws fixed. Each 160-unit, two-layer GRU head received 2,500 updates over eight masked history slots and seventeen features; timestamps represented gaps without dense imputation.

M0 retained the native mass vector. M1 preserved mass per sequence group while allocating TTC bands in the ratio 0.5/0.3/0.1/0.1. M2 also balanced crucial and long bands, bounding each row between 0.25 and four times its M1 weight. Uniform sampling with replacement kept the number of episodes unchanged. L0 retained the phase, quantile, and auxiliary objectives; L1 added

$$\mathcal{L}_1 = \mathcal{L}_0 + 0.25\,\mathbb{E}_{i\sim\text{batch}}\left[Nm_i\,\text{SmoothL1}\left(\frac{\widehat{T}_i - T_i}{\max(|T_i|, 0.5\,\text{s})}, 0\right)\right], \tag{46}$$

where m_i is the registered sample mass. Both losses kept the native TTC mapping, clipping, and flat-gradient regions; neither used a straight-through estimator.

No loss-reweighting candidate met all selection criteria. M1–L0 reduced crucial-band MiD from 156.8 to 134.5, while large-band error rose from 52.2 to 76.4 and negative-band error from 61.4 to 84.2. M0–L1 achieved the best aggregate and MAE in the matrix, but missed the required crucial-band improvement and other safeguards. All outputs were finite; the failures came from measured regressions.

Only 76 V queries fell in the urgent interval $(0.1, 1]$ seconds. Across the factorial, 81.6–100% received late warnings; negative TTC predictions did not count as positive warnings. Such miss rates preclude a safety claim regardless of false-alert rate.

Table 32. V14 factorial study on exposed V: apparent crucial-range gains can coexist with non-crucial regressions. Source [L02].

Model	MiD	Crucial	Negative	MAE (s)
RT-M0-L0-S7	106.21	156.83	61.38	2.51
RT-M0-L1-S7	101.94	149.59	61.01	2.03
RT-M1-L0-S7	102.23	134.47	84.21	2.98
RT-M1-L1-S7	104.04	136.19	80.04	2.34
RT-M2-L0-S7	104.32	134.87	87.62	3.37
RT-M2-L1-S7	104.83	134.59	79.93	2.38

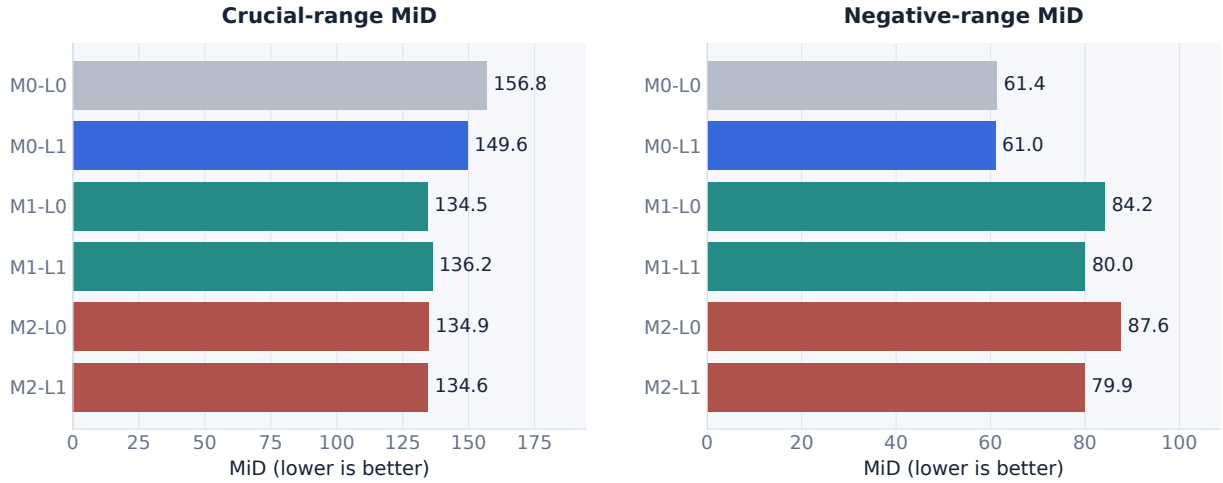


Figure 26. V14 MiD changes across crucial, small, large, and negative TTC bands.

No loss-reweighting variant met the joint selection criteria, so the conditional replication and transfer stages were not run. The factorial exposed a tradeoff between urgent-range accuracy and the other signed-TTC ranges without producing an admissible improvement. [L02]

O.3 V15 and V17: testing readout and geometry explanations

Neither readout intervention met the joint selection criteria. Sampling proportional to sample mass, corrected by w_i/p_i, preserved V14's cross-band tradeoff. Expanding the A5 residual range through an equal-slope hyperbolic tangent left crucial-band performance almost unchanged. The modified producer received regenerated PHASE17 fields and H normalizers.

V17 tested frozen geometry from several angles. A bypass exposed geometry and transport tokens to the temporal head; other arms examined learned NLL precision, grouped ridge probes, directional differences, height calibration and supervision, vertical position, gradient clipping, and nonlinear spatial-height readout. Each had a dedicated control. Directional differences remove static levels while also reducing dimensionality, so they do not isolate appearance memorization.

The geometry bypass carried a useful but unstable downstream signal. On V, its three-seed mean improved crucial MiD from 157.1 to 148.4 and aggregate MiD from 106.5 to 102.8, while MAE worsened. The effect was stronger on H than V, and the three heads shared encoders rather than repeating representation training.

The first height-regression arm emitted predictions outside the admissible MiD domain. Those invalid rows were retained, leaving the strict aggregate unavailable. Temporal supervision restored coverage but not competitive TTC accuracy. Preserving vertical positions improved training height fit while worsening H TTC, and the nonlinear readout also failed. The tested frozen-feature calibration was insufficient, though the bypass result still showed usable geometric signal.

Table 33. V18 primary and adaptive experiments on development subset H (18,422 rows). Sources [L10--L14,L20--L25].

Model	MiD	Crucial	Negative	MAE (s)
A5-P2-FROZEN	112.29	157.27	65.52	5.64
PAIR-DIRECT-S7	103.87	117.98	148.17	5.00
PAIR-CORR-S7	103.51	115.56	156.00	5.50
PAIR-CORR-FROZEN-S7	106.72	123.61	146.99	5.40
NATIVE-P2-FT-S7	104.05	145.12	61.26	4.65
PAIR-CORR-GEOMETRY-S7	90.95	114.35	106.23	6.95
PAIR-CORR-WIDE-S7	115.12	124.48	206.42	6.79
B1-TTC-CONTROL-STAGE2-S7	85.98	106.71	96.63	6.11
B1-CORRESPONDENCE-STAGE2-S7	81.44	100.22	84.12	5.48
A1-LAST-BLOCK-S7	90.64	111.03	97.96	7.11
A1-FULL-S7	78.92	95.85	87.09	5.45

Removing gradient clipping changed little. Learned precision improved crucial-band error in every head seed, but missed the required magnitude and safeguards. The recorded gradient norms describe one loss component rather than the full-dataset objective. None of V17's nine research lines produced a candidate. [L50,L51,L52]

O.4 V18: adaptation of the measurement stage

V18 moved adaptation into the measurement stage. Its primary comparison covered direct and correspondence-based pair readouts, trained and frozen encoders, and a native-system control. C1 tested geometric readout, D1 predictor width, B1 auxiliary correspondence supervision, and A1 adaptation depth. Some arms also updated correspondence, so equal TTC-update counts did not mean equal computation.

C1 improved bandwise geometry over an unconstrained pair-correction readout but failed as a complete TTC estimator. D1 found no benefit from greater predictor width. B1 beat a TTC-only control at the same number of TTC updates, yet its negative-band error blocked promotion over A5. A1 then compared full adaptation with partial retention of the correspondence-trained representation.

A1-FULL delivered V18's largest TTC gain and its clearest tradeoff. On H, aggregate MiD fell from 112.3 for A5-P2 to 78.9, and crucial MiD from 157.3 to 95.8; negative MiD rose from 65.5 to 87.1. It passed the global sign-error safeguard and beat last-block adaptation overall, but did not preserve accurate negative magnitude or demonstrate retention of the correspondence representation.

C1 also tested event information after the RGB context estimate. TRUE fusion and an equal-capacity ZERO-event control used seeds 7, 13, and 23. Each seed repeated C1 adaptation and head fitting, while all shared the historical A5 initialization and RGB context parent. The comparison isolates event input within this configuration, not variability from complete pretraining.

TRUE fusion improved on ZERO within V18. Across seeds, aggregate MiD was 87.3 ± 0.9 for TRUE and 92.5 ± 0.2 for ZERO; TRUE's crucial MiD was 126.6 ± 0.9. A paired whole-group bootstrap favored TRUE in aggregate loss. The crucial-band gain missed the preregistered 10% threshold, and the interval against the preserved V16 PHASE-ONLY family crossed zero at about $[-3.7, 0.1]$. Thus, events complemented RGB in this model without establishing superiority over V16.

No joint-measurement variant satisfied all accuracy and mechanism criteria. A later portability replay verified archived metrics and exports without adding training or performance evidence. [L10,L24,L25,L03,L31,L32,L33]

Table 34. V18 C1 equal-capacity TRUE/ZERO event controls on exposed V (16,432 rows), using adaptation/head seeds 7, 13, and 23. Source [L03].

Head / seed	Overall MiD	Crucial MiD	MAE (s)
FUSION-TRUE-S7	88.270	127.661	1.817
FUSION-ZERO-S7	92.638	136.989	1.842
FUSION-TRUE-S13	86.630	125.976	1.807
FUSION-ZERO-S13	92.455	136.500	1.837
FUSION-TRUE-S23	86.961	126.069	1.818
FUSION-ZERO-S23	92.315	136.314	1.836

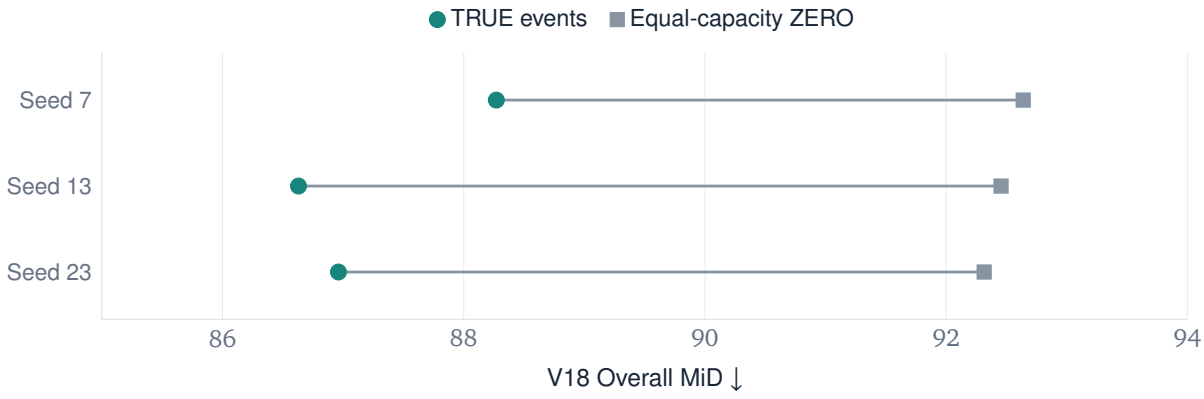


Figure 27. V18 measurement adaptations and paired TRUE/ZERO event-fusion controls.

Table 35. V19 decision recovery on exposed H. LW trades a small overall MiD increase against improved MAE and negative-range behavior. Source [L30].

Model	MiD	Crucial	Negative	MAE (s)
A5	112.29	157.27	65.52	5.64
L0	78.92	95.85	87.09	5.45
LW	79.87	100.42	70.78	4.61

O.5 V19: repairing negative motion changes the tradeoff

V19 tested loss weighting as the source of A1's negative-band regression. L0 and LW shared the correspondence-trained B1 parent, full adaptation, seed 7, uniform sampling, and 12,000 updates. L0 assigned crucial/small/large/negative masses of 50/30/10/10%. LW doubled negative mass to 20% and set the positive bands to 44.44/26.67/8.89%, preserving their relative proportions. Both arms kept the same encoder architecture and B1 initialization, while FULL adaptation updated encoder and readout weights; only the loss-band masses differed between arms. The training implementation requires a nonzero encoder-weight change after updates. [P03]

LW repaired part of the negative-motion error but worsened the primary aggregate. Relative to L0, it reduced negative MiD by 18.7%, MAE by 15.3%, and sign error by 1.55 percentage points, while aggregate and crucial MiD increased. The same pattern appeared on P training data, ruling out an H-only generalization failure. Because the replication trigger required lower aggregate error, LW did not advance.

Later two-expert selectors also did not meet the system-level selection criteria. They required both fitted experts and the measured feature path, rather than forming a jointly trained sparse mixture. Grouped validation limited fitting leakage, while reused H data and fixed experts constrained the conclusion to these selectors. [L30]

P. Representation and physical-measurement follow-ups

The new experiments extend the historical controls with narrower questions: whether future-state prediction preserves useful expansion increments, whether a scalar motion quantity is observable under the fitted model, and whether a sensor or clock residual predicts a better TTC estimate. Each experiment retains its own population and reference model. An improvement in an auxiliary metric does not substitute for a TTC comparison. [N04, N05]

P.1 Direct event-expansion solver and its negative result

An earlier physical follow-up fits causal event expansion and translation directly. Its three parameters are inverse TTC q and translation coefficients b_x, b_y, with endpoint warp $\mathbf{p}_{\text{ref}} = \mathbf{p} + (q\mathbf{p} - \mathbf{b})(t_{\text{ref}} - t)$. A free expansion focus and separate polarities avoid imposing a box-centered focus. An 81-candidate grid and three Powell starts, each limited to 240 evaluations, precede explicit validity checks. Histogram-energy objectives admit spurious contraction in stationary fixtures. The retained objective instead compares early and late warped images through zero-mean normalized cross-correlation, with a minimum correlation of 0.8. It passes twelve confirmation fixtures with previously unused seeds, but accepts incorrect rates in both new fixtures with 50% static-background events and abstains in both at 75%. Ideal contour fixtures therefore provide only a limited measurement check. The implementation is an original bounded solver, not a reproduced EvTTC CMax baseline. [A06]

The Slider-500 evaluation seals predictions on 77 queries before joining targets. It uses a causal 100 ms window, a supplied current ROI expanded by 1.2×, at most 20,000 events per query, and distortion correction. Slider-500 had already informed development and the reference adapter's fit. The event solver returns 62/77 estimates: RAE 15.69% and MAE 0.2690 s. On exactly those same 62 queries, box geometry gives 14.71% and 0.2073 s. The current adapter returns 77/77 with 13.58% and 0.2050 s; replacing it whenever the event solver returns a value, with adapter fallback otherwise, worsens full-population performance to 13.88% and 0.2561 s. The solver abstains on all first ten queries. Winning 36 of 62 paired individual errors does not overcome its larger extreme errors or its startup gap. [A06]

The float64 CUDA-graph implementation initially passes only 76/77 strict parameter-parity comparisons; both paths reject the remaining case, but their internal parameters differ beyond tolerance. A later hybrid policy recalculates GPU abstentions on CPU and passes 77/77 comparisons with fifteen CPU returns in 34.272 s. In that second pass, the GPU estimates before fallback also satisfy all comparisons, so the result does not identify a deterministic fallback repair. No contemporaneous CPU reference establishes a controlled acceleration factor. The experiment remains separate from the delivered estimator and motivates no promotion. [A06]

P.2 Future-state prediction and expansion increments

The event student learns against spatial features from a frozen, distilled V-JEPA 2.1 ViT-B teacher [8]. For each of seeds 7, 13, and 21, the pilot runs static distillation, one-step prediction, and a cross-sequence shuffled-future control for 1,000 updates with batch size 16. TTC labels do not enter these student losses. The final checkpoint is fixed before evaluation. After training, a ridge probe with regularization parameter one reads temporal differences of the frozen encoder, with the predictor removed. It fits on 512 P samples from twenty-four sequences and evaluates 256 H samples from eight different sequences. Box-based localization remains privileged information. [N04]

The primary target is annotated log-height change. Its error is dimensionless: it is neither a time-normalized rate nor TTC error in seconds. The criterion requires the one-step arm to beat both static distillation and shuffled futures, with the upper endpoint of each sequence-bootstrap difference interval below zero. The bootstrap resamples eight complete sequences while keeping the three seeds together, so its interval remains conditional on those seeds and on historically exposed H sequences.

Table 36. Frozen-encoder probe after the fixed-teacher pilot, averaged over seeds 7, 13, and 21. The target is annotated log-height change (dimensionless), not TTC in seconds. A ridge probe fits 512 samples from 24 sequences and evaluates 256 samples from eight other, historically exposed sequences. Neither prespecified superiority test passes.

Training arm	Probe MAE
Static distillation	0.03763239
One-step prediction	0.03564004
Shuffled future	0.03506189
Random initialization	0.03719641

The candidate's probe MAE is 0.03564, compared with 0.03763 for static distillation and 0.03506 for shuffled futures. Its candidate-minus-static difference is −0.001992, with interval $[-0.004962, 0.0008190]$; candidate-minus-shuffled is 0.0005782, with interval $[-0.002170, 0.002926]$. Neither superiority criterion passes. The random-initialization control gives 0.03720. The intervals leave the superiority of one-step prediction unresolved under these controls.

The auxiliary latent metrics clarify the limitation. Future teacher-state MSE improves relative to both controls, but teacher-increment MSE worsens. The corresponding candidate-minus-static changes are −0.01719 for state and +0.03848 for increment; candidate-minus-shuffled changes are −0.05317 and +0.004474. Predicting a future state can compensate for present-state errors or model variation irrelevant to expansion, but the experiment does not identify either explanation causally. Its results establish that lower state-prediction error was insufficient for the prescribed physical probe. The pilot therefore stopped at this probe comparison. [N04]

P.3 Target-specific observability and model error

For a linearized observation model $\mathbf{y} = H\boldsymbol{\beta} + \boldsymbol{\epsilon}$ and target $q = \mathbf{c}^\top\boldsymbol{\beta}$, noiseless identifiability requires $\mathbf{c}$ to lie in the row space of H. In the affine image-motion parameterization, isotropic expansion uses $\mathbf{c} = (1/2, 0, 0, 1/2, 0, 0)^\top$. The target can be estimable without every affine coefficient being identifiable. Conversely, a pseudoinverse always returning a number does not certify that the target is supported. This algebraic condition is classical and is used here as a diagnostic. [N05]

On the 149-query screen, the diagnostic marks q estimable in 63 queries. Among those estimates, 26 have an incorrect sign, ten lie outside the canonical phase domain, and eight have zero residual degrees of freedom. A small fitted residual in the last group cannot demonstrate reliability. The target itself also need not equal local isotropic image expansion when rotation, parallax, or changing visible surfaces violate the measurement model.

The gain-prediction experiment compares physical descriptors against simpler support descriptors with leave-one-acquisition-out fitting. Physical-feature MSE is 0.9193, versus 0.4920 for support features; their difference is 0.4272 with interval $[0.08055, 1.010]$. This is MSE for predicting correction gain, not TTC MAE. The result does not support adding the proposed descriptors. A historical robust readout returns only eight usable estimates, four with the wrong sign; after preserving F0 fallback, its MAE is 0.7795 s versus 0.7660 s for the matched R0 baseline. The comparison records unavailable measurements and the original missing output; the error of the eight supported measurements is not presented as complete-population performance. [N05]

P.4 Sensor ambiguity and the missing reference

The sensor-identifiability bank constructs 24 pairs of matching event streams, with pixel, polarity, and timestamp agreement to a tolerance of one picosecond. Twelve pairs have different inverse TTC. Under the bank's log-affine texture model, changing motion and inversely scaling the texture gradient can preserve the increments that generate events; the distinct initial references are absent from the event stream. Providing a known reference helps distinguish the restricted conditions. This construction diagnoses an ambiguity in the tested model, not a general impossibility theorem for event-based estimation. [N05]

The real-data proxy uses 72 road queries from twenty-four acquisitions. Slider is excluded from the photometric fit because its alignment remains provisional. On common support, observed RGB change gives sensor-residual MSE 96.66 versus 145.6 for the physical proxy. Its improvement over an activity control is less decisive: the corresponding descriptive difference interval includes zero. Sensor-residual reductions cannot be interpreted as TTC improvements, and the inventory does not establish calibrated contrast thresholds, gain, initial reference, or refractory response.

The direct rate readout tests that implication. The affine photometric model permits translation, deformation, and global illumination variation and returns 72 valid rate estimates. With the original fallback retained, its macro TTC MAE is 9.169 s, compared with 0.7660 s for R0. The spatially permuted control is worse still, at 45.63 s. Beating that control is insufficient when the matched estimator deteriorates so strongly. The experiment opens no producer or JEPA training and supplies no calibrated confidence signal for fusion. [N05]

P.5 Surfaces and dynamic clocks

For positive depth Z, signed inverse TTC is $q = -\dot{Z}/Z$. Differentiation gives

$$\dot{q} = q^2 - \ddot{Z}/Z. \tag{47}$$

If axial relative velocity is constant in this coordinate, its finite-time evolution is

$$q(t+\delta) = \frac{q(t)}{1-\delta q(t)}, \qquad T(t+\delta) = T(t) - \delta. \tag{48}$$

The transport cannot cross the contact singularity. Acceleration, camera rotation, or a change in the visible surface can invalidate the assumptions even when an object has constant velocity in world coordinates. These kinematic identities motivate transport and observer rules whose utility is tested against the matched event-context estimator. [N05]

A three-frame surface route supplies 133 of 149 measurements and falls back on the other sixteen. Its system MAE is 3.104 s versus R0's 0.7660 s, with a paired difference interval $[0.9342, 4.198]$ s. The crop audit finds no changing crop within the 146 auditable histories; 53 contain a constant anisotropic resize. This rules out one proposed changing-zoom explanation for that cohort, not errors from occlusion, backgrounds, or parallax.

The constant-axial-velocity transport law above is exact only under its model assumptions. Reapplying its composition identity inside an exact implementation adds no new measurement. The follow-up compares fixed persistence, smoothing, transport, and observer rules against their matched event context after canonical PAIR. Context macro MAE is 1.942 s; EWMA gives 1.953 s, the observer 1.981 s, and persistence/transport 3.104 s. These comparisons do not establish a benefit. Their baseline is the event context, which must remain distinct from the multimodal R0 baseline used above.

Across these experiments, the limiting factor is measurement quality: algebraic support, low sensor residuals, and exact transport identities do not yield more accurate TTC under the tested conditions. Further model development should target the observed sign errors, missing reference information, and motion-model mismatch before adding these measurements to fusion. [N05]

Q. Detection, geometric consistency, and the localization boundary

The TTC evaluations in this paper condition on supplied object regions. A separate eAP study develops a portable RGB MonoDGP-R50 variant, a proposal ranker, and an interface from tracked detections to PHASE-TTC. This study examines the localization dependency. Its detection scores, geometric diagnostics, and integration checks do not establish improved end-to-end TTC accuracy. The detector is also distinct from the event representation and predictive-learning experiments. [N06]

Q.1 Fixed proposals and an aligned evaluator

On 23,958 development-exposed frames, the aligned eight-class evaluator gives mAP 0.05702 for the retained Dfull detector and 0.06204 after a TRAIN-fitted shared logistic ranker. Both evaluations use the same 1,197,900 proposals and 190,469 raw targets. Ranking changes the scores while preserving proposal geometry, classes, and velocities. Bicycle and other-vehicle AP remain zero; the aggregate gain does not imply uniform class improvement. The study uses one detector seed and a selection population reused during development. [N06]

Evaluation reconciliation matters here. Earlier AP101 reports used a restricted target population and a precision envelope. The aligned calculation uses greedy class matching in ego-XY coordinates, strict distance thresholds of 0.5, 1, 2, and 4 m, recall interpolation without the envelope, and the declared minimum-recall and minimum-precision rules. The two calculations are different estimands. Their historical records retain the distinct evaluation definitions. The ranked submission obtains official AP 0.05130 on 35,940 test frames; no paired baseline submission exists, so the local ranking gain cannot be transferred to the official test. [N06]

Q.2 Ideal cuboid inversion and predicted geometry

The geometric audit asks whether a height-only depth prior has a bias even with ideal inputs. Let $\boldsymbol{\delta}_i$ be a rotated cuboid-corner offset, $\mathbf{u}$ the projected center, and K the camera intrinsic matrix. Define

$$\mathbf{a}_i = K_{1:2,:}\boldsymbol{\delta}_i - \mathbf{u}\delta_{iz}. \tag{49}$$

For a positive center-to-edge distance t_e, cuboid containment gives

$$z_e = \max_i \left(\frac{s_e(\mathbf{a}_i)}{t_e} - \delta_{iz} \right), \tag{50}$$

where s_e selects the signed horizontal or vertical numerator for that edge. The implementation takes a median over non-clipped edges; a clipped image boundary is not treated as an equality. The calculation enforces pinhole geometric consistency. [N06]

With target-derived dimensions, yaw, and projected edges, the admitted set contains 25,424 objects from 4,096 TRAIN frames. On the same admitted objects, the height prior has depth MAE 4.923 m, whereas cuboid-edge inversion reaches 2.1085×10^{-6} m. The near-zero error verifies consistency with the ideal target generator. It does not measure sensor calibration or learned depth accuracy.

A second experiment substitutes predicted geometry on a *different* 4,096-frame TRAIN panel. It preserves the proposal pool, class, score, size, and yaw, but replaces the learned depth output and removes its learned residual. Baseline mAP 0.2106 falls to 0.1648 with cuboid-edge depth and to 0.08308 with the height-only prior. Invalid geometric cases use the declared fallback. Thus the better ideal inversion does not improve this direct substitution. No model is promoted, and the experiment does not rule out retraining a compatible residual or estimator. [N06]

Q.3 Negative screens and integration scope

The later range, class-budget, query-ranking, KITTI-transfer, and MonoCoP campaigns do not justify replacing the retained detector. Short, single-seed screens and differing upstream training exposure restrict their interpretation; they are not evidence that every related architecture or transfer recipe must fail. The unpromoted candidates remain in the experimental record. [N06]

An eight-frame integration replay produces five available signed TTC outputs and one unconfirmed track in the last frame. This demonstrates execution of the detector–tracker–TTC interface. Frozen TRAIN40 producers had already seen these sequences, and the replay supplies no independent end-to-end TTC accuracy estimate. The admission protocol also distinguishes target time from availability: the target is anchored at RGB exposure start, while inference can consume observations through exposure end. The resulting estimate has a measured availability lag and must not be described as causal at the earlier target instant. These boundaries remain necessary when replacing supplied boxes with predictions. [N06]

R. Evolution of experimental methodology and integrity controls

PHASE-TTC developed from predictive event representations into geometric motion estimation, phase-domain temporal refinement, and event–RGB fusion. The methodological changes below explain how evaluation controls evolved alongside the model. Historical experiment identifiers and configurations remain in the evidence registry; Table 38 collects the associated code revisions.

R.1 Predictive representations and causal measurements

Initial experiments established indexed EvTTC access, voxel representations, a supervised TinyCNN, and geometric and event-rate references. Multihorizon predictive learning, frozen-encoder probes, and low-label comparisons tested the original E-JEPA-TTC hypothesis. Token transformers, tubelets, navigation conditioning, and rollout probes extended the temporal models.

Causal validity required changes to both the measurements and evaluation procedure. Centered geometric derivatives used future observations and were replaced with past-only derivatives. This annotation-assisted reference was evaluated on its supported queries, with coverage reported separately from methods using different inputs. The same distinction later informed the separation of sensor support, prediction availability, and complete-system accuracy.

R.2 Sequence isolation and initialization dependence

The early CPLA-high comparisons reused inspected sequences and shared one self-supervised initialization across three fine-tuning seeds. They consequently measure downstream optimization variability on development data. Checkpoint provenance, cache versions, nested label subsets, and matched-scratch comparisons were added to the protocol, and these runs were excluded from independent-test conclusions.

A separate validation-path error recorded completion without executing the intended evaluation matrix. The corrected path executes the matrix and checks its results; the earlier completion markers provide no experimental evidence. This correction distinguishes verified evaluation outputs from the status of a software run.

The first grouped geometry study also initialized held-out folds from a representation trained on all nine development sequences. Fold-specific producer training removed that overlap before downstream fitting. Checkpoint selection and earlier sequence exposure still define these studies as development comparisons. [E03]

R.3 Geometry, transport, and controlled model selection

Object-centred experiments separated absolute geometry from changes in apparent scale. Foreground supervision, RGB-derived segmentation masks, and relational distillation tested spatial information; cross-time transport tested motion correspondence. Their supervision sources are recorded separately from their inference modalities.

Scaling the relationally distilled student from 2,048 to 8,192 training examples while increasing its train-selected distillation weight reduced MiD from 322.7 to 262.8 at unchanged capacity. The missing same-size $\lambda = 4$ control prevents separate attribution to data and loss weight. This single-seed result preceded the fold-local recovery studies. [H_A4_S1]

The fold-local transport studies compared accuracy and geometric retention under matched samples and optimization budgets. No model in the two initial recovery campaigns met all selection criteria. Complementary errors between the baseline and coarse-to-fine producers then motivated a prospective nested-routing comparison, accompanied by frozen, partial-finetuning, random, scratch, and shuffled-future predictive-learning controls. [E03--E11,E16--E24]

R.4 Temporal context and multimodal estimation

Phase-domain studies examined incremental transport information, raw timing, expert routing, risk estimation, and context with fixed optimization budgets. The original residual-context campaign completed 72 fits and 180,000 saved updates. Its eight-slot history improved on the current-observation head by 11.3 MiD across three head seeds, with a hierarchical 95% difference interval of [−16.2, −6.6] on reused development data. Trained reversal, repeated-current, selection, and latent-input controls constrain the mechanism interpretation (Table 28). A later extension completed 24 fits and 60,000 updates, reusing original H8 endpoints; the two totals describe distinct campaigns. Short- and long-history heads shared upstream producers. Capacity comparisons retained incomplete training endpoints as partial records. [H_T6,M_SIMPLEX]

Efficient input preparation and the RGB port added native sensor clocks and transfer evaluation.

The intervening multimodal selection campaign completed 32 fits and 80,000 updates. Its phase-only correction passed the original 2% band safeguards in three head seeds, retaining seed 7. This historical selection used in-sample stacking on H; a separate four-fold H cross-fitting experiment failed the MAE criterion. Both differ from the subsequent five-rotation TRAIN40 refit. Section N.5 records their roles and the different event controls used in V16 and V18. [H_V16]

Later range and representation studies tested clipping, nonlinear readouts, geometry bypasses, encoder adaptation, and loss weighting. Several variants improved individual TTC bands while worsening receding-motion estimates or another selection metric. None of the joint-measurement variants met the complete selection criteria. The earlier phase-correction recipe remained the reference for the final refit, while these unsuccessful variants motivated separate negative-range controls. [M_SIMPLEX,L02,L10,L24,L25,L30]

R.5 Final refitting, adaptation, and execution

The submitted TRAIN40 lineage follows the phase-correction recipe, using group-separated auxiliary models for out-of-fold fitting and all forty sequences for deployed producers. The official GarlTTC Test evaluation covers the resulting complete system (Section 4).

Runtime studies examined persistent histories, exact normalization, graph execution, crop reuse, and multiple objects. Timing separates initialization from measured arrivals and successful outputs from failed requests. Smaller event packets preserved numerical parity in the tested setting; throughput requires a separate paced measurement.

Subsequent work recovered all 434 local EvTTC targets, fitted the quality-conditioned adapter, and tested canonical cadence throughout the frozen estimator. The adapter and later cadence refits remain separate variants. Predictive-learning and physical-measurement controls, together with a separate detection study, examine the remaining representation and localization dependencies. Sections 7 and D, and Appendices P and Q, report their protocols and outcomes.

S. Open science and reproducibility

The manuscript sources, analysis scripts, result ledgers, and selected query-level predictions are available in Kripta-Studios/phase-ttc. The source package supports regeneration of the reported tables and figures. Repeating training or sensor inference additionally requires the datasets, checkpoints, and configurations identified in the evidence registry.

S.1 Manuscript and result regeneration

Install Python with NumPy, Matplotlib, Pillow, and PyMuPDF, together with pdfLaTeX and BibTeX. From the manuscript source root, run:

```
python scripts/build_expanded.py
python scripts/verify_publication.py
```

The builder verifies the distributed inputs, preserves historical figures, regenerates the expanded result tables, and compiles both modular and consolidated sources. After dependency installation, this process uses bundled data without network access or raw sensor files. A publication receipt binds the scientific inputs, compiled PDF, consolidated source, and build validation by SHA-256. Packaging rejects changed inputs or artifacts until the manuscript is rebuilt. Earlier editorial receipts identify their historical PDF hashes and do not validate the corrected delivery.

Each result ledger identifies its evaluation population and measured quantities. Verification scripts recompute the distributed query-level metrics and check their source identities. Hardware-specific parity requires the corresponding inputs, checkpoints, and runtime environment.

S.2 Numerical and software verification

The source checks cover thirteen data-file hashes, eight transfer summaries, twelve official-scorer fields, and weighted Overall MiD. The competition verifier recomputes coverage, signed ranges, disagreement, and timing summaries across 1,736 model–query rows. Official scores reconcile with the dated leaderboard after rounding.

Table 37. Additive decomposition of the official Overall MiD difference. Each band contributes $w_b(\mathrm{MiD}_b^{\mathrm{Garl}} - \mathrm{MiD}_b^{\mathrm{PHASE}})$. The 0.0010-point reconciliation residual comes from Garl-TTC's published two-decimal band scores and appears separately below. [B01, B02]

Band	Weight	MiD difference	Weighted contribution
Crucial	0.5	9.6110	4.8055
Small	0.3	9.3232	2.7970
Large	0.1	14.8921	1.4892
Negative	0.1	1.3586	0.1359
Band subtotal	1.0	–	9.227548419
Published-score rounding residual	–	–	0.0010
Official Overall difference	–	–	9.228548419

The historical transfer analysis under `revision/` uses 468 eligible Extended-new rows under the original target-source assignment. Correcting the CCRs1 low/medium sources and recomputing eligibility yields 460 eligible rows from the same 508 candidates; the corrected analysis and source map are retained separately under `audit_20261104/`. Both analyses retain the saved predictions and use complete sequence groups for paired bootstrap resampling. Their fixed seed, quantile convention, paired differences, and source digests distinguish corrected analysis from the historical snapshot.

The recorded software checks cover event representations, target updates, causal prefixes, label rejection, source tracking, fusion fallbacks, and paced-runtime accounting. With Python 3.11.15, PyTorch 2.11.0+cu128, and pytest 9.1.1, two batches passed 178 tests with four skips. The original M21 XML receipts record 70 cases with four skips and 112 cases without skips, respectively, with no errors or failures; the corresponding logs and XML files are distributed as `data/verification_architecture.*` and `data/verification_protocol.*`. These files retain the SHA-256 identities in the architecture source index and report the historical runs of 28 October 2026. Rebuilding the paper does not rerun these tests. Four further tests passed against the fixed official evaluator, covering four-band perfect predictions, imminent-contact weighting, missing bands, and invalid outputs. [M21, M24]

The competition adapter and verifier passed 41 focused tests; the CPU suite skipped one CUDA-dependent case. A separate CUDA check confirmed strict loading and native-model forward parity for all three public Garl checkpoints. Validation also checked the 1,302 Garl outputs, 434 PHASE predictions, query identities, input timing, signed height-ratio conversion, and result completeness. These are the archived execution results, separate from rebuilding the manuscript. [M21, M24, T01]

S.3 Source revisions and methodological corrections

Table 38 centralizes code revisions associated with the scientific record. The evidence registry identifies 95 original result snapshots by SHA-256 and records their available numerical projections. Entries containing only source identity require the corresponding original artifacts for metric recomputation. The archived scorer JSON and public leaderboard identify the official result; the scorer JSON does not contain a submitted-prediction digest.

Table 38. Source revisions for reproduction and protocol corrections. Historical identifiers remain unchanged in the repository.

Scope	Git revision
Scientific source snapshot	`0219a84f0fc3b07c171295a8a65176a7e0f631c1`
Competition adapter	`1d8de7d95b2ff8ccba884dbd40141c48745a1e95`
CPLA-high exposure classification	`26480260`
Evaluation-matrix execution correction	`1f9e2f99`
Fold-specific producer correction	`de3349fd`

The competition projection includes predictions, measurement boundaries, checkpoint identities, and source digests for coverage, disagreement, and timing analysis. Raw sensor data remain subject to their dataset distribution terms. `ASSETS.md` records image attribution and display transformations; `data/real_sensor_excerpt/` preserves the real sensor excerpt's provenance, timestamps, and original asset hashes.

S.4 Runtime configuration

The L40 receipts specify the final component identities and the separate paced-replay input and clock. Model loading, synthetic shape priming, and history initialization precede measured arrivals; unexpected preparation during updates remains in latency. The records identify the laptop GPU and execution settings but omit the CPU model and GPU-driver version. [L40,R02]

Table 39. Paced-replay hardware, configuration, and initialization timings. Startup stages were timed separately and excluded from the arrival clock; unexpected shapes were processed within the measured update. [L40,R02]

Item	Recorded value
GPU	NVIDIA GeForce RTX 5070 Ti Laptop GPU
Software	Python 3.11.15; PyTorch 2.11.0+cu128; CUDA build 12.8
Precision	FP32; TF32 disabled for matrix multiplication and cuDNN
CPU / preparation	2 PyTorch CPU threads; 8 RGB workers; queue capacity 2
Parity tolerance	Absolute and relative tolerance both 10^{-5}; all 250 outputs checked
Outside the measured arrival clock	
RAW preload	2.769 s
Model and service initialization	0.516 s
Synthetic shape priming	12.095 s; 7 layouts; synthetic tensors only, no RAW/future sensor data
History initialization	0.132 s for 600 ms of preceding sensor history
Source warmup	0.051 s; 13 already-available RGB frames
Unrecorded hardware fields	CPU model and GPU driver version

S.5 Expanded edition

The expanded result tables regenerate from the copied JSON ledgers under `updates/`. The build preserves original sections and figures, updates conclusions where later measurements supersede them, and includes the source index in Appendix U. Rebuilding the document reuses recorded model outputs. The source manifest binds the final manuscript, figures, ledgers, and validation receipts; the source ZIP includes the assets needed to compile the consolidated LaTeX file.

T. Evidence index

Source keys map each reported measurement to its result artifact. Numerical ledgers drive table and figure regeneration, while `data/public_evidence_registry.json` records full digests and field selectors. The index separates original-artifact hashes from distributed-file hashes. *Digest only* entries preserve source identity without distributing the fields required for metric recomputation. B02 denotes the official scorer JSON.

T.1 Result-source identities

Source key	Distributed ledger	Original SHA-256
E01	data/early_results.json	55b5494b40f97f1b
E02	data/early_results.json	27ee1c9a21cca0ec
E03	data/early_results.json	aa46f6d199bcf5a4
E04	data/early_results.json	02c215aa4678f307
E05	*Digest only*	6fe8c8761d0a14fb
E06	*Digest only*	d38021db5b3b5b62
E07	data/early_results.json	d53b6ce406fa7b80
E08	data/early_results.json	d69f8b9082c883da
E09	data/early_results.json	9bf197c3d64635a3
E10	data/early_results.json	d47face1e36d1d2c
E11	data/early_results.json	6db419fd7c7fe536
E12	data/early_results.json	f33dc9abbad8fd08
E13	*Digest only*	ac26c1b2ad87c6c9
E14	*Digest only*	f947b32f05e655a1
E15	*Digest only*	7c60bc0fa9f1c5f0
E16	data/early_results.json	be5fdbc0b2e9773d
E17	data/early_results.json	d3fb5a5feb848ac7
E18	data/early_results.json	a3c4765f5278a3f4
E19	data/early_results.json	f1471599d502f2de
E20	data/early_results.json	3e02fd81751e3f86
E21	data/early_results.json	5c7b1eb24c07bd08
E22	*Digest only*	2e2eef6698ef28a4
E23	data/early_results.json	bc540cc21ddb317f
E24	*Digest only*	dab563a02450abe6
E25	data/early_results.json	06619299d1ceb785
E26	data/early_results.json	06dc45cf3ea7a3e0
M_SIMPLEX	data/middle_results.json	93f83e71f86dfcad
M_HEAD_COST	data/middle_results.json	8d96bbc9bd66d8fb
M_ROUTE_COST	data/middle_results.json	d1d834ab36d14fbd
M_WIDE	*Digest only*	c15e32a8dea0eece
M_EWMA	*Digest only*	47cca03ccda6f054
M_STAGE61	data/middle_results.json	20f1990bb70e5de7
M_STAGE62	data/middle_results.json	c4a0e1ae20d79094
M_GEOMETRY	*Digest only*	fa917902b5696e4b
M_STAGE67_DECISION	*Digest only*	ab1d3095955477f4
M_STAGE69_DECISION	*Digest only*	dfd1956775492584
M_ENGINEERING	*Digest only*	e756ba5c7a433efd

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Source key	Distributed ledger	Original SHA-256
M_SCIENCE_AUDIT	*Digest only*	06073f4182de1a41
M_STAGE63_REPAIR	*Digest only*	f96b7fe795637cc5
M_CAPACITY_PROTOCOL	*Digest only*	1be3a4c9f9b64a08
M_CAPACITY_AUDIT	*Digest only*	07ef97a56b4159fa
M_STAGE70_LEDGER	*Digest only*	62597e162571def0
M_TRANSFER_REPORT	*Digest only*	d92c9074516354bd
M_ROUTE_REPORT	*Digest only*	1f29a459c91f6b68
M_TEMPORAL_MODEL	*Digest only*	4c26ad5ecf9e8354
M_PHASE_CONTRACT	*Digest only*	7ee447190ef7b218
M_STAGE64	*Digest only*	68409fb88499899f
M_X0	*Digest only*	61d8a0e40b6d87ad
M_X05	*Digest only*	c4aa55dd6a944d01
M_X05_GATE	*Digest only*	61eae7d38fb6ea5a
M_STAGE65	*Digest only*	4f88f676eecc7e90
M_STAGE67_DIAGNOSTICS	*Digest only*	885ac3604160cf27
M_S67_SIMPLEX17	*Digest only*	969cf5db29b2a251
M_S67_UNCONSTRAINED17	*Digest only*	1419522be60fc952
M_S67_SIMPLEX8GATE	*Digest only*	e094512164179869
M_S67_SIMPLEX17_PAIRPERM	*Digest only*	451821bb98c1d0c2
M_S65_RISK17	*Digest only*	b964e12180efbcbc
M_S65_RISK8	*Digest only*	75d98c9e0be9ddc3
M_V12_EVTTC_946	data/middle_results.json	03c57b4503ca95f4
M_V12_EVTTC_946_PAIRED	*Digest only*	5fee507ffe8a4aca
M_V12_EVTTC_946_ROWS	*Digest only*	d284e52a0c25a897
M_V12_EVTTC_3640	data/middle_results.json	792a5042dfa8be8b
M_V12_EVTTC_3640_PAIRED	*Digest only*	2619cc3d11be7320
M_V12_EVTTC_3640_ROWS	*Digest only*	20f58c1200b3834d
M_V12_FCWD_597	data/middle_results.json	a1c0c4a741b08a36
M_V12_FCWD_597_PAIRED	*Digest only*	2581bd042f29a466
M_V12_FCWD_597_ROWS	*Digest only*	68f4cd5610f50c8b
L01	data/late_results.json	9136babd4a07411e
L02	data/late_results.json	844e0771bab79717
L03	data/late_results.json	4af996900a99c585
L10	data/late_results.json	f5bf12b2873a2ee4
L11	data/late_results.json	22e82a376b18c9f0
L12	data/late_results.json	fafd9cefbeddcd49
L13	data/late_results.json	86ce201ca25bcc12
L14	data/late_results.json	0588956e96dab44f
L20	data/late_results.json	6d15d1acfad38592
L21	data/late_results.json	2b33ba213186f3c5
L22	data/late_results.json	b1ad71b126b20e87
L23	data/late_results.json	0698e433731da0d3
L24	data/late_results.json	5f3789bd7efebce3
L25	data/late_results.json	774c751e4aaa91d7
L30	data/late_results.json	1073ab21e307ae26
L31	data/late_results.json	54ebca7a19d5a9eb
L32	data/late_results.json	d4fbc7494769a5d1
L33	data/late_results.json	025a058d846a7ceb
L40	data/late_results.json	c27d5d980f2bb1c1
L41	*Digest only*	d8df9ca6444c6dc0
L42	*Digest only*	06ab49e521db7255
L43	*Digest only*	aed0bdd40c58d827
L44	*Digest only*	e447c5147e4e1c25
L45	*Digest only*	4f643a15cbf3d963
L46	data/transfer_queries.csv	1fbff40f000c879b
L50	*Digest only*	3069072fb5180272
L51	*Digest only*	ad4a2a0660bd2da9
L52	*Digest only*	46e5516c40511837
B01	data/official_benchmark.json	0c7114c5acfded5e
B02	data/official_scoring_scores.json	e06e99ba1e5d6c4e
T01	data/evttc_competition_summary.json	96c849f29383c35f

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Source key	Distributed ledger	Original SHA-256
T02	data/evttc_competition_summary.json	ceb5c6b5f3ef1b93

T.2 Implementation and verification references

Implementation keys map source revisions, method descriptions, and selected verification results to their distributed manifests.

Source key	Distributed manifest	Ledger selector
M01	research/architecture_sources.json	id=M01
M02	research/architecture_sources.json	id=M02
M03	research/architecture_sources.json	id=M03
M04	research/architecture_sources.json	id=M04
M05	research/architecture_sources.json	id=M05
M06	research/architecture_sources.json	id=M06
M07	research/architecture_sources.json	id=M07
M08	research/architecture_sources.json	id=M08
M09	research/architecture_sources.json	id=M09
M10	research/architecture_sources.json	id=M10
M11	research/architecture_sources.json	id=M11
M12	research/architecture_sources.json	id=M12
M13	research/architecture_sources.json	id=M13
M14	research/architecture_sources.json	id=M14
M15	research/architecture_sources.json	id=M15
M16	research/architecture_sources.json	id=M16
M17	research/architecture_sources.json	id=M17
M18	research/architecture_sources.json	id=M18
M19	research/architecture_sources.json	id=M19
M20	research/architecture_sources.json	id=M20
M21	research/architecture_sources.json	id=M21
M22	research/architecture_sources.json	id=M22
M23	research/architecture_sources.json	id=M23
M24	research/architecture_sources.json	id=M24

T.3 Distributed data integrity

Digest prefixes identify the numerical files and manifests used for regeneration; the build verifies each full hash before analysis.

File ID	Distributed file	SHA-256 prefix
D01	data/early_results.json	bbc3e4b30c44e6af
D02	data/early_tables.json	e7c769877f14ecf2
D03	data/git_summary.json	40b8eac8566ed3de
D04	data/late_results.json	e7093f1b75a4e96b
D05	data/middle_results.json	236eb5dab7db1932
D06	data/official_benchmark.json	f7bc1a8bd786f814
D07	data/official_scorer_verification.json	98ccc01371304d3a
D08	data/official_scoring_scores.json	e06e99ba1e5d6c4e
D09	data/transfer_queries.csv	2fd4c33716a6c48e
D10	data/visual_source_manifest.json	7cd53750ff473479
D11	research/architecture_sources.json	a544f7aab9d12e34
D12	data/evttc_competition_summary.json	9ad728d4168c0b41
D13	data/evttc_competition_predictions.csv	5e58c4155d90cf5e

T.4 Post-hoc paired transfer analysis

R01 maps the historical `revision/derived_transfer_analysis.json` and its paired-bootstrap differences to `scripts/prepare_revision.py`. It retains the L46 predictions under the former 468-query target assignment. GT01 and GT02 identify the corrected 460-query projection and analysis under `audit_20261104/`. Appendix G gives the current sequence-resampling procedure and distinguishes the archived historical figures and numerical cases.

T.5 Deployment metadata and output map

R02 maps `revision/deployment_contract.json` to the native phase-output transformation and L40 paced-run configuration. It records original source digests, the executed code revision, model identities, hardware, precision, and initialization boundaries. R02 contains metadata from the recorded run rather than a new performance measurement. The build verifies its full digest before regenerating Table 39.

U. Expanded evidence: 29 October–2 November 2026

The preceding index preserves the original article's source identifiers and historical measurements. New identifiers N01–N07 refer to the later work; they do not overwrite E01–E08 or any original experiment key. The source package includes the new ledgers under `updates/evidence/`, their full SHA-256 digests, and branch and document-coverage manifests.

Source	Scope	Distributed source and interpretation
N01	Final quality adapter; corrected 434-query comparison	`updates/evidence/adapt.json`; corrected local evaluator with historical exposure and unresolved Slider-750 equivalence. Model description in the copied 1 November delivery report.
N02	Canonical PAIR, complete development population	`updates/evidence/canonical.json`; recording-macro errors, explicit availability, and paired recording-bootstrap intervals.
N03	Rejected paired refit; native output sensitivity	`updates/evidence/refit_result.json` and `updates/evidence/native_result.json`; the refit fails selection and the native comparison measures output changes without Test12 targets.
N04	Fixed-teacher predictive-learning pilot	`updates/evidence/jepa.json`; physical-increment probe and static/shuffled controls, with neither superiority criterion passed.
N05	Physical measurement diagnostics	The copied `updates/evidence/support/` records; source paths, exact field selectors, and original hashes accompany the interpretation.
N06	eAP detection follow-up	`updates/evidence/detection/source_manifest.json`; six branch-pinned JSON ledgers, including distinct ideal and predicted geometry panels.
N07	Runtime and local execution boundaries	`updates/evidence/runtime/summary.json`; the original successful-output paced population. The later raw-path comparison has its own copied receipts and a different workload and timing boundary.

U.1 Source coverage and numerical checks

The source inventory covers 241 commits across 72 refs in the period beginning 28 October, with experiment branches preserved unchanged. The 411 directly touched Markdown paths comprise 406 semantic or source-verified documents and five generated historical inventories checked structurally. The latter include older records outside the requested period. The manifest distinguishes these categories and preserves divergent branch versions.

Numerical checks cover transfer and fusion, runtime, eAP detection, and physical measurement. Equation checks, metric units, evaluation populations, and causal input boundaries constrain the interpretation of these records. These checks do not constitute independent experimental replications. The preparation of that 2 November evidence expansion involved no new training or official benchmark submission.

V. Scientific revision: sources and remaining experiments

The source audit of 4 November, incorporated into the corrected delivery of 11 November 2026, separates corrections to recorded measurements from new model experiments. Transfer targets are reassociated across all 508 registered Extended-new candidates using the corrected public low/medium sources, the original native query clocks, and the existing nearest-label rule. The resulting 460-query analysis retains every saved prediction and preserves the historical 468-query ledger under its original assignment. The complete derived candidate projection includes source hashes, neighboring label timestamps, eligibility changes, and the frozen model outputs. [GT01]

The additional records below restore the original temporal campaign, multimodal selection, adaptation lineage, physical solver, and runtime measurements. Commit-pinned copies are distinguished from local-cache receipts and derived checks in the source index. Recomputing saved metrics or reading a verified training receipt does not repeat training or sensor inference.

Table 44. Additional sources for the scientific revision. Complete commits, original paths and byte hashes are distributed in `audit_20261104/source_index.json`.

Source	Scope	Files
H_T6	Original temporal-context campaign and trained controls	1
H_V16	Multimodal selection and separate context cross-fitting	7
H_A4_S1	Relational-distillation scaling and unexecuted matched control	1
H_V16_ZERO	Historical phase-only zero-feature intervention	4
H_V18_ZERO	Later joint-measurement zero-event intervention	1
P01	Fold-local feature generation and deployed context fitting	3
P02	Study-specific bootstrap units, cache access and sequence-group limits	23
P03	Full encoder adaptation in the negative-band loss comparison	1
P04	Temporal-field construction and phase-only input slicing	5
A01	Head adaptation lineage and affected all-32 supervised cache	13
A02	Grouped adaptation, interval coverage and controlled comparisons	2
A03	Earlier quality gates and geometry-aware controls	3
A04	Startup campaign and its changed development population	3
A05	Subsequent small-head campaigns without accepted replacements	1
A06	Direct event-expansion solver and complete-population fallback	1
A07	Adapter submission, weight bindings and original latency records	8
A08	Historical prediction-difference distribution	1
N01	Corrected local evaluator, signed and offline-scoring views	1
L40	Original per-arrival paced-runtime measurements	1
GT01	Transfer target reassociation, source identity and unchanged native clocks	5
M21	Original archived architecture and protocol test receipts	4
GT02	Corrected transfer statistics, sequence bootstrap and derived figures	1

V.1 Artifact availability by experiment family

The source archive contains the reported projections, source snapshots, and their identities. Full training or sensor inference additionally requires the recorded datasets, model weights, and runtime dependencies; the manuscript archive does not bundle all of those inputs.

Table 45. Available evidence and requirements for further reproduction. Recomputing saved outputs is distinct from repeating a fit.

Variant	Distributed evidence	Additional requirement
Official TRAIN40	Score, component identities, code snapshots, training accounting	Original checkpoints and sensor inputs for inference or refitting
Historical and corrected transfer	Fixed predictions, both target assignments, corrected eligibility, bootstrap builder	Pinned public curves and query manifest for repeating the full target join
T6, V16 and A4-S1	Original reports, accounting, controls and source revisions	Original features, splits and initializations for repeated training
All-32 F1	Cache ancestry and affected target-source identities	Corrected supervised cache and three new fits
Grouped F1 and quality gates	Population, fit results, calibration summaries and weight lineage	Per-variant inputs and checkpoints; matched final geometric controls now appear in Section 7
Runtime and test checks	Original timing summary, derived timing checks and original M21 receipts	Exact hardware/runtime for a new latency run; test environment for a fresh execution

V.2 Experiments that remain outstanding

The all-32 F1 checkpoints depend on 239 admitted low/medium targets under the old assignment. Their supervised cache and eligibility must be rebuilt before refitting seeds 7, 13, and 21. The source audit verifies that the delivered quality router uses the original F0 experts instead of those F1 weights; the two weight lineages remain distinct. [A01, A03, A04, A07]

Section 7 now supplies the matched 434-query comparison of geometry alone, fallback, a fixed mixture, and the learned router. Independent recordings, a reconciled Slider-750 evaluator, calibrated intervals for the delivered all-data router, and sustained physical-camera execution also remain separate experimental requirements. Historical results from earlier gates and different populations are not substituted for these comparisons. [N01, A03, A04, A07]

W. Reproduction of the matched revision experiments

The new comparisons use frozen source features and saved causal predictions. Their protocols, complete query outputs, source snapshots, seed traces, and result bindings are distributed under `data/consolidation/`. The nine source-only head fits are additional experiments; they do not replace the released weights or receive a new official benchmark score. Running `scripts/build_consolidation.py` regenerates their main-text tables.

W.1 Identity and missing-feature recovery

We join transfer rows by query identity, requiring exactly 460 eligible queries with no duplicates or omissions. The original local feature cache contains 457 of them. Its obsolete target prefilter excluded three queries before feature extraction. We recover those three from their sensor observations with the unchanged sealed nine-component graph, capturing the final head inputs without reading TTC labels. The released head reproduces all 460 saved predictions within absolute and relative tolerance 10^{-5}; the maximum absolute difference is 2.8610×10^{-6} s. This validates the features for the head comparison and does not resolve the separate historical full-inference discrepancy.

W.2 Training and coordinate control

The source cache contains 88,744 eAP out-of-fold rows from forty recordings plus separate local EvTTC rows. Only the eAP rows enter these fits; domain, identities, group counts, cache hash, and original feature ancestry are checked before training. Zero-event heads erase all ten event descriptor fields both during training and evaluation. They retain the same twenty-input network, RGB skip connection, widths, parameter count, objective, schedule, and seeded minibatch order as the real-event arm.

For the inverse-TTC variant, each of the ten phase inputs becomes $(1 - e^{-\phi})/0.03$, with the native phase clamp applied first. Timing and support fields are unchanged. Its point rate is $q_R + 0.3c_0$, where $q_R = (1 - e^{-\ell_R})/0.1$; the native rate bounds are applied before conversion back to phase. Quantile-width corrections and the loss remain in phase coordinates. This compares a specified inverse-rate parameterization at the original budget, rather than tuning each coordinate family for its own best score.

All three head arms receive 12,044 updates with AdamW, batch size 128, weight decay 0.001, gradient-norm clip 1, and a learning-rate schedule warming to 3×10^{-4} then decaying to 3×10^{-5}. The final affine layer starts at zero. Every seed's final checkpoint is evaluated; there is no EvTTC-based early stopping or seed selection. Seed means describe separately scored models. For paired intervals, we average each query's loss difference over seeds, resample all rows of complete recordings, and recompute the query-weighted mean. We never average TTC predictions to obtain the head-comparison scores.

The scalar mixture uses 101 equally spaced event weights between zero and one. Its selected weight minimizes source-training recording-balanced absolute phase error. Evaluation targets do not enter calibration. The fixed mixtures and geometric comparison use their specified equal weights without fitting.

W.3 Coverage and local scoring

The geometric comparison includes 434 queries, 424 with a valid geometric estimate. Missing geometry stays unavailable in the geometry-only arm; fallback and fixed-mixture arms return the original fused output on those rows. Primary comparisons retain signed runtime predictions. A separate saved analysis applies offline negative interpolation only where its stated endpoint and finite-support requirements hold, recording unavailable rows rather than inventing a scoring rule. The published primary table uses raw signed outputs throughout.

W.4 Corrections and unresolved reproduction questions

The V18 replication record identifies its evaluation cohort as V. Table 13 corrects the previous H label while preserving the archived numbers and bootstrap protocol. The H normalization and fitting population belongs to a different stage.

The V8 result archive supplies signed predictions for all five controls and three folds. Its aggregate matches the manuscript's original E23 digest; recomputation from associated predictions differs from the cached D2/D3 values by at most 0.1772 MiD. The complete recomputed table exposes that discrepancy and retains the original aggregate. Neither this small inconsistency nor the 504 historical native-replay tolerance failures is treated as resolved by current-platform repeatability.

W.5 Requirements for independent confirmation

The author reports no known unused recordings with synchronized events, RGB, and compatible TTC labels. The public FCWD data have already appeared in the project's development record. The public T^2CEF tooling derives dense TTC and ego-flow from M3ED poses and depth; this does not directly supply the object-track TTC and localization contract used here. Adapting that data would require a new target-association protocol before evaluating predictions. It is a potential separate study, not an independent confirmation included in these results. The public source assessment is recorded with links in the revision disposition file. No new independent recording is claimed by the present revision.